20 July 2017
EMA/496529/2017
Committee for Medicinal Products for Human Use (CHMP)

Assessment report

Bavencio

International non-proprietary name: avelumab

Procedure No. EMEA/H/C/004338/0000

Note

Assessment report as adopted by the CHMP with all information of a commercially confidential nature deleted.

30 Churchill Place ● Canary Wharf ● London E14 5EU ● United Kingdom
Telephone +44 (0)20 3660 6000 **Facsimile** +44 (0)20 3660 5555
Send a question via our website www.ema.europa.eu/contact

An agency of the European Union

Table of contents

List of abbreviations

1L	first line
2L	second line
2L+	second line or later
ADA	anti-drug antibody
ADCC	antibody-dependent cell-mediated cytotoxicity
ADR	adverse drug reaction
AE	adverse event
AESI	adverse event of special interest
AEX	anion exchange chromatography
ALT	alanine aminotransferase
AST	aspartate aminotransferase
AUCss	area under the serum concentration-time curve at steady state
BOR	best overall response
CEOI	concentration at the end of the infusion
CHO	chinese hamster ovary cells
CL	total systemic clearance
CLL	chronic lymphocytic leukemia
CMA	conditional marketing authorization
Cmax	maximum plasma concentration observed postdose
CPI	Relative Cluster pI
CPP	critical process parameter
CR	complete response
CT	computed tomography
Ctrough	concentration at the end of the dosing interval
DDI	drug-drug interaction
DMRIE	1,2-dimyristyloxy-propyl-3-dimethyl-hydroxy ethyl ammonium bromide (transfection reagent)
DOR	duration of response
DP	drug product
DRR	durable response rate
DS	drug substance
ECG	electrocardiogram
ECOG	Eastern Cooperative Oncology Group
FT4	free thyroxine
HCP	host cell proteins
HIV	human immunodeficiency virus
HMW	high molecular weight
HRQoL	health-related quality of life
ICI	immune checkpoint inhibitor
IERC	Independent Endpoint Review Committee
IFNγ	interferon-γ
IgG1	immunoglobulin G1
IIV	interindividual variability
IL	Interleukin
IPC	in process control

IQR	interquartile range
irAE	immune-related adverse event
IRR	Infusion-related reaction
ITT	intent-to-treat
LMW	low molecular weight
mBC	metastatic breast cancer
MCB	master cell bank
MCPyV (MCV)	Merkel cell polyomavirus
MM	mixed mode chromatography
mMCC	metastatic Merkel cell carcinoma
NAb	neutralizing antibody
NCA	non-compartmental analysis
NK	natural killer
NSCLC	non-small cell lung cancer
ORR	objective response rate
OS	overall survival
PBMC	peripheral blood mononuclear cell
PD-1	programmed death 1
PD-L1	programmed death ligand 1
PFS	progression-free survival
PK	pharmacokinetic(s)
Pop PK	population pharmacokinetic(s)
PR	partial response
QTc	QT interval corrected for heart rate
QTcF	QT interval corrected for heart rate by Fridericia's formula
QTcP	QT interval corrected for heart rate by a project specific factor
RECIST 1.1	Response evaluation criteria in solid tumors version 1.1
t1/2	terminal elimination half life
TEAE	treatment-emergent adverse event
TNF- α	tumor necrosis factor-α
TO	target occupancy
TTP	time to progression
UF/DF	ultrafiltration/diafiltration
V1	central volume of distribution
V2	peripheral volume of distribution
Vss	volume of distribution at steady state
WCB	working cell bank

1. Background information on the procedure

1.1. Submission of the dossier

The applicant Merck Serono Europe Limited submitted on 6 October 2016 an application for marketing authorisation to the European Medicines Agency (EMA) for Bavencio, through the centralised procedure falling within the Article 3(1) and point 4 of Annex of Regulation (EC) No 726/2004. The eligibility to the centralised procedure was agreed upon by the EMA/CHMP on 15 September 2016.

Bavencio was designated as an orphan medicinal product EU/3/15/1590 on 14 December 2015 in the following condition: Treatment of Merkel cell carcinoma.

The applicant applied for the following indication: Bavencio is indicated for the treatment of adult patients with metastatic Merkel cell carcinoma (MCC).

Following the CHMP positive opinion on this marketing authorisation, the Committee for Orphan Medicinal Products (COMP) reviewed the designation of Bavencio as an orphan medicinal product in the approved indication. The outcome of the COMP review can be found on the Agency's website: ema.europa.eu/Find medicine/Human medicines/Rare disease designation.

The legal basis for this application refers to:

Article 8.3 of Directive 2001/83/EC - complete and independent application

The application submitted is composed of administrative information, complete quality data, non-clinical and clinical data based on applicants' own tests and studies and/or bibliographic literature substituting/supporting certain test(s) or study(ies).

Information on Paediatric requirements

Pursuant to Article 7 of Regulation (EC) No 1901/2006, the application included an EMA Decision(s) P/0319/2015 on the granting of a product-specific waiver.

Information relating to orphan market exclusivity

Similarity

Pursuant to Article 8 of Regulation (EC) No. 141/2000 and Article 3 of Commission Regulation (EC) No 847/2000, the applicant did not submit a critical report addressing the possible similarity with authorised orphan medicinal products because there is no authorised orphan medicinal product for a condition related to the proposed indication.

Applicant's request(s) for consideration

Conditional marketing authorisation

The applicant requested consideration of its application for a Conditional marketing authorisation in accordance with Article 14(7) of the above mentioned Regulation.

New active Substance status

The applicant requested the active substance avelumab contained in the above medicinal product to be considered as a new active substance, as the applicant claims that it is not a constituent of a medicinal product previously authorised within the European Union.

Protocol Assistance

The applicant received Scientific Advice from CHMP on 22 May 2014, 18 December 2014, 23 July 2015, 24, September 2015, 22 October 2015, 17 December 2015, 28 April 2016, and 26 May 2016. As a follow-up the applicant received on 23 June 2016 Protocol Assistance from the CHMP. The Scientific Advice and Protocol Assistance pertained to quality, non-clinical and clinical aspects of the dossier.

1.2. Steps taken for the assessment of the product

The Rapporteur and Co-Rapporteur appointed by the CHMP were:

Rapporteur: Filip Josephson Co-Rapporteur: Daniela Melchiorri

- The application was received by the EMA on 6 October 2016.
- The procedure started on 27 October 2016.
- The Rapporteur's first Assessment Report was circulated to all CHMP members on 17 January 2017. The Co-Rapporteur's first Assessment Report was circulated to all CHMP members on 16 January 2017. The PRAC Rapporteur's first Assessment Report was circulated to all PRAC members on 23 January 2017.
- During the meeting on 23 February 2017, the CHMP agreed on the consolidated List of Questions to be sent to the applicant.
- The applicant submitted the responses to the CHMP consolidated List of Questions on 20 April 2017.
- The following GCP inspection(s) were requested by the CHMP and their outcome taken into consideration as part of the Quality/Safety/Efficacy assessment of the product:

  - A GCP inspection at 2 clinical investigator sites in United States on 07 February 2017 and 14 February 2017.

  - The outcome of the inspection carried out was issued on 10 April 2017, the date of distribution of IIR to CHMP.
- The Rapporteurs circulated the Joint Assessment Report on the applicant’s responses to the List of Questions to all CHMP members on 30 May 2017.
- During the PRAC meeting on 9 June 2017, the PRAC agreed on the PRAC Assessment Overview and Advice to CHMP.

- During the CHMP meeting on 22 June 2017, the CHMP agreed on a list of outstanding issues to be sent to the applicant.
- The applicant submitted the responses to the CHMP List of Outstanding Issues on 27 June 2017.
- The Rapporteurs circulated the Joint Assessment Report on the applicant's responses to the List of Outstanding Issues to all CHMP members on 5 July 2017.
- During the meeting on 20 July 2017, the CHMP, in the light of the overall data submitted and the scientific discussion within the Committee, issued a positive opinion for granting a marketing authorisation to Bavencio on 20 July 2017.

2. Scientific discussion

2.1. Problem statement

2.1.1. Disease or condition

MCC is a very rare cutaneous neoplasm belonging to the group of neuroendocrine tumours. MCC is an aggressive disease with frequent locoregional recurrences, visceral metastatic evolution, and a high mortality rate.[1, 2]

2.1.2. Epidemiology and risk factors

Incidence rates for Merkel cell carcinoma from the European Union (EU) are in the range of 0.1 to 0.4 per 100,000. Its incidence is 0.2-0.4 cases/100 000 individuals / year in Europe, while in the US it is 0.79 and in Australia 1.6 (where it is most linked to ultraviolet exposure). The median age at diagnosis is around 75 years. A minority of cases are metastatic at presentation, 5-12%. The overall 5-year survival for node-negative disease is 64%, in regional nodal disease at presentation is 39% and drops to 18% in the metastatic setting.

Merkel cell carcinoma is associated with UV exposure, Merkel cell polyomavirus, immunosuppression (8-10% of the MCC patients, mainly in relation to CLL, organ transplant, HIV infection and elderly Caucasians (≥65 yo). A history of extensive sun exposure is a major risk factor for MCC. Most MCC tumours are located on sun exposed areas, with 36% being diagnosed on the face. The incidence of MCC is higher in geographic areas with a greater solar ultraviolet (UV) B index.[3, 4] Immunosuppression has been determined as an important risk factor, although the majority of patients with MCC are immunocompetent.[5] MCC risk is increased by ~10-fold after solid organ transplantation[6], by ~13-fold

[1] Becker J. Merkel cell carcinoma. Ann Oncol 2010;21(Suppl 7):vii81-vii85.
[2] Boccara O, Girard C, Mortier L, et al. Guidelines for the diagnosis and treatment of Merkel cell carcinoma - Cutaneous Oncology Group of the French Society of Dermatology. Eur J Dermatol 2012;22(3):375-379.
[3] Agelli M and Clegg LX. Epidemiology of primary Merkel cell carcinoma in the United States. J Am Acad Dermatol 2003;49(5):832-841.
[4] Miller RW, Rabkin CS. Merkel cell carcinoma and melanoma: Etiological similarities and differences. Cancer Epidemiol Biomarkers Prev 1999;8:153-158.
[5] Asgari MM, Sokil MM, Warton EM, et al. Effect of host, tumor, diagnostic, and treatment variables on outcomes in a large cohort with Merkel cell carcinoma. JAMA Dermatol 2014;150(7):716-723.
[6] Penn I and First MR. Merkel's cell carcinoma in organ recipients: report of 41 cases. Transplantation 1999;68:1717-1721.

among human immunodeficiency virus (HIV) positive patients[7], and by ~12-fold among patients with chronic lymphocytic leukemia (CLL)[8]. Consistent with the association of MCC with immunosuppression, viral involvement in the etiology of MCC has been shown recently. The literature describes compelling evidence for a causal link between Merkel cell polyomavirus (MCPyV or MCV) and MCC[9] . MCPyV, a deoxyribonucleic acid (DNA) virus, is detected in approximately 80% of patients with MCC[10] . MCC tumour regression has been reported following improvement in immune system function in immune compromised individuals[11, 12, 13]. However, the viral-negative subtypes have high mutational burdens characterised by UV signature events, supporting sun damage etiology[14].

2.1.3. Biologic features

The biologic features of MCC are dependent on the etiology. Patients which have a history of extensive sun exposure have been shown to have a high mutational burden and increased expression of neoantigens. The majority of MCV negative MCCs have p53 mutations as well as other mutations present in varying degrees (including NOTCH, NF1, FGF receptor 2 and the PI3K/AKT pathway). The role of polyomavirus integrated into the DNA of the Merkel tumour tissue (identified in 2008) in the pathogenesis of MCC is still unclear. MCV is reported in approximately 80% of MCC tumours. MCV is ubiquitous in the general population and is believed to be acquired in childhood as an asymptomatic primary infection[15, 16, 17]. However, the value of baseline Merkel cell polyoma virus serology as prognostic factor and to assess disease recurrence is currently unclear.

2.1.4. Clinical presentation, diagnosis and stage/prognosis

The diagnosis of MCC is rarely clinically suspected, since the primary tumour lacks distinguishing characteristic features. Therefore, the diagnosis of MCC is based on histologic features of the tumour itself [18]. The American Joint Committee on Cancer has proposed the following anatomic staging framework for MCC: Stage 0 - in situ tumour; Stage I and II - negative lymph nodes; Stage IIIa includes both occult nodal disease and unknown primary disease; Stage IIIbincludes those with a known primary tumour and clinically detected regional metastatic disease; and Stage IV - distant metastasis beyond regional lymph nodes.

[7] Engels EA, Frisch M, Goedert JJ, Biggar RJ, Miller RW. Merkel cell carcinoma and HIV infection. Lancet 2002;359(9305):497-498.
[8] Kaae J, Hansen AV, Biggar RJ, et al. Merkel cell carcinoma: incidence, mortality, and risk of other cancers. J Natl Cancer Inst 2010;102(11):793-801.
[9] Spurgeon ME and Lambert PF. Merkel cell polyomavirus: a newly discovered human virus with oncogenic potential. Virology 2013;435(1):118-30.
[10] Feng H, Shuda M, Chang Y, Moore PS. Clonal integration of a polyomavirus in human Merkel cell carcinoma. Science 2008;319(5866):1096-1100.
[11] Burack J and Altschuler EL. Sustained remission of metastatic Merkel cell carcinoma with treatment of HIV infection. J R Soc Med 2003;96:238-239.
[12] Muirhead R, Ritchie DM. Partial regression of Merkel cell carcinoma in response to withdrawal of azathioprine in an immunosuppresion-induced case of metastatic Merkel cell carcinoma. Clin Oncol (R Coll Radiol) 2007;19:96.
[13] Bhatia S, Afanasiev O, Nghiem P. Immunobiology of Merkel cell carcinoma: implications for immunotherapy of a polyomavirus-associated cancer. Curr Oncol Rep 2011;13(6):488-497.
[14] Harms PW, Vats P, Verhaegen ME, et al. The distinctive mutational spectra of polyomavirus-negative Merkel cell carcinoma. Cancer Res 2015;75(18):3720-3727.
[15] Feng H, Shuda M, Chang Y, Moore PS. Clonal integration of a polyomavirus in human Merkel cell carcinoma. Science 2008;319(5866):1096-1100.
[16] Santos-Juanes J, Fernandez-Vega I, Fuentes N, et al. Merkel cell carcinoma and Merkel cell polyomavirus: A systematic review and meta-analysis. Br J Dermatol 2015;173(1):42-49.
[17] Rodig SJ, Cheng J, Wardzala J, et al. Improved detection suggests all Merkel cell carcinomas harbor Merkel polyomavirus. J Clin Invest 2012;122(12):4645-4653.
[18] National Comprehensive Cancer Network Clinical Practice Guidelines in Oncology, Merkel cell carcinoma, Version 1, 2018.

The prognosis for mMCC is poor, with a median survival of 9.6 months from the time of diagnosis of distant mMCC[19].

2.1.5. Management

There are currently no approved therapies for recurrent, non-resectable or metastatic Merkel cell carcinoma, a clearcut unmet medical need. In clinical practice chemotherapy and radiotherapy is used in the first-line setting of mMCC (NCCN 2017). In June 2015, a collaborative group of multidisciplinary experts form the European Dermatology Forum (EDF), The European Association of Dermato-Oncology (EADO) and the European Organization of Research and Treatment of Cancer (EORTC) published a European consensus based interdisciplinary guideline on diagnosis and treatment of Merkel Cell Carcinoma – the European Consensus Guideline[20] . It made recommendations on MCC diagnosis and management, based on a critical review of the literature, existing guidelines and expert's experience. The European Consensus Guideline recommends a multidisciplinary treatment approach with lymph node dissection and/or radiation therapy (RT), and consideration for adjuvant chemotherapy in the presence of metastatic MCC. The most common chemotherapy regimen used is a platinum compound ± etoposide, with short term control of the disease and no durable responses (recurrence in 4-15 months) and high toxicity. CAV is the second most used regimen, with response rates of 76%, with significant toxicities. The median PFS for chemotherapy is about 2 months.

Table 1: Stage IV MCC responses reported in the literature

MCC population	PFS (months)	ORR (%)	Median Durability of Response (months)
1st line chemotherapy (N = 62)	5.6	53	2.8
2nd line chemotherapy (N = 30)	2	23	4.3

Iyer, et al, 2014

A small phase II trial of pembrolizumab in the first-line setting of Stage IIIb/IV reported a 71% initial response rate in 17 evaluable patients[21] . The published study reported a total of 26 patients receiving at least one dose of pembrolizumab. The ORR among the 25 patients with at least one post-baseline evaluation was 56% (95% CI 35 to 76), with 4 CR and 10 PR. During a 33 weeks' follow-up (range 7-53), 2/14 relapsed. The DOR varied between 2.2 and 9.7 months. The rate of PFS at 6 months was 67%. 17/26 patients were MCV positive; the RR in MCV+ patients was 62% (10/16) and 44% for MCV- tumors (4/9). With regard to other immune-checkpoint inhibitors, there are published case reports with nivolumab in MCC[22] .

Responses to molecularly targeted therapy (e.g. pazopanib) and somatostatin analogues have been reported in literature, and some phase II trials with this class of compounds are ongoing[23] .

[19] Iyer JG, Blom A, Doumani R, et al. Response rates and durability of chemotherapy among 62 patients with metastatic Merkel cell carcinoma. Cancer Med 2016 Jul 19. doi: 10.1002/cam4.815.
[20] Lebbe C, Becker JC, Grob JJ, et al. Diagnosis and treatment of Merkel cell carcinoma. European consensus-based interdisciplinary guideline. Eur J Cancer 2015;51:2396-2403.
[21] Nghiem PT, Bhatia S, Lipson EJ, et al. PD-1 blockade with pembrolizumab in advanced Merkel-cell carcinoma. N Engl J Med. 2016;374:2542-2552.
[22] Walocko FM, Scheier BY, Harms PW, Fecher LA, Lao CD. Metastatic Merkel cell carcinoma response to nivolumab. J Immunother Cancer. 2016 Nov 15;4:79.
[23] Tothill R, Estall V, Rischin D. Merkel cell carcinoma: emerging biology, current approaches, and future directions. Am Soc Clin Oncol Educ Book. 2015:e519-26.

About the product

Programmed death ligand 1 (PD-L1) is the ligand to PD-1 receptor and may be expressed on the surface of either tumour cells and/or tumour-infiltrating immune cells. It has been shown to contribute to the inhibition of the anti-tumour immune response in the tumour microenvironment.

Avelumab is a human immunoglobulin G1 (IgG1) monoclonal antibody directed against programmed death ligand 1 (PD L1). Avelumab binds PD L1 and blocks the interaction between PD L1 and the programmed death 1 (PD 1) and B7.1 receptors. This removes the suppressive effects of PD L1 on cytotoxic CD8+ T cells, resulting in the restoration of anti-tumour T cell responses.

Avelumab has also shown to induce natural killer (NK) cell mediated direct tumour cell lysis via antibody dependent cell mediated cytotoxicity (ADCC).

Type of Application and aspects on development

The applicant applied for the following indication:

- Bavencio is indicated for the treatment of adult patients with metastatic Merkel cell carcinoma (MCC).

The final agreed indication was as follows:

- Bavencio is indicated as monotherapy for the treatment of adult patients with metastatic Merkel cell carcinoma (MCC).

Treatment should be initiated and supervised by a physician experienced in the treatment of cancer.

Posology
The recommended dose of Bavencio is 10 mg/kg body weight administered intravenously over 60 minutes every 2 weeks.

Bavencio has to be diluted with either sodium chloride 9 mg/mL (0.9%) solution for injection or with sodium chloride 4.5 mg/mL (0.45%) solution for injection. It is administered over 60 minutes as an intravenous infusion using a sterile, non-pyrogenic, low-protein binding 0.2 micrometre in-line or add-on filter.

For instructions on the preparation and administration of the medicinal product, see SmPC section 6.6.

Conditional marketing authorisation

This application falls under the scope of the regulation No507/2006 for medicinal products designated as orphan medicinal product and to treat a life-threatening disease in accordance with Article 3 of Regulation (EC) No 141/2000.

The applicant requested consideration of its application for a Conditional Marketing Authorisation in accordance with Article 14(7) of the above mentioned Regulation based on the following claim(s):

- The risk-benefit balance of the medicinal product, as defined in Article 1(28a) of Directive 2001/83/EC, is positive.

The applicant claims that given the durability of response with features such as responses in large tumours or visceral sites and prolonged responses after discontinuation of treatment, together with a safety profile that is manageable and consistent with the overall avelumab program, the pivotal clinical data suggest that avelumab has a favourable benefit/risk profile in patients with mMCC.

- It is likely that the applicant will be in a position to provide comprehensive clinical data.

The applicant conducted a single arm study in treatment naïve patients with metastatic Merkel cell carcinoma in a highly similar population as with the pivotal and observational studies, all in Stage IV disease, to serve as a confirmatory study relative to the initial conditional application in patients with treatment experience. The design of this study has been subject to discussions at the SAWP. As the treatment landscape had changed with the early results of efficacy with pembrolizumab in naive patients with mMCC(IIIb/IV), it was no longer feasible to conduct a comparative phase III trial of avelumab vs chemotherapy as problems with recruitment would be anticipated in the chemotherapy arm. Therefore, it was agreed that confirmatory data will be provided from:

  - Study EMR100070-003 (Part B) in 1st line therapy which is ongoing. The applicant anticipated that approximately 10-15 subjects would reach 3 months of observation so as to enable submission of preliminary efficacy data during the procedure.
  - In addition supportive data from the Observational Study 100070-Obs001 (Part A) – cohort in 1st line therapy which is completed – and the 1L literature based data from Iyer et al, 2016 would support the efficacy data.

- Unmet medical needs will be fulfilled.

The unmet medical need will be addressed, as there are no authorised medicinal products for mMCC and current treatment options have limited efficacy for metastatic disease.

- The benefits to public health of the immediate availability on the market of the medicinal product concerned outweighs the risk inherent in the fact that additional data are still required.

The applicant considers that the benefits to public health (the durable response rate compared to chemotherapy) outweigh the risks.

2.2. Quality aspects

2.2.1. Introduction

The finished product is presented as a clear, colourless to slightly yellow concentrate for solution for infusion, containing 20 mg/ml of avelumab as active substance.

Other ingredients are mannitol, glacial acetic acid, polysorbate 20, sodium hydroxide and water for injections.

The product is available in a 10 ml vial (Type I glass) with a halobutyl rubber stopper and an aluminium seal fitted with a removable plastic cap.

2.2.2. Active Substance

General Information

Avelumab is a fully human monoclonal antibody based on a human immunoglobulin G1 (IgG1 λ) framework. The recombinant antibody is produced in Chinese hamster ovary (CHO) cells and consists of two heavy chains (HC) of 450 amino acid residues each and two light chains (LC) of 216 amino acid residues each with typical IgG1 inter and intra chain disulfide bonds.

Avelumab has primarily a β-sheet structure, consistent with the structure of an IgG-1 antibody. The typical nine disulphide bonds of IgG were confirmed as:

- Intra-chain disulfide bonds: Cys138-Cys197; Cys22-Cys96; Cys147-Cys203; Cys264-Cys324; Cys370-Cys428; Cys22-Cys90;
- Inter-chain disulfide bonds: Cys215–Cys223; Cys229-Cys229; Cys232-Cys232. The molecule contains one N-glycosylation site on Asn-300 of the heavy chain.

N-glycan structures identified were complex, biantennary type core fucosylated oligosaccharides with zero (G0F), one (G1F), or two galactose (G2F) residues. The molecular weight of the intact avelumab molecule, calculated on the basis of amino acid composition and predicted disulfide bonding of the predominant isoform is 143’832 Da (approximately 147’000 Da with glycosylation).

Avelumab mechanism of action is based on the inhibition of the interaction between PD-L1 and its receptors programmed death 1 (PD-1) and B7.1. This removes the suppressive effects of PD-L1 on anti-tumor CD8+ T cells, resulting in the restoration of cytotoxic T cell response.

The biological activity (potency) of avelumab is evaluated through a cell based assay able to measure its capability to bind the PD-L1 receptor over-expressed on the recombinant HEK-293 (hPD-L1) cell line.

Antibody dependent cell mediated cytotoxicity (ADCC) represents an additional mechanism of action of avelumab and was confirmed by in vitro testing.

Manufacture, process controls and characterisation

Description of manufacturing process and process controls

Information about the manufacturing, storage and control facilities for the active substance has been provided in the dossier. GMP compliance for the manufacturers has been demonstrated.

The avelumab active substance is manufactured at Merck Serono SA, Corsier-sur-Vevey, Switzerland. The manufacturing process is a cultivation process with nutritive feeds. One vial of the working cell bank (WCB) is thawed and the cell culture is expanded in shake tubes, wave bags and seeding bioreactors. The production bioreactor is harvested after a defined production period and a clarification is performed.

The purification process includes three chromatography steps as well as viral inactivation/clearance steps, ultrafiltration/diafiltration (UF/DF), final formulation and final filtration (0.22 μm).

The purification process has been described in sufficient detail, providing lists of process parameters and their acceptance criteria, for each step. Typical elution profiles have been provided for the three chromatography steps.

Avelumab active substance is stored in polyethylene bags at 2-8°C and shipped to the finished product manufacturing facility at controlled conditions.

Dimensions and specifications of the bags have been provided in the dossier. The compatibility of the container for storage of active substance has been evaluated. A summary of an extractable and leachable study was presented and concluded that the risk for patients due to substances leaching into avelumab active substance is negligible.

Control of materials

Raw materials used in the active substance manufacturing process are sufficiently described and controlled. Tests for adventitious agents are performed according to regulatory guidelines.

A description of the generation of cell substrate and the cell banking system has been provided. The construction of the expression vector, the host cell line used, the transfection and the isolation of cell line have been described in an acceptable way. The preparation of the cell banks have been described in sufficient detail.

An expression vector, coding for both antibody chains of avelumab, was generated and stably introduced into the CHO-S host cell line, grown in animal component-free medium. Genotypic and phenotypic characterization of the Master Cell Bank (MCB), Working Cell Bank (WCB) and Extended Cell Bank (ExCB) was performed to confirm identity, purity and stability of the cell lines, according to ICH Q5B and ICH Q5D.

A protocol for production and qualification of future WCBs has been presented.

Control of critical steps and intermediates

A list of In-process Controls (IPCs), including a test for amino acid misincorporation, and the corresponding acceptance criteria or action limits for the cell cultivation and purification process have been provided (viability, bioburden, endotoxin, filter integrity and step yield). The control of critical steps and intermediates has been sufficiently described and is found acceptable.

Process validation

The validation of the avelumab active substance manufacturing process was performed using several consecutive full scale batches, manufactured using the proposed commercial process. The production reactor step and the purification process was appropriately validated. Numerical results for all Critical Process Parameters (CPPs) and IPCs were presented, together with profiles from chromatograms.

Removal of process and product related impurities have been evaluated in the frame of the process validation studies. The levels of impurities have been assessed and the impurity control strategy justified.

Impurities were tested at one or several purification steps, to demonstrate their clearance (DNA, Host-Cell Protein (HCP), residual protein A, High Molecular Weight (HMW) species, Low Molecular Weight (LMW) species). The removal of process-related impurities (HCP, protein A and DNA) was further supported by batch analysis results, showing low levels of impurities in all clinical trial batches. The results, regarding removal of impurities, from the validation data together with the batch analysis data from clinical trials batches were found acceptable.

Manufacturing process development

A traditional process characterization approach was used for the characterization range studies (one factor at a time; OFAT). No multivariate experiments were performed or design space claimed. The process characterization results were presented in sufficient detail and were considered acceptable. A comprehensive control strategy was presented, describing the strategy for each individual critical quality attribute (CQA).

The nonclinical studies (including the pivotal toxicology study) and initial clinical studies were conducted using avelumab material derived from the initial manufacturing process (also denominated Process A) produced at Merck Bio development facility, Martillac (MBD), France. The avelumab manufacturing process has undergone one major change that aimed at developing an optimized process with higher performance in order to mainly fulfil clinical development program and product launch needs. The new process was denoted process B.

An extensive comparability exercise between processes A and B was performed. The Applicant provided background information and details about the process differences and a rationale for the proposed changes.

The results demonstrated comparability for most of the quality attributes except for some minor quantitative differences. However, no impact of these differences was observed on biological activity.

Since the differences are considered minor, the Applicant's conclusion that the two materials can be considered comparable is endorsed.

Characterization

Avelumab is a fully human immunoglobulin (Ig) G1 monoclonal antibody directed against PD-L1. Avelumab binds PD-L1 and blocks the interaction between PD-L1 and the programmed death 1 (PD- 1) and B7.1 receptors. This is the main mode of action.

An extensive list of analytical methods has been applied for structural and biological characterization of avelumab. The majority of the characterization has been performed on one batch only; the currently used reference material. As much of the same tests were performed in the comparability studies, the use of only one batch is acceptable.

In general, the substance demonstrated characteristics that are typical of a human IgG1. The biological functional analyses evaluated during the characterization are well chosen with respect to the mode of action.

Amino acid substitution was detected in avelumab reference material in the frame of the characterization studies.

With regards to the characterization of impurities, this topic has been adequately addressed. Low levels of process-related impurities (HCP, DNA and residual protein A) were demonstrated during batch analysis.

The product related impurities HMW (aggregates) and LMW (fragments) were discussed.

Specification

Avelumab quality control testing for batch release includes appearance (Ph. Eur.), pH (Ph. Eur.), purity, identity, quantity, potency, process-related impurities and endotoxins/bioburden (Ph. Eur.). The biological activity is tested using a cell-based bioassay.

The proposed release specification for the active substance is found acceptable, with respect to test methods chosen. The proposed specification limits are based on batch analysis and stability study results. This approach is considered acceptable.

Analytical methods

All analytical methods used for testing of the active substance have been described in the dossier. Biological activity is determined using an in vitro cell binding assay.

Antibody dependent cell mediated cytotoxicity (ADCC) represents an additional mechanism of action of avelumab. Both PD-L1 binding and ADCC have been classified as CQAs. The Applicant has confirmed a correlation between afucosylated glycan species of avelumab and its ADCC activity.

Batch analysis

Batch analysis from an extensive number of batches used in clinical studies, stability studies and process validation, all manufactured using process B, has been provided. The results confirm consistency of the manufacturing process.

Reference materials

A one-tiered approach is used with respect to reference standards for avelumab. The currently used reference originates from a GMP batch manufactured at Merck Serono S.A Vevey, Switzerland, according to the manufacturing process proposed for commercial purpose, used in clinical studies, and has been qualified. The Applicant plans to replace the current standard with a Primary Reference.

The suggested protocols for establishment of a primary reference standard and working standard are in general considered acceptable.

The comparison between two historical reference standards demonstrated in general similar characteristics. However, a few differences were noted. The differences observed are further discussed in the comparability assessment, and are considered acceptable.

Stability

Stability has been demonstrated by real time data from both primary batches (manufactured using process B and stored in a representative container closure system) and supportive batches (manufactured using process A).

Stability data for the active substance stored under long-term (5±3°C), accelerated and stressed conditions was provided.

For the primary batches, long-term data was provided for up to 24 months (3 batches) under long-term (5°C±3C). Data from active substance stored under accelerated conditions was provided for up to 6 months and stressed conditions for up to 3 months.

For the supportive batches, long-term data was provided for up to 36 months and accelerated data for up to 6 months.

The data submitted on support the proposed shelf life of 24 months for the active substance when stored at the recommended temperature of 5 ± 3°C.

All stability studies will be tested and controlled using the final, agreed limits in the release specification.

2.2.3. Finished Medicinal Product

Description of the product and Pharmaceutical Development

Avelumab finished product is a concentrate for solution for infusion presented at the concentration of 20 mg/mL. The product is available in a vial (Type I glass) with a halobutyl rubber stopper and an aluminium seal fitted with a removable plastic cap. One vial of 10 mL contains 200 mg of avelumab, mannitol and polysorbate 20 in a preservative-free acetate-buffered solution pH 5.0-5.6.

The formulation development has been adequately described and the final formulation intended for marketing was used in the phase III clinical trials.

All excipients are well known pharmaceutical ingredients and their quality is compliant with Ph. Eur standards. There are no novel excipients used in the finished product formulation.

To investigate leachable compounds a stability study was conducted in the commercial primary container closure system. The conclusion from the Applicant that the risk for patients treated with avelumab finished product due to leachables is negligible seems reasonable based on data and worst case calculations presented in the dossier. The container closure is considered suitable for its intended use and a risk assessment for the potential glass delamination has been included, as requested.

Manufacture of the product and process controls

The finished product is manufactured at Merck Serono SA, Aubonne, Switzerland.

The avelumab finished product manufacturing process consists of active substance pooling, filtration through a 0.2 µm sterilizing grade filter, aseptic filling of the pre formulated active substance and visual inspection. All filled vials are visually checked, discarding those with defects. After the inspection process, the vials are stored at 5° ± 3°C pending labelling and packaging.

The description of the manufacturing process has been provided in sufficient detail.

To ensure that the finished product meets high quality standards, its manufacturing process was developed with defined manufacturing procedures, CPPs, IPCs, and release specifications.

Process validation

For the process validation studies minimum, medium and maximum size batches were produced. All validation batches complied with the established in-process and release specifications as well as additional process monitoring data. No critical deviations were observed.

For the sterile filtration step appropriate filter validation studies have been performed by the filter manufacturer at worst case conditions. The aseptic filling has been sufficiently validated with media fills covering the maximum duration of filling. No growth was detected.

All proposed maximum process hold times have been covered in the process validation.

Product specification

The finished product Quality Control for batch release includes identity, potency, purity, impurities, sterility (Ph. Eur.), bacterial endotoxin (Ph. Eur.) and several other general tests.

For many tests the same acceptance criteria are proposed for active substance and finished product. The initially proposed specification based on the use of process capability analysis was not endorsed. For parameters related to efficacy and safety it is normally expected that the main basis for setting limits is the actual levels qualified through clinical studies. Upon request the Applicant has tightened the limits for some CQAs. The applicant also tightened the specifications for the active substance as appropriate to be able to fulfil the revised finished product specifications.

Analytical methods

All analytical methods used for testing of the finished product are satisfactorily described in the dossier and non-compendial methods have been validated. Many test methods used for release testing and stability testing of the finished product are the same as those used for release testing and stability testing of the active substance.

Batch analysis

Batch analysis data for an extensive number finished product batches manufactured at Merck Serono SA, Aubonne and using avelumab active substance from Process B (commercial process) have been provided. The results demonstrate a satisfactory batch to batch consistency.

Reference materials

The reference standard used in the testing and release of avelumab finished product is the same as the one used for the testing and release of avelumab active substance.

Stability of the product

The proposed shelf life for the finished product is 24 months when stored at 2- 8ºC protected from light.

The stability program was designed in accordance with relevant ICH guidelines and included primary stability batches as well as supportive stability batches. The primary stability program included finished product batches to support long term storage at 5±3°C. Accelerated stability studies and stressed stability studies were also performed. All primary finished product stability batches were produced using active substance from the manufacturing process intended for commercial use (Process B). The container closure system used in the primary stability studies is identical to the one intended for the storage of the commercial finished product.

For the primary batches, long-term data was provided for up to 24 months (3 batches). Data from finished product stored under accelerated conditions was provided for up to 6 months and stressed conditions for up to 3 months.

At long term storage all test parameters remained within the specification limits and no significant changes were observed for any of the parameters tested.

A photostability study was conducted on one finished product batch in accordance with ICH Q1B. The study was conducted on vials (immediate pack), directly exposed to light without the secondary packaging and on vials stored in their secondary packaging. Results support the recommended storage condition for the finished product vials protected from light and demonstrate that the secondary packaging is suitable to provide sufficient light protection throughout the shelf-life.

A cumulative stability study (covering 24 months of storage for the active substance and 24 months of storage for the finished product) is currently ongoing and any unexpected trends or out of specification results will be reported.

Data has been presented to support the claimed in-use stability of the diluted solution (up to 24 hours at 20 - 25°C) and the compatibility with infusion solution containers. It is also noted that the use of a filter is recommended during clinical administration.

Based on the data provided the proposed shelf-life of 24 months at 2-8ºC is considered acceptable.

Comparability exercise for Finished Medicinal Product

The initial avelumab finished product was formulated at a protein concentration of 10 mg/mL using avelumab active substance from the initial manufacturing process (process A) with a fill volume of 8 mL. This formulation was used throughout the early development program, e.g., nonclinical studies, Phase I/II clinical trials and part A of the Phase II mMCC study.

To support clinical development and commercial use, an optimized formulation of avelumab at a higher concentration (20 mg/mL) was designed. This formulation was prepared using avelumab active substance from an optimised manufacturing process (process B). This formulation (current composition) has been used in all Phase III clinical trials as well as in the expansion cohorts of Phase I trial and part B of the Phase II mMCC study. It is identical to the to-be-marketed formulation.

Finished product comparability studies were conducted in order to demonstrate that the quality of the commercial manufacturing process is comparable to the pre-change product. All quantitative batch analysis results post-change were within the 95% and 99% tolerance intervals for pre-change product except for pH, avelumab content, and the extractable volume that were intentionally adjusted to different values based on formulation development.

Electrophoretic purity was comparable among tested samples in terms of purity level and aligned with expectations and the target product profile. Comparable results were also obtained for potency. In addition, the currently available stability data show no difference between pre- and post-change product.

Adventitious agents

No substances of animal origin were used in the manufacturing process for Bavencio. Neither the culture media nor the raw materials used in its manufacture were derived from bovine or other animal sources.

The unprocessed bulk harvests are routinely checked for the absence of bacteria and fungi, mycoplasma as well as in vitro detection of virus. All harvests tested were found to be negative.

The retroviral like particles (rVLP) of crude harvest samples were quantified by density gradient centrifugation and negative stain Transmission Electron Microscopy (TEM). Based on the results of the virus clearance studies and TEM results for unprocessed bulk harvest, the estimated maximum number of rVLP particles per clinical dose is considered acceptable.

Overall, the safety of avelumab in relation to adventitious agents is considered adequate.

2.2.4. Discussion on chemical, pharmaceutical and biological aspects

Information on development, manufacture and control of the active substance and finished product has been presented in a satisfactory manner. The results of tests carried out indicate consistency and uniformity of important product quality characteristics, and these in turn lead to the conclusion that the product should have a satisfactory and uniform performance in clinical use.

2.2.5. Conclusions on the chemical, pharmaceutical and biological aspects

The quality of this product is considered to be acceptable when used in accordance with the conditions defined in the SmPC. Physicochemical and biological aspects relevant to the uniform clinical performance of the product have been investigated and are controlled in a satisfactory way. Adequate data has been presented which give reassurance on viral and TSE safety.

2.3. Non-clinical aspects

2.3.1. Introduction

All in vitro and in vivo primary and secondary pharmacology studies were not conducted under GLP conditions. In vivo studies were performed in mice, rats and monkeys. Most repeat-dose toxicity studies in cynomolgus monkeys were conducted in compliance with GLP regulations.

Basic pharmacokinetic (PK) properties of avelumab were investigated in nonclinical PK studies in mice and cynomolgus monkeys after single iv administration. Multiple dose pharmacokinetic/toxicokinetic (TK) data were obtained during the course of repeat-dose toxicity studies in mice, rats, and cynomolgus monkeys. Immunogenicity was evaluated in the single dose PK studies in mice and cynomolgus monkeys, in the 4-week toxicity studies in mice, rats and cynomolgus monkeys and during the course of the pivotal 13-week toxicity study in cynomolgus monkeys.

Avelumab (also known as MSB0010718C and MSB0010718) is a fully human monoclonal antibody. During the development of avelumab, different versions of the antibody have been used in the nonclinical pharmacology studies. Two other closely-related antibodies were used where indicated (MSB0010608H and MSB0010294). MSB0010608H has the same sequence as avelumab but was produced by transient transfection of HEK293 cells. MSB0010294 was the parental version of avelumab.

2.3.2. Pharmacology

Primary pharmacodynamic studies

In vitro primary pharmacodynamics

Kinetics of PD-L1 occupancy in non-tumor-bearing C57BL/6 mice (IONC03082013RT)

A PK study was performed in non-tumour bearing mice study to support pharmacokinetic (PK) modeling and human dose predictions for avelumab. A FACS-based assay was used to determine the percentage of PD-L1 target occupancy on splenocytes (left figure) or peripheral blood leukocytes (right figure).

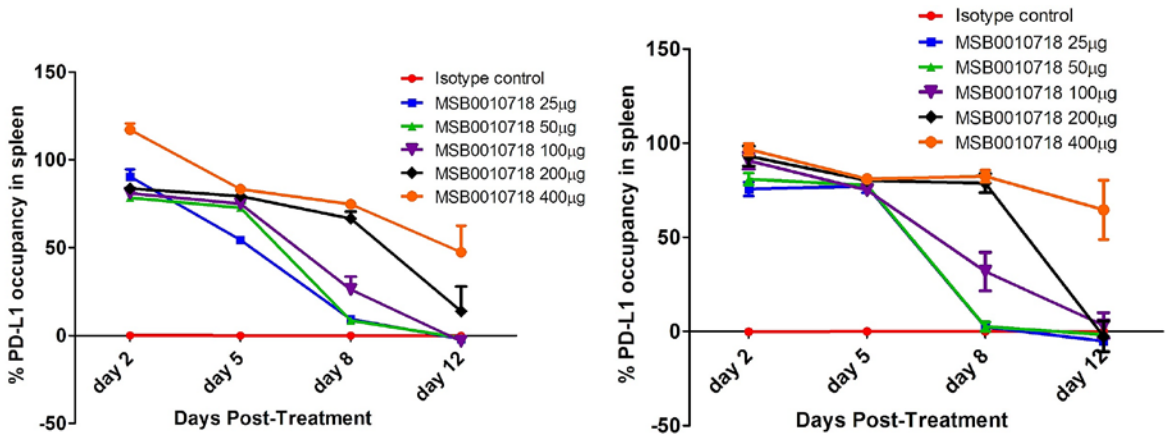


Figure 1: Kinetics of PD-L1 Occupancy by Avelumab in Splenocytes

All doses of avelumab tested showed similar levels of PD-L1 target occupancy (~75-100%) on peripheral blood and splenocytes on days 2 and 5. Dose-dependent decreases in target occupancy were observed on days 8 and 12. By Day 12, only the highest dose level of 400 µg still showed significant occupancy (~50%).

<u>Study PEAT18052012BJW - Complement-Dependent Cytotoxicity against Human Cancer Cells</u>

The CDC potential of avelumab was investigated in vitro against the A431, A549, and M21 human tumour cell lines. Target cells were radiolabeled with 51Cr and then incubated with various concentrations of avelumab (ranging from 2.43 to 10,000 ng/ml) for 45 minutes in the presence of complete human complement. The amount of 51Cr released into supernatants was measured as an indicator of CDC activity. As a positive control, a previously characterized CDC competent antibody, 14.18.IL2, was tested against the M21 cell line. Two negative control antibodies were used: the inactive version of avelumab and the anti-EGFR antibody, Erbitux (cetuximab). The results are shown below:

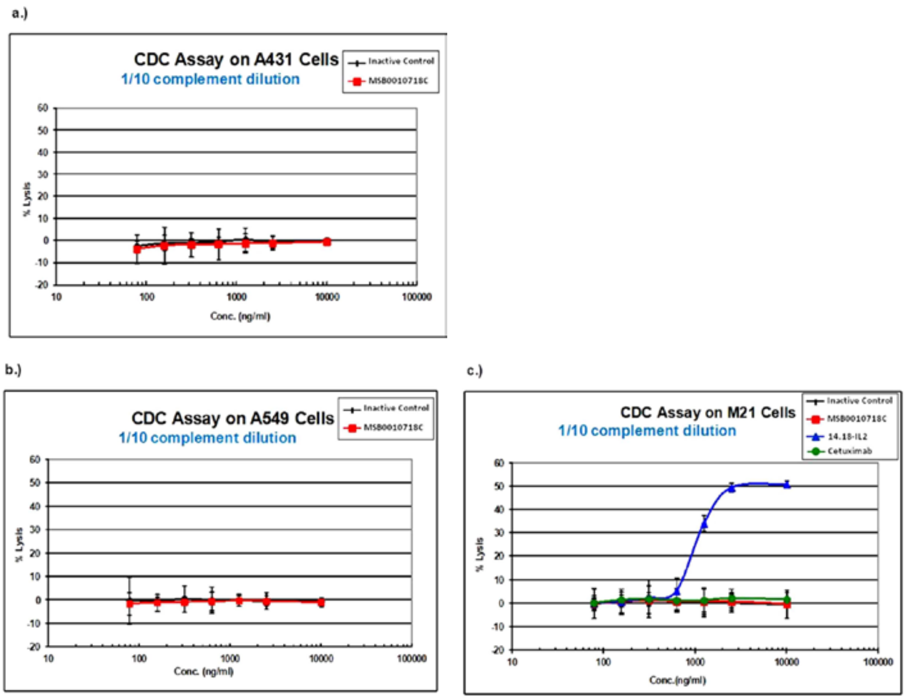


Figure 2: Complement-Dependent Cytotoxicity potential of avelumab against Human Cancer Cells

In vivo primary pharmacodynamics

Dose-dependent anti-tumor activity as a monotherapy in the MC38 colon carcinoma model (IONC20042011AKH)

In all of the in vivo pharmacology studies, 3 administrations of avelumab were given, with three days between each treatment (e.g., treatment on days 0, 3, and 6) in order to maximize drug exposure prior to the onset of an ADA response.

Avelumab showed activity against MC38 tumors and the anti-tumor effect was associated with consistent modulation of T cell phenotypes, including increased levels of splenic $CD8^+PD\text{-}1^+$ T cells and increased levels of $CD8^+$ T cells with an effector memory phenotype. MC38 tumors express an endogenous murine retroviral protein, p15E, which has been identified as a tumor-associated $CD8^+$ T-cell antigen. Frequencies of p15E-specific T cells were determined using a fluorescently-labeled synthetic MHC class I pentamer loaded with a p15E-derived antigenic peptide epitope. Treatment with avelumab was associated with an increased frequency of p15E antigen-specific CD8+ T cells

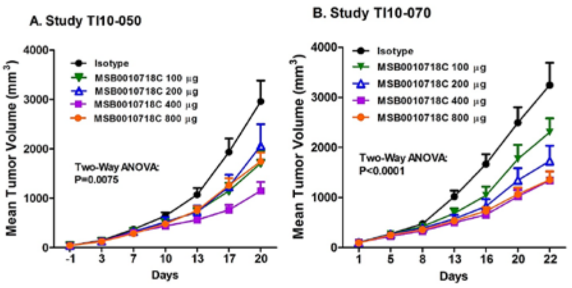


Figure 3: Dose-Dependent Inhibition of MC38

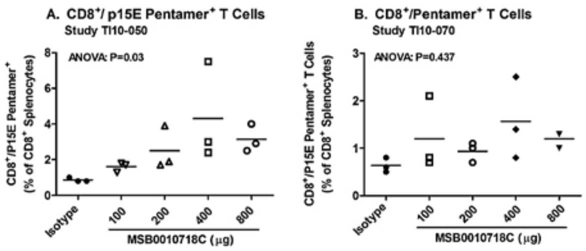


Figure 4: Monotherapy Dose-Response Studies: GrowthTumor Pentamer Analysis of Tumor Antigen-Specific T cells in Spleens

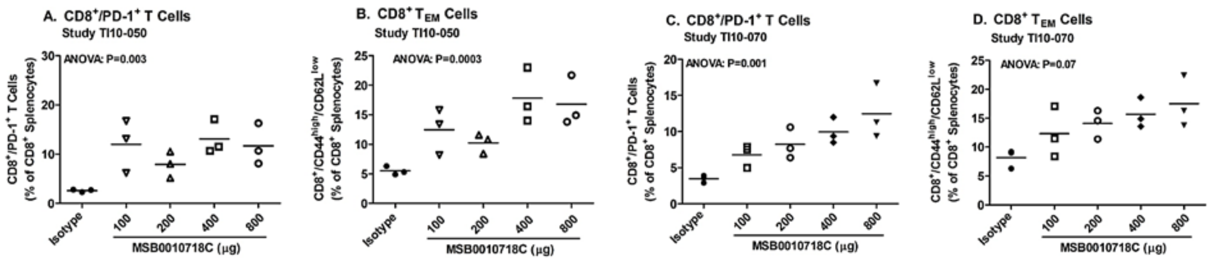


Figure 5: Monotherapy Dose-Response Studies: Changes in Splenic CD8+ T cell Phenotypes

Secondary pharmacodynamic studies

A secondary pharmacology study was performed to investigate the potential off-target effect of avelumab to induce ADCC against human peripheral blood mononuclear cells. The data demonstrated that, under conditions of in vitro immune stimulation that upregulated PD-L1 expression by immune cells, avelumab failed to induce detectable killing of any of the immune cell subsets analysed, including the CD8+ cytotoxic T cells. Tumor cells used as positive control displaying a higher PD-L1 expression level were subject to ADCC-mediated killing.

Safety pharmacology programme

As per guideline ICH S6(R1), the investigation of safety pharmacologically relevant parameters was included in the pilot 4 week iv repeat-dose toxicity study and in the pivotal 13-week iv repeat-dose toxicity study in cynomolgus monkeys. In both studies the heart rate, electrocardiogram, arterial blood pressure, respiratory rate, CNS parameters, and body temperature were unaffected by treatment with avelumab at the high dose level of 140 mg/kg weekly.

Pharmacodynamic drug interactions

2.3.3. Pharmacokinetics

The applicant did not submit pharmacodynamic drug interactions studies (see non-clinical discussion).

2.3.1. Pharmacokinetics

Table 2: Study B-09-009- Single Dose PK Study in CD-1 Mice

Dose (µg)	Dose (mg/kg)	C_{max} (µg/mL)	AUC_{last} (h*mg/mL)	CL (mL/h/kg)	V_z (mL/kg)	V_{ss} (mL/kg)	$t_{1/2}$ (h)	MRT (h)
500	20	675	14.1	1.20	79.5	73.9	46.1	61.8
100	4	96.1	2.74	1.29	73.3	66.9	39.4	51.9
10	0.4	9.91	0.079	5.04	100	73.3	13.7	14.6

Source: Table 3, Section 3.1 of Study B-09-009

C_{max}: maximum observed concentration; AUC_{last}: area under the concentration-time curve till the last observed concentration; CL: total systemic clearance; V_z: terminal phase volume of distribution; V_{ss}: volume of distribution at steady state; $t_{1/2}$: terminal half-life; MRT: mean residence time

Table 3: Study IONC03082013RT - Single Dose PK/PD Study in C57BL/6 Mice

Dose (µg)	Dose (mg/kg)	Assumed C_0 (µg/mL)	AUC_{last} (h*mg/mL)	CL (mL/h/kg)	V_z (mL/kg)	V_{ss} (mL/kg)	$t_{1/2}$ (h)	MRT (h)
400	20	400	30.9	0.644	36.9	45.6	39.7	69.6
200	10	200	12.3	0.814	21.6	42.7	18.4	52.5
100	5	100	5.64	0.887	15.3	37.4	11.9	42.2
50	2.5	50	2	1.25	25.4	32.9	14.1	26.2
25	1.25	25	0.88	1.42	32.2	22.4	15.8	15.7

Table 4: Study RF2120 - PK/PD Study After Single Intravenous Administration in Monkeys

Group	Dose (mg/kg)	C_0 (µg/mL)	AUC (hr*µg/mL)	$t_{1/2}$ (hr)	CL (mL/hr/kg)	V_z (mL/kg)	V_{ss} (mL/kg)	MRT (hr)
1	0.80	19.0 ±7.02	807 ±31.8	32.3 ±3.37	0.992 ±0.0392	46.3 ±6.58	54.0 ±2.80	54.4 ±1.06
2	4.0	97.8 ±17.3	3270 ±273	33.1 ±5.76	1.23 ±0.101	59.0 ±13.7	70.2 ±22.5	56.7 ±15.0
3	20	474 ±32.9	31100 ±17000	64.2 ±31.7	0.766 ±0.346	60.6 ±2.93	74.1 ±13.0	109 ±41.1

No specific studies have been submitted on the metabolism and excretion of avelumab (see non-clinical discussion)

No specific nonclinical in vitro or in vivo drug-drug interaction (DDI) studies have been conducted (see non-clinical discussion).

2.3.2. Toxicology

Single dose toxicity

The applicant did not submit single dose toxicity studies (see non-clinical discussion).

Repeat dose toxicity

Table 5: Summary of repeat-dose toxicity studies performed with avelumab

Study ID /GLP/ Duration	Species/Sex/ Number/Group	Dose (mg/kg/day/ Route	NOAEL/NOEL
RF2740 / non-GLP/ 4 weeks	CD-1 mouse/ 10/sex/group Satellites for TK	0 (vehicle), 20, 40, 140 iv injection (once weekly) vehicle: 10 mM sodium acetate, 140 mM sodium chloride, 0.05% polysorbate 20, pH 6.0	**NOAEL:** no NOAEL

TK: mean values for ♂ on Day 29 were, from lowest to highest dose: 20700, 30000, 111000 µg · h/mL (AUC_{168}). For ♀, corresponding values were 22600, 38600, 117000 µg · h/mL (AUC_{168}).

Mortality: deaths occurred within 30 min after the 3rd-5th injection in 46%, 34%, and 14% of the mice at an avelumab dose of 20, 40, and 140 mg/kg, respectively. Clinical signs immediately after injection included paralysis of the hind limbs, sternal recumbency, sedation and dyspnea. Histopathological evaluation did not establish the cause of death; however, it was assumed that death occurred as a consequence of a hypersensitivity reaction to foreign protein.

<table>
<tr><th>Study ID
/GLP/
Duration</th><th>Species/Sex/

Number/Group</th><th>Dose (mg/kg/day/
Route</th><th>NOAEL/NOEL</th></tr>
<tr><td colspan="4">Clinical signs: 140 mg/kg: none. 20 and 40 mg/kg: transient sternal recumbency within 2 hr of the 3rd administration. One ♀ at 20 mg/kg showed hypomotility after the 2nd injection. One ♂ at 20 mg/kg showed piloerection on Days 23-25.
Body weight, food consumption, urinalysis: no effects.

Serum chemistry: ≥ 20 mg/kg : AST ↑, ALT ↑, albumin/globulin ratio ↓ (slight), chloride ↑.

Hematology: ≥ 20 mg/kg: lymphocytes ↓, neutrophils ↑, MCH ↓.

Immunophenotyping: ≥ 20 mg/kg: lymphocytes ↓ due to marked reduction in cytotoxic T cells.

Antigenicity: 14/36 (38.9%) mice treated with avelumab tested positive for anti-drug antibodies (ADA). The highest incidences were in the low and intermediate dose groups. ADAs were detected in 9 ♀ and 5 ♂.
IgE and IgG in serum, gross pathology, organ weights: no clear treatment-related changes.

Histopathology: Liver: ≥20 mg/kg: ↑ sinusoidal lining cells (mainly Kupffer cells), ↑ microgranuloma (mononuclear inflammatory cell foci), ↑ hepatocellular necrosis (single cell or foci). Blood vessels: 20 and 140 mg/kg: granulomatous inflammation (vasculitis) in various organs in occasional animals. Heart: 40 and 140 mg/kg: slight myocarditis in occasional ♂. Brain: 20 and 140 mg/kg: minimal perivascular mononuclear cell cuffing in occasional animals. Lymph nodes, spleen and thymus: 140 mg/kg: lymphoid atrophy. Bone marrow: 20 and 140 mg/kg: hypocellularity, increased myeloid/erythroid ratio in occasional animals. Injection site: haemorrhage and inflammation in all groups including controls.</td></tr>
<tr><td>T16228/
non-GLP/
4 weeks
investigative</td><td>CD-1 mouse/
50/sex/group</td><td>0 (vehicle), 20

iv injection (once weekly)

vehicle: 280 mM mannitol, 10 mM sodium acetate, 1.4 mM methionine, 0.05% polysorbate 20, pH 5.5</td><td>NOAEL: not applicable</td></tr>
<tr><td colspan="4">Mortality: one ♂ at 20 mg/kg was found dead on Day 14. Histopathological evaluation did not reveal the cause of death. Seven ♂ and 7 ♀ were euthanized on Day 22 (after the 4th injection) due to marked clinical signs. Histopathology: see below.
Clinical signs: 20 mg/kg: the majority of animals showed staggering movements and reduced activity within 10 min after administration, following the 2nd-4th dose. Other clinical signs in occasional drug-treated animals included piloerection, decreased respiratory rate, apathic reaction to touch, incomplete eyelid closure and wet coat.
Body weight, food consumption: no clear treatment-related effects.

Body temperature: an attempt at checking the body temperature as an indicator for anaphylactic reaction failed due to the rapid appearance of clinical signs.

Planned investigations of cytokines, immunogenicity and serum chemistry could not be performed due to technical</td></tr>
</table>

Study ID /GLP/ Duration	Species/Sex/ Number/Group	Dose (mg/kg/day/ Route	NOAEL/NOEL

problems (limited blood volumes).

Gross pathology: no treatment-related findings.

Histopathology (limited sampling of organs): ***Liver**:* 20 mg/kg: minimal to moderate multifocal microgranuloma, focal hemorrhage. ***Kidney:*** 20 mg/kg: minimal to mild tubular basophilia. ***Lung:*** minimal to moderate focal or multifocal granuloma formation (macrophages containing brown pigment).
The histopathological findings were qualitatively the same in mice euthanized on Day 22 and Day 29, respectively, but tended to be more pronounced in the latter.

RF3310/ non-GLP/ 4 weeks	Han Wistar rat/ 6/sex/group Satellites for TK	0 (vehicle), 20, 40, 140 iv injection (once weekly) vehicle: 280 mM mannitol, 10 mM sodium acetate, 1.4 mM methionine, 0.05% polysorbate 20, pH 5.5	**NOAEL**: 140 mg/kg

TK: mean values for ♂ on Day 29 were, from lowest to highest dose: 39236, 100600, 333865 µg · h/mL (AUC_{168}). For ♀, corresponding values were 52209, 89748, 274237 µg · h/mL (AUC_{168}).

Mortality: none.
Clinical signs: none.
Body weight, food consumption: no effects.

Serum chemistry, urinalysis, immunophenotyping, IgG and IgE in serum: no treatment-related effects.
Coagulation: $\geq$ 40 mg/kg: PT and APTT ↓ in ♀. 140 mg/kg: PT ↓ in ♂.

Antigenicity: 9/36 (25%) rats treated with avelumab tested positive for anti-drug antibodies (ADA). There was no relevant difference in incidence between the dose groups. At 20 and 40 mg/kg, a decrease of exposure was observed in ADA-positive animals.

Gross pathology, organ weights: no treatment-related findings.

Histopathology: ***Liver**:* ≥20 mg/kg: ↑ sinusoidal lining cells (mainly Kupffer cells) in ♂; slightly increased severity of mononuclear inflammatory cell infiltration (microgranuloma) as compared to controls (equivocal finding).

Study ID /GLP / Duration	Species / Sex / Number / Group	Dose (mg / kg / day / Route	NOAEL / NOEL
RF2710 / partly GLP/ 4 weeks	Cynomolgus monkey/ 2/sex/group	0 (vehicle), 20, 60, 140 iv infusion (once weekly) vehicle: 280 mM mannitol, 10 mM sodium acetate, 1.4 mM methionine, 0.05% polysorbate 20, pH 5.5	**NOAEL:** 140 mg/kg

TK: mean values (both sexes) on Day 22 were, from lowest to highest dose: 26400, 115000, 227000 µg · h/mL (AUC_{168}).
Mortality: none.
Clinical signs, body weight, food consumption, ophthalmology: no effects.

Heart rate, ECG, arterial blood pressure, respiratory rate, functional observational battery, body temperature: no treatment-related findings.

Hematology: 140 mg/kg: ↓ lymphocytes.
Immunophenotyping: 140 mg/kg: ↓ lymphocytes and NK cells.
Coagulation, serum chemistry, urinalysis: no treatment-related findings.

Antigenicity: 3/12 (25%) monkeys treated with avelumab tested positive for anti-drug antibodies (ADA). Two were in the low dose and one in the high dose group. At 20 mg/kg, a decrease of exposure was observed in ADA-positive animals.

Panel of cytokines and chemokines in serum (multi analyte profile, MAP[a]): no obvious effects.
Gross pathology: 140 mg/kg: ↑ severity of subcutaneous reddish area at the injection site.
Organ weights: no treatment-related effects.
Histopathology: ***Injection site:*** 140 mg/kg: ↑ severity of haemorrhage, vascular necrosis and haemorrhage as compared to other dose groups and controls.

RF4990 / GLP/ 13 weeks + 8 weeks recovery	Cynomolgus monkey/ 3/sex/group 2/sex/group for recovery (0 and 140 mg/kg)	0 (vehicle), 20, 60, 140 iv infusion (once weekly) vehicle: 280 mM mannitol, 10 mM sodium acetate, 1.4 mM methionine, 0.05% polysorbate 20, pH 5.5	**NOAEL:** no NOAEL (Applicant's proposed NOAEL at 140 mg/kg not agreed with due to perivascular cell infiltration in the brain and spinal cord)

Study ID /GLP / Duration	Species / Sex / Number / Group	Dose (mg/kg/day / Route	NOAEL / NOEL

TK: mean values for ♂ on Day 85 were, from lowest to highest dose: 33785.2, 110702.3, 356513.8 µg · h/mL (AUC_{168}). Corresponding values for ♀ were, from lowest to highest dose: 33980.2, 140645.1, 303661.8 µg · h/mL (AUC_{168}).
Mortality: none.
Clinical signs, body weight, food consumption, ophthalmology: no treatment-related effects.

Heart rate, ECG, respiratory rate, functional observational battery, body temperature: no treatment-related findings.

Arterial blood pressure: 140 mg/kg: Slight and transient, not statistically significant, ↑ in mean, systolic and diastolic arterial blood pressure on Days 8 and 29.

Hematology, urinalysis, coagulation, immunphenotyping: no treatment-related findings.
Serum chemistry: 140 mg/kg: slight ↑ in C-reactive protein, minimal ↑ in AST and ALT.

Antigenicity: no ADA were detected.
Panel of cytokines and chemokines in serum (multi analyte profile, MAP[b]): no obvious effects.
Gross pathology: > 20 mg/kg: focal subcutaneous reddish areas at the injection site.
Organ weights: 140 mg/kg: ↓ adrenal wt (abs) in ♀ (no microscopic correlation).
Histopathology: ***Injection site:*** > 20 mg/kg: ↑ severity of haemorrhage, subcutaneous fibroplasia and mononuclear cell infiltration as compared to controls. ***Brain and spinal cord:*** > 20 mg/kg: minimal to slight mononuclear perivascular cell infiltration (3/6 animals at 140 mg/kg, 4/6 animals at 60 mg/kg, 3/6 animals at 20 mg/kg). ***Spleen:*** > 60 mg/kg: slight hyalinization in germinal centers.
Recovery: all findings were fully reversible with the exception of increased ALT in ♀.

[a] MCP-1, IP-10, TGFβ, IL-2, IL-4, IL-10, IL-6, IL-12, IFN-γ, IL-1β, IL-17, IL-7, IL-8, IL-15, TNFα, TNFβ, IFNα, MIP-1α, IL-5 and IL-1RA
[b] MCP-1, IP-10, TGFα, IL-2, IL-4, IL-10, IL-6, IFN-γ, IL-1-β, IL-17 and IL-1RA

Genotoxicity and Carcinogenicity

The applicant did not submit genotoxicity and carcinogenicity studies (see non-clinical discussion).

Reproduction Toxicity

The applicant did not submit reproduction toxicity studies (see non-clinical discussion).

Toxicokinetic data

Toxicokinetics and exposure margins
In all three species evaluated (mice, rats, monkeys), exposure to avelumab increased roughly dose-proportionally. There was no gender difference in exposure. Accumulation was observed in rats (marked) and monkeys (moderate). The $T_{1/2}$ was 60-80 hrs in mice, > 100 hrs in rats and 58-70 hrs in monkeys.

Plasma exposure (AUC) achieved in the repeat-dose toxicity studies in monkeys exceeded the human therapeutic exposure by 10- to 15-fold. Due to the finding of perivascular mononuclear cell infiltrates in the brain and spinal cord at all dose levels in the 13-week monkey study, there was no NOAEL in this study and accordingly only a LOAEL, which was 1.5-fold above the human therapeutic exposure.

Local Tolerance

The local tolerance of avelumab was evaluated within repeat-dose toxicity studies.

At the infusion site in cynomolgus monkeys, there was an increased severity of haemorrhage, subcutaneous fibroplasia and mononuclear cell infiltration at $\geq$ 20 mg/kg as compared to controls, suggesting local irritative properties of avelumab.

Other toxicity studies

Cytokine release in whole blood and peripheral blood mononuclear cells (PBMCs) of humans and cynomolgus monkeys

Multiple Th1 and Th2 cytokines were measured in non-stimulated human or cynomolgus whole blood and PBMCs, following exposure to avelumab for 6 and 24 hours. In human whole blood, IL-8, IL-6, IL-1β and IFNγ were mildly increased after 24 hours. Two female donors showed a marked induction of IL-6 release. In cynomolgus whole blood, minor increases in IL-6, IL-8, TNFα and IFNγ were seen in females. Larger increases in the release of these same cytokines (plus IL-1β) were seen in male whole blood. In human PBMCs, a significant increase in TNFα release was observed in both males and females after 24 hours (at avelumab concentrations $\geq$ 14.38 ng/mL). Only minor effects on cytokines were observed in cynomolgus PBMCs.

2.3.3. Ecotoxicity/environmental risk assessment

The applicant submitted a justification for the lack of studies concerning environmental assessment. According to the guideline on the environmental risk assessment of medicinal products for human use (CHMP/SWP/4447/00), proteins are unlikely to result in significant risk to the environment.

2.3.4. Discussion on non-clinical aspects

All in vitro and in vivo primary and secondary pharmacology studies were not conducted under GLP conditions. This is acceptable as the pharmacology of PD-L1 has been extensively studies and well known through the literature.

Avelumab shows similar binding affinity for human, mouse and cynomolgus PD-L1.

In *in vitro* assays, utilizing primary T cells of human or mouse origin, avelumab enhanced T-cell activation in a dose-dependent manner. Avelumab was also shown to exhibit ADCC activity against human tumour cell lines, but no CDC activity was observed. When combing tumour cells with peripheral blood mononuclear cells, ADCC activity was only directed against tumour cells, with no killing of any immune cell subsets. An in vivo contribution of ADCC to the anti-tumour effect of avelumab was demonstrated in the mouse tumour MC38 model. When using n ADCC-incompetent form of avelumab, generated through enzymatic deglycosylation, or by depleting NK cells (the main mediators of ADCC), a similar modest reduction in anti-tumor response was observed. These data suggest a role for ADCC in the anti-tumour response. A decrease in peripheral blood lymphocytes, due to marked reduction in cytotoxic T cells, was observed in mice treated with avelumab at > 20 mg/kg, once weekly injection, for 4 weeks. In a cynomolgus monkey 4-week, once weekly infusion, study decreased total lymphocytes and NK cells occurred at 140 mg/kg.

Cardiovascular, respiratory and central nervous system safety pharmacology parameters were incorporated in the cynomolgus monkey toxicity studies instead of being conducted as stand-alone studies, which is considered acceptable. Apart from a transient, not statistically significant, increase in arterial blood pressure at 140 mg/kg, no treatment-related findings were observed.

Repeated administration of avelumab to CD-1 mice caused severe anaphylactic reactions, resulting in the death of up to 46% of the animals. Anti-drug antibody (ADA) formation was detected in 38.9% of the mice, with highest incidence in the low and intermediate dose groups (where also most of the deaths occurred). A mechanistic study supported the hypothesis that the mortalities were caused by an immune-mediated anaphylactic reaction (IgG/IgE mediated), primarily driven by IgG isotype antibodies.

It is unlikely that anaphylactic reactions like this, triggered by an immune reaction against human protein in mice, would occur in the clinical setting. A low percentage (3.5%) of patients treated with avelumab has been reported to test positive for ADA (see clinical pharmacology). Nevertheless, immunogenicity is included as an important potential risk in the RMP, and is described under section 4.8 of the SmPC. Furthermore, infusion-related reactions including drug hypersensitivity and immune-related adverse reactions are included as important identified risks in the RMP and as warnings in sections 4.4 and 4.8 of the SmPC.

Inflammatory changes in the liver, and/or increased AST and ALT levels, were observed in surviving mice as well as in cynomolgus monkeys. In the latter species, there was also a slight increase in C-reactive protein (a marker for inflammation) at 140 mg/kg. Avelumab's mode of action removes the suppressive effects of PD-L1 on T cells, which may lead to decreases in self-tolerance and immune-related effects. Autoimmune hepatotoxicity has been reported in connection with PD-1 inhibitors[24]. Immune-related hepatitis is included as an Important identified risk in the RMP, and is described in sections 4.4 and 4.8 of the SmPC. From a non-clinical perspective, no further action is considered needed.

The applicant did not submit single dose toxicity studies. The lack of studies is acceptable as the results in cynomolgus monkeys following the 1st administration of avelumab up to 140 mg/kg (AUC exposure approximately 15-fold above human therapeutic exposure) indicate that the acute toxicity of avelumab is low. Non-clinical data reveal no special hazard for humans based on conventional studies of repeated dose toxicity in Cynomolgus monkeys administered intravenously doses of 20, 60 or 140 mg/kg once a week for1 month and 3 months, followed by a 2-month recovery period after the 3-month dosing period. Perivascular mononuclear cell cuffing was observed in the brain and spinal cord of monkeys treated with avelumab at ≥ 20 mg/kg for 3 months. Although there was no clear dose-response relationship, it cannot be excluded that this finding was related to avelumab treatment.

No dedicated metabolism studies were carried out with avelumab, as the expected consequence of the metabolism of biotechnology-derived pharmaceuticals is the degradation to small peptides and individual amino acids. Guideline ICH S6 Preclinical Safety Evaluation of Biotechnology-Derived Pharmaceuticals states that classical biotransformation studies need not be done.

It is unknown whether avelumab is excreted in human milk. Since it is known that antibodies can be secreted in human milk, a risk to the newborns/infants cannot be excluded. It is known that antibodies of

[24] Villadolid J, Amin A. Immune checkpoint inhibitors in clinical practice: update on management of immune-related toxicities. Transl Lung Cancer Res. 2015 Oct;4(5):560-75.

the IgG class can be secreted into breast milk[25], therefore it is predicted that this may occur with avelumab. Therefore a statement has been included in section 4.6 of the SmPC that breast feeding women should be advised not to breast feed during treatment and for at least 1 month after the last dose due to the potential for serious adverse reactions in breast fed infants.

No studies have been conducted to assess the potential of avelumab for carcinogenicity or genotoxicity. Genotoxicity and carcinogenicity studies are generally not required for biotechnology-derived monoclonal antibodies such as avelumab (ICH S9).

No reproductive and developmental toxicity studies have been performed. Animal reproduction studies have not been conducted with avelumab. The PD 1/PD L1 pathway is thought to be involved in maintaining tolerance to the foetus throughout pregnancy. Blockade of PD L1 signalling has been shown in murine models of pregnancy to disrupt tolerance to the foetus and to result in an increase in foetal loss. These results indicate a potential risk that administration of avelumab during pregnancy could cause foetal harm, including increased rates of abortion or stillbirth.

Fertility studies have not been conducted with avelumab. In 1-month and 3-month repeat-dose toxicology studies in monkeys, there were no notable effects in the female reproductive organs. Many of the male monkeys used in these studies were sexually immature and thus no explicit conclusions regarding effects on male reproductive organs can be made (SmPC section 4.6 and 5.3).

Animal reproduction studies have not been conducted with avelumab. However, in murine models of pregnancy, blockade of PD L1 signalling has been shown to disrupt tolerance to the foetus and to result in an increased foetal loss (see section 5.3). These results indicate a potential risk, based on its mechanism of action, that administration of avelumab during pregnancy could cause foetal harm, including increased rates of abortion or stillbirth (see section 4.6). There are no or limited data from the use of avelumab in pregnant women. Women of childbearing potential should be advised to avoid becoming pregnant while receiving avelumab and should use effective contraception during treatment with avelumab and for at least 1 month after the last dose of avelumab.

Avelumab is a protein that undergoes breakdown into amino acids, and as such, does not pose a significant risk to the environment.

2.3.5. Conclusion on the non-clinical aspects

The pharmacologic, pharmacokinetic, and toxicological characteristics of avelumab have been well characterised in the non-clinical aspects. The non-clinical aspects are considered to be appropriately addressed.

2.4. Clinical aspects

2.4.1. Introduction

GCP

The clinical trials were performed in accordance with GCP as claimed by the applicant.

[25] Hurley W, Theil P. Perspectives on Immunoglobulins in Colostrum and Breast Milk. Nutrients 2011;3:442-74.

The applicant has provided a statement to the effect that clinical trials conducted outside the community were carried out in accordance with the ethical standards of Directive 2001/20/EC.

- Tabular overview of clinical studies

Table 6: Overview of clinical studies with clinical pharmacology components

Protocol No	Study population (N)	PK/ADA sampling	Treatments
EMR100070-001 (Phase I)	Adult subjects with metastatic or locally advanced solid tumors and expansion to selected indications. (1490 enrolled as of 20 November 2015)	Serial PK sampling during the first dose interval of the dose escalation cohorts (53 subjects) and expansion cohorts (25 subjects) in CRC and CRPC expansion cohorts: prior to and at the end of the 1-hour infusion, and at 0.5, 1, 2, 4, 6, and 12 hours, and 24,36, 48 hours post the first infusion Sparse PK sampling at trough and/or the end of infusion throughout the study on multiple visits in all cohorts including the aforementioned dose escalation, CRC, and CRPC cohorts on visits beyond the first dose interval ADA sampling at baseline, prior to dosing throughout the study and at the End-of-Treatment visit	Avelumab administered via iv infusion once every 2 weeks Dose escalation phase: 1.0 mg/kg (n = 4) 3.0 mg/kg (n =13) 10.0 mg/kg (n =15) 20.0 mg/kg (n =21) Treatment expansion phase: 10.0 mg/kg (n=1437)
EMR100070-002 (Japan Phase I)	Adult Japanese subjects with metastatic or locally advanced solid tumors, with expansion in subjects with gastric cancer. (51 enrolled as 20 November 2015)	Serial PK sampling during the first dose interval of the dose escalation cohorts (17 subjects): prior to and at the end of the 1-hour infusion, and at 0.5, 1, 2, 4, 6, and 12 hours, and optional 24, 36, 48 hours post the first infusion Sparse PK sampling at trough and/or the end of infusion throughout the study in gastric cancer expansion cohort and on multiple visits beyond first dose interval in the dose escalation cohorts ADA sampling at baseline, prior to dosing throughout the study and at the End-of-Treatment visit	Avelumab administered via iv infusion once every 2 weeks Dose escalation phase: 3.0 mg/kg (n = 5) 10.0 mg/kg (n = 6) 20.0 mg/kg (n = 6) Treatment expansion phase:10.0 mg/kg (n=34)
EMR100070-003 (Phase II)	Adult subjects who have received at least 1 line of previous chemotherapy for the treatment of mMCC. (88 enrolled and treated with 6 m minimum follow-up as of Mar 3rd, 2016)	Sparse PK sampling including predose, at the end of infusion and 2 to 8 hours after the end of infusion on multiple visits throughout the study ADA sampling for baseline, prior to dosing throughout the study and at the End-of-Treatment visit	Avelumab administered via iv infusion once every 2 weeks 10 mg/kg (n=88)

CRC: colorectal cancer; CRPC: castrate-resistant prostate cancer; mMCC: metastatic Merkel cell carcinoma

Table 7: Overview of Pivotal Study in Metastatic Merkel Cell Carcinoma and Key Supportive Studies

Study No.	Study Design	Subject Population	No. of Subjects
Pivotal Study in mMCC			
EMR100070-003 Conducted in US, Australia, Austria, France, Germany, Italy, Japan, Spain, and Switzerland	Part A: Phase II, open-label, single-arm study Objectives: Efficacy, safety, biomarkers, PK Primary endpoint: Confirmed BOR per IERC	Part A: Adult subjects who have progressed after receiving at least 1 line of previous chemotherapy for the treatment of mMCC	Part A: 88 subjects (enrollment complete) 10 mg/kg every 2 weeks
	Part B: Phase II, open-label, single-arm study	Part B: Adult, systemic chemotherapy-naïve	Part B: 112 subjects planned 10 mg/kg every 2 weeks

Study No.	Study Design	Subject Population	No. of Subjects
	Objectives: Efficacy, safety, biomarkers, PK Primary endpoint: Durable response	subjects with mMCC	
Key Supportive Studies			
100070-Obs001 Part A – US Part B – EU	Retrospective observational study to evaluate outcomes under current clinical practice in mMCC Objectives: To assess ORR, DOR, PFS, and OS of standard of care therapy Primary endpoint: BOR	Subjects treated for mMCC in a real-world clinical practice setting	Part A (1L chemotherapy): 67 subjects (51 immunocompetent) Part A (2L+ chemotherapy): 20 subjects (14 immunocompetent) Part B (2L+ chemotherapy): 34 subjects (29 immunocompetent)
EMR100070-001 Conducted in US, Belgium, Czech Republic, France, Germany, Hungary, Korea, Poland, Taiwan, Republic of China, and UK	Phase I, open-label, 2-phase (dose escalation and treatment expansions) in solid tumors Objectives: Safety/tolerability, MTD, efficacy (treatment expansion phase only), and PK	Adult subjects with metastatic or locally advanced solid tumors and expansion to selected indications	Dose escalation: 53 subjects (completed) 4 at 1 mg/kg, 13 at 3 mg/kg, 15 at 10 mg/kg, and 21 at 20 mg/kg every 2 weeks Dose expansion: (ongoing) 1437 subjects 10 mg/kg every 2 weeks
EMR100070-002 Conducted in Japan	Phase I, open-label, 2-phase (dose escalation and treatment expansion) in solid tumors Objectives: Safety/tolerability, MTD, efficacy, and PK	Adult subjects with metastatic or locally advanced solid tumors, with expansion in subjects with gastric cancer	Dose escalation: (completed) 17 subjects 5 at 3 mg/kg, 6 at 10 mg/kg, and 6 at 20 mg/kg every 2 weeks Dose expansion: (ongoing) 34 subjects 10 mg/kg every 2 weeks

1L = first line, 2L+ = second line or later, EU = European Union, BOR = best overall response, DOR = duration of response, IERC = Independent Endpoint Review Committee, mMCC = metastatic Merkel cell carcinoma, MTD = maximum tolerated dose, ORR = objective response rate, OS = overall survival, PFS = progression-free survival, PK = pharmacokinetics, UK = United Kingdom, and US = United States

2.4.2. Pharmacokinetics

Absorption

Bavencio is for intravenous infusion only. It must not be administered as an intravenous push or bolus injection.

Distribution

The volume of distribution during terminal phase (V_z) was estimated using non-compartmental analysis and was determined to 50.9 mL/kg (mean after single dose 10 mg/kg). The volumes of the central and peripheral compartments were also estimated in the Pop PK analysis and were 2.84 L and 1.21 L in the typical subject, respectively. The geometric mean Vss (calculated from individual V1 and V2 parameter values) for a subject receiving 10 mg/kg was 4.72 L.

Elimination

Following IV administration of a 10 mg/kg dose the mean clearance determined by non-compartmental analysis was 0.36 mL/h/kg (study EMR100070-001). The corresponding mean half-life was 95 h (~4 days). The Applicant points out that the t½ is shorter compared to other IgG1 antibodies (8-20 days) and endogenous IgG1 (30 days). According to the Applicant the shorter t½ could be due to the higher isoelectric point of avelumab (pI 8.5 – 9.3). From the PopPK analysis the estimated t½ was approximately 6 days in subjects receiving 10 mg/kg every 2 weeks.

No mass balance or metabolism studies of avelumab have been submitted.

Dose proportionality and time dependencies

The dose-normalized C_{max} and $AUC_{0\text{-}336hr}$ after first dose were approximately similar across 3 to 20 mg/kg. C_{trough} increased proportionally with doses between 10 to 20 mg/kg, but more than proportionally for doses between 1 to 10 mg/kg, likely due to the presence of target mediated drug disposition (TMDD) at these lower dose levels.

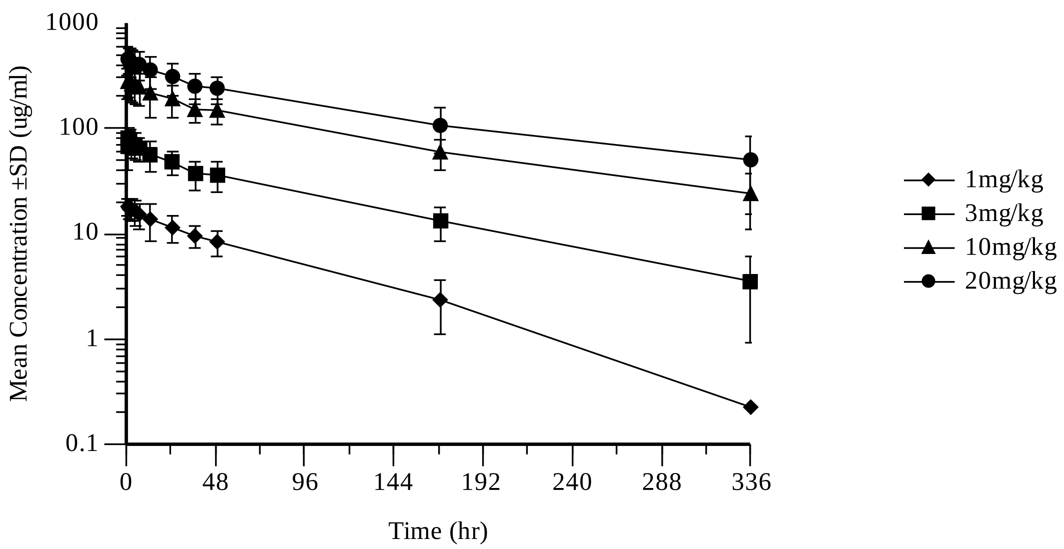


Figure 6: Study EMR100070-001: Serum Avelumab Concentrations Mean (± s.d.) vs. Time following first 1-hour IV Infusion to subjects with solid tumours

Table 8: Selected Pharmacokinetic Parameters after the First Dose of Avelumab for subjects with Solid Tumors – Non Compartmental Analysis

Variable	Dose Group	N[a]	GM[b]	CV% GM[b]	Mean	Median	s.d.
C_{max} (µg/mL)	1 mg/kg	4	18.4	21.8	18.7	18.9	3.96
	3 mg/kg	13	78.9	29.5	81.9	82.6	22.1
	10 mg/kg	48	294	32.5	310	291	109
	20 mg/kg	21	470	29.9	489	494	140
C_{trough} (µg/mL)	1 mg/kg	3	NC	NC	0.227	0.00	0.393
	3 mg/kg	13	NC	NC	3.55	2.98	2.61
	10 mg/kg	39	20.9	57.4	24.0	19.8	13.3
	20 mg/kg	19	36.9	112.0	50.1	44.3	34.6
AUC_{0-t} (hr*µg/mL)	1 mg/kg	4	976	40.7	1040	889	443
	3 mg/kg	13	5740	37.6	6080	7000	1970
	10 mg/kg	48	22800	48.2	24800	24100	9150
	20 mg/kg	21	42200	40.8	45100	45600	15600
$AUC_{0-336hr}$ (hr*µg/mL)	1 mg/kg	2	1180	52.2	1250	1250	591
	3 mg/kg	12	6080	32.1	6340	7020	1800
	10 mg/kg	39	25200	25.3	25900	24300	6680
	20 mg/kg	15	38300	42.0	41100	45400	14800
$AUC_{0-\infty}$ (hr*µg/mL)	1 mg/kg	2	1200	56.7	1290	1290	650
	3 mg/kg	12	6520	34.8	6850	7570	2100
	10 mg/kg	39	27700	27.0	28600	27400	7700
	20 mg/kg	15	42700	47.6	46600	48400	18500
$t_{1/2}$ (hr)	1 mg/kg	2	59.5	37.4	61.4	61.4	21.7
	3 mg/kg	12	81.2	30.3	84.3	89.1	21.8
	10 mg/kg	39	94.6	22.0	96.6	97.7	18.6
	20 mg/kg	15	99.1	29.9	103	109	27.4

a $AUC_{0-336hr}$, $AUC_{0-\infty}$, C_{trough}, and $t_{1/2}$ were not calculated for subjects whose last concentration was not collected after 169 and for subjects with more than 20% AUC extrapolation beyond the last sampling time within the first dosing interval. The full terminal phase may not have been captured as 336 hours (14 days) is less than 6 $t_{1/2}$ (36 days) and subjects with longer $t_{1/2}$ are more likely to have more than 20% AUC extrapolation beyond 336 h

b GM and CV%GM were not calculated for the group where values below quantitative limit are present

Special populations

<u>Impaired renal function</u>

The PopPK analysis did not detect any relation between clearance of avelumab and renal function.

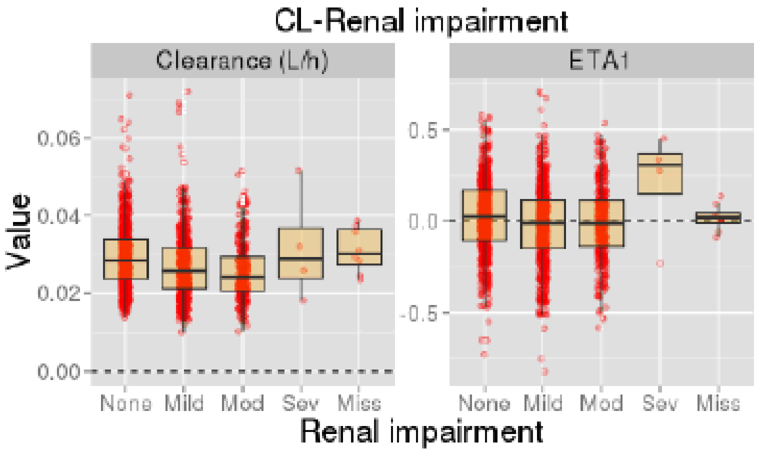


Figure 7: Influence of renal impairment on CL and inter-individual variability (IIV) CL

Impaired hepatic function

There was no clinical relevant influence of hepatic function in mild HI on avelumab CL in the final PopPK model (see Figure below).

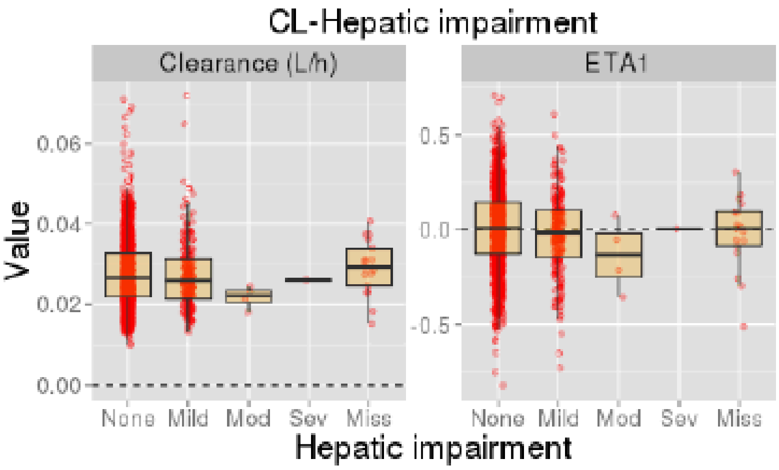


Figure 8: Influence of hepatic impairment on CL and inter-individual variability (IIV) CL

Weight

Body weight was identified to be a significant covariate for both CL and V1 (see Figure below for CL from the final PopPK model).

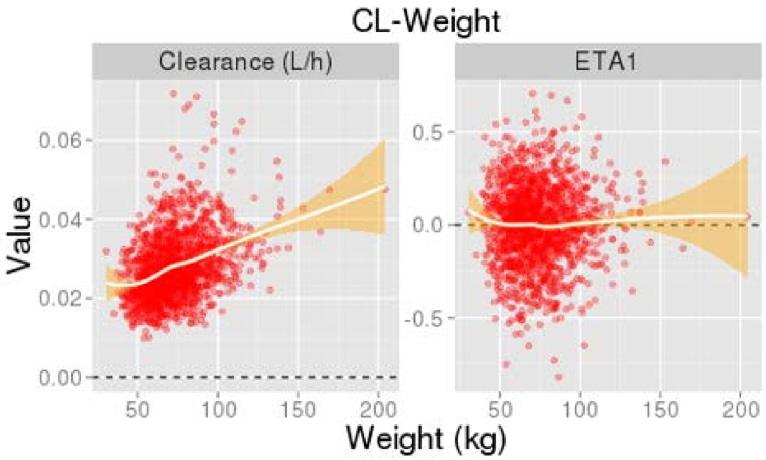


Figure 9: Influence of weight on CL and inter-individual variability CL (ETA1)

<u>Gender</u>

The influence of gender was evaluated in the PopPK analysis including 811 male and 818 female subjects. After accounting for body weight, gender was found to influence both CL (male subjects have a 19.9% higher CL relative to female subjects) and V1 (male subjects have a 20.3% higher V1 relative to female subjects). The final Pop PK model was used to perform simulations to compare exposure for each sex. The expected AUCss for female is slightly higher than that for male (see Figure below). The changes in avelumab PK due to gender after accounting for body weight appear to be small.

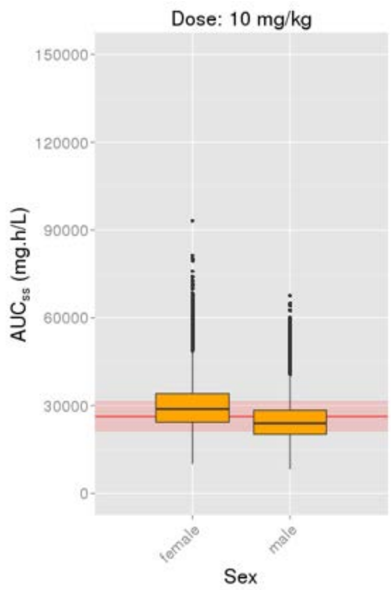


Figure 10: Simulated Steady State AUC for male and female subjects after repeated dosing with 10 mg/kg

N=100,000 simulated datasets. Black horizontal lines are medians. Boxes are interquartile range. Solid red horizontal line represents median exposure. Red shaded area represents 80-120% range. Box widths represent the amount of observations in each category.

Elderly

The influence of age was evaluated in the population PK analysis where the geometric mean age was 63 years and ranged between 20 to 91 years (n=1629). No influence of age was detected on avelumab clearance (see Figure below for final popPK model).

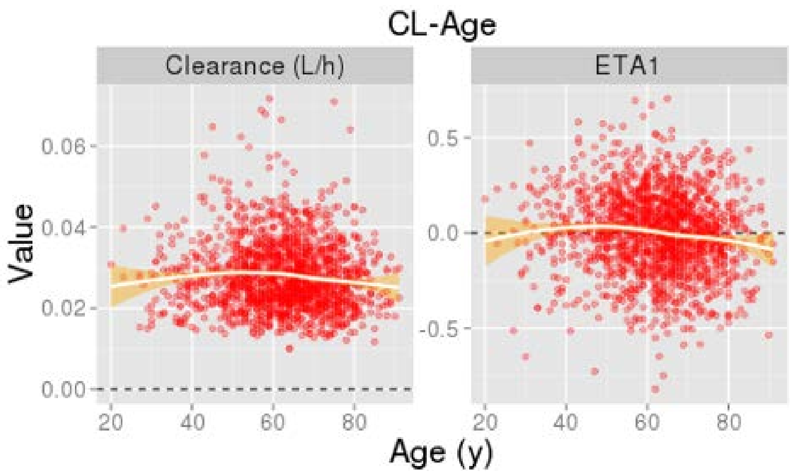


Figure 11: Influence of age on CL and inter-individual variability CL (ETA1)

Children

The applicant did not submit data in children or adolescents. The safety and efficacy of Bavencio in children and adolescents below 18 years of age have not been established.

Pharmacokinetic in the target population

Pharmacokinetics has been documented and analysed in patient populations and not in healthy volunteers. For PK sparse blood samples were collected prior to each infusion through Week 15, then Week 25, and then at 12 week intervals while on treatment. Post infusion samples were collected at the end of infusion and 2 to 8 hours after the end of infusion at Weeks 1, 7, 13, and 25, and then at 12 week intervals while on treatment.

The observed coefficients of variation (CV%) of geometric mean ranged from ca. 14% to 35% for the concentration at end of infusion (C_{EOI}, Table 9) and from ca. 13% to 102% for C_{trough} (Table 10). The mean avelumab concentration was 252 µg/mL (range 107 to 1108 µg/mL) at the end of infusion and 23.8 µg/ml (range 1.58 to 245 µg/mL) at trough after the first dose. Both CEOI and C_{trough} appeared to increase gradually over time (Table 9 and Table 10). Individual C_{trough} over time is given in Figure 12).

Table 9: Summary Table of Avelumab Serum Concentration at End of Infusion over Nominal Time

Dose Group	Day (Week)	N	GM (µg/mL)	CV% GM	Mean (µg/mL)	Median (µg/mL)	s.d. (µg/mL)
10 mg/kg	1 (1)	59	235	34.7	252	236	129
	43 (7)	48	258	24.4	266	261	74.2
	85 (13)	30	268	22.1	274	271	57.7
	169 (25)	23	308	22.7	315	326	65.0
	253 (37)	15	311	23.3	318	322	70.1
	337 (49)	6	371	13.8	373	389	48.3
	421 (61)	4	448	15.8	453	445	71.5

C_{EOI}: concentration at end of infusion; CV%: percent coefficient of variation; GM: geometric mean

Table 10: Summary Table of Avelumab Serum Ctrough over Nominal Time

Dose Group	Day (Week)	N	GM (µg/mL)	CV% GM	Mean (µg/mL)	Median (µg/mL)	s.d. (µg/mL)
10 mg/kg	15 (W3)	77	18.5	73.8	23.8	19.4	28.4
	29 (W5)	69	NC	NC	26.4	25.2	13.7
	43 (W7)	59	24.6	88.3	32.3	25.7	35.8
	57 (W9)	56	27.6	68.6	32.7	29.1	18.8
	71 (W11)	52	28.0	70.8	33.5	29.8	21.1
	85 (W13)	42	33.9	92.7	45.5	36.0	53.6
	99 (W15)	37	33.7	78.6	40.3	35.2	24.0
	169 (W25)	24	40.1	42.8	43.6	40.8	19.6
	211 (W31)	1	57.2	NC	57.2	57.2	NC
	253 (W37)	11	34.8	54.9	38.4	37.1	15.7
	337 (W49)	4	36.0	101.6	43.9	49.8	23.7
	421 (W61)	2	61.2	12.8	61.4	61.4	7.79

NC: not calculable (due to existence of values below quantitative limit in the groups)

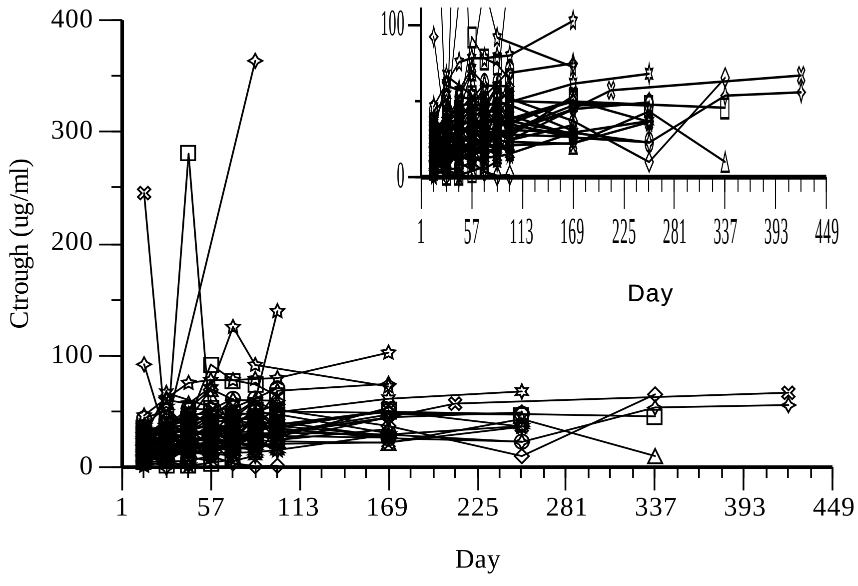


Figure 12: Individual Serum Trough Concentrations of Avelumab over Nominal Time Following Repeated iv Infusion

Pharmacokinetic interaction studies

The applicant did not submit studies on PK interaction studies (see clinical pharmacology discussion).

Pharmacokinetics using human biomaterials

The applicant did not submit PK using human biomaterials (see clinical pharmacology section).

2.4.3. Pharmacodynamics

Mechanism of action

The applicant did not submit any studies on the mechanism of action (see clinical pharmacology section).

Primary and Secondary pharmacology

Exposure-response (E-R)

The efficacy endpoints included in the exposure-efficacy analyses were BOR, PFS and OS. The exposure-efficacy analyses used data from 88 subjects with mMCC treated with avelumab in the pivotal study EMR100070-003.

The adverse event (AE) categories analyzed in the exposure-safety analyses included irAE, IRR (infusion-related reactions), and TEAE. The exposure-safety analyses used data from all 1629 subjects treated with avelumab and who had PK data in Studies EMR100070-001, EMR100070-002 and EMR100070-003.

A number of the presented analyses showed a positive relation between efficacy/safety variables and exposure. However, due to confounding factors related to disease status no conclusion is drawn about exposure-response for avelumab.

Exposure-QTc analysis

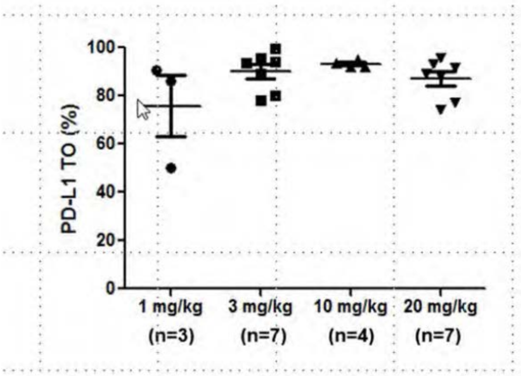
The QTc endpoints in the exposure-QTc analyses included QTcP and QTcF. A total of 2194 time-matched singlet locally-read 12-lead ECG and avelumab concentrations were collected from a total of 689 study subjects from Studies EMR100070-001, EMR100070-002, and EMR100070-003.

The effect of avelumab concentration on ΔQTcP or ΔQTcF was minimal with a small and nonsignificant slope.

Pharmacodynamic biomarkers

Pharmacodynamic biomarkers included target occupancy (TO) of PD-L1 in peripheral blood mononuclear cells (PBMCs) and cytokine measurements in serum:

- *In vitro* TO: after the first infusion, TO was predicted to reach or exceed 95% throughout the entire dosing interval for 10 of 13 subjects at 3 mg/kg, and for all 12 subjects in the 10 mg/kg group from the dose-escalation cohort of Study EMR100070-001. A median target occupancy of 86.3% was observed prior to the second infusion on Day 15 in the 1 mg/kg cohort. In the 3 mg/kg cohort, the median target occupancy on Day 15 was 93.5%. A median target occupancy of 92.9% was observed in 10 mg/kg cohort on Day 15, while the median target occupancy was 89.1% in the 20 mg/kg cohort on Day 15. Overall, 3 mg/kg bi-weekly dose was sufficient to achieve high target occupancy in the circulation throughout the entire dosing period for most of the subjects tested. The figure below reports TO in 1,3,10 and 20 mg/mL cohorts on day 15 prior to the second avelumab infusion.



Source: Table 15.5.2.1V.
Data represents the mean ± standard deviation

Figure 13: Target occupancy in 1, 3, 10 and 20 mg/mL cohorst assessed on Day 15 prior to the second avelumab infusion

- *Ex vivo* TO in PBMCs: 93.2 ± 1.29% TO throughout the dose interval at doses of 10 mg/kg once every 2 weeks thus supporting in vitro observations.
- Levels of several major circulating cytokines were measured in Study EMR100070-001 over period of 6 weeks, including interleukin (IL)-1β, IL-2, IL-4, IL-6, IL-10, interferon-γ (IFNγ) and tumor necrosis factor-α (TNFα). After repeated administration of avelumab at 10 mg/kg once every 2 weeks, IFNγ and TNFα concentrations exhibited transient and mild change but remained low overall (e.g., 1.9 ± 1.6 pg/mL on Day 15 for IFNγ). No apparent dose response was observed based on data collected from the 1 to 20 mg/kg cohorts.

The extent of TO in the tumor was not assessed in the dose-escalation cohort of Study EMR100070-001. No PD or biomarker information for the Expansion phase is available for this report.

Immunogenicity

For study EMR100070-001, post-treatment samples for evaluation of anti-avelumab antibody (HAHA) responses were available for 45 of 53 subjects from the dose-escalation cohorts. The table below reports the HAHA incidence:

Table 11: Human anti-human antibody: the incidence and percentage - dose escalation safety analysis set

	Dose Cohort				
	1.0 mg/kg N=4 n/N (%)	3.0 mg/kg N=13 n/N (%)	10.0 mg/kg N=15 n/N (%)	20.0 mg/kg N=21 n/N (%)	Overall N=53 n/N (%)
Pre-existing, n/N1[a]	0/4	0/13	0/7	0/19	0/43
Treatment boosted, n/N2[b]	0/3	0/13	0/6	0/18	0/40
Treatment emergent, n/N3[c]	0/3	2/13 (15.4)	0/9	0/20	2/45 (4.4)
Transient positive, n/N3	0/3	1/13 (7.7)	0/9	0/20	1/45 (2.2)
Persistent positive, n/N3	0/3	1/13 (7.7)	0/9	0/20	1/45 (2.2)

Source: Table 15.5.3.1V.
a N1 = The number of subjects with valid Baseline result.
b N2 = The number of subjects with valid Baseline and at least one valid post-Baseline result.
c N3 = The number of subjects with at least one valid post-baseline result and without positive Baseline results (including missing, NR).

No subject had pre-existing HAHAs. Across **the dose-escalation cohorts**, 2 subjects (4.4 %), both in the 3 mg/kg Cohort, were positive for treatment-emergent HAHAs, including 1 subject who was persistent positive, with persistent defined as treatment emergent subjects with duration between first and last positive result ≥ 16 weeks or a positive evaluation at the last assessment.

No subjects with hypersensitivity were identified from dose escalation at the time of analysis, so the IgE analysis was performed solely with samples from subjects in the expansion cohorts.

As of the data cutoff date (20 November 2015), samples for evaluation of HAHA response were available for 1298 of 1437 subjects. Across **all expansion cohorts**, 7 of 1218 subjects (0.6 %) were positive at Baseline for HAHA. Of these pre-existing positive subjects, none had titer increases ≥ 8 × relative to Baseline (ie, treatment-boosted) after treatment with avelumab. The table below reports the HAHA incidence.

Table 12: Human anti-human antibody: the incidence and percentage - expansion cohort safety analysis set

Number of Subjects with	Expansion Cohorts N=1437 (100%) n/N (%)
Pre-existing, n/N1[a]	7/1218 (0.6)
Treatment boosted, n/N2[b]	0/1132
Treatment emergent, n/N3[c]	46/1291 (3.6)
Transient positive, n/N3	16/1291 (1.2)
Persistent positive, n/N3	30/1291 (2.3)

Source: Table 15.5.3.1W.

a N1 = The number of subjects with valid Baseline result.

b N2 = The number of subjects with valid Baseline and at least one valid post-Baseline result.

c N3 = The number of subjects with at least one valid post-baseline result and without positive baseline results (including missing, NR).

Across all expansion cohorts, 46 of 1291 subjects (3.6 %) who were not positive for HAHA at Baseline had subsequent positive HAHA results after treatment with avelumab. The persistent treatment-emergent antibody incidence was 30 of 1291 subjects (2.3 %), with persistent defined as treatment emergent subjects with duration between first and last positive result ≥ 16 weeks or a positive evaluation at the last assessment

For study EMR100070-003, immunogenicity of avelumab was correlated to exposure in subjects with MCC and as exploratory objectives was also correlated to the immunogenicity of avelumab with clinical results (ORR and adverse events [AEs]). Subjects were categorised as either never positive or ever positive (a positive result at any time point, including Baseline).

Among the 88 subjects with at least one HAHA sample available, 3 subjects were regarded as ever positive with HAHA status. Although no within-subject change on PK was observed in the subjects after positive HAHA results, no firm conclusion can be drawn given the small incidence of seroconversion.

Table 13: Human anti-human antibody: the incidence and percentage - safety analysis set

Number of Subjects With	Expansion Cohorts N=88 (100%) n/N (%)
Pre-existing, n/N1[a]	0/85
Treatment boosted, n/N2[b]	0/79
Treatment emergent, n/N3[c]	3/82 (3.7)
Transient positive, n/N3	1/82 (1.2)
Persistent positive, n/N3	2/82 (2.4)

Source: Table 15.5.1.1.

a N1 = The number of subjects with valid Baseline result.

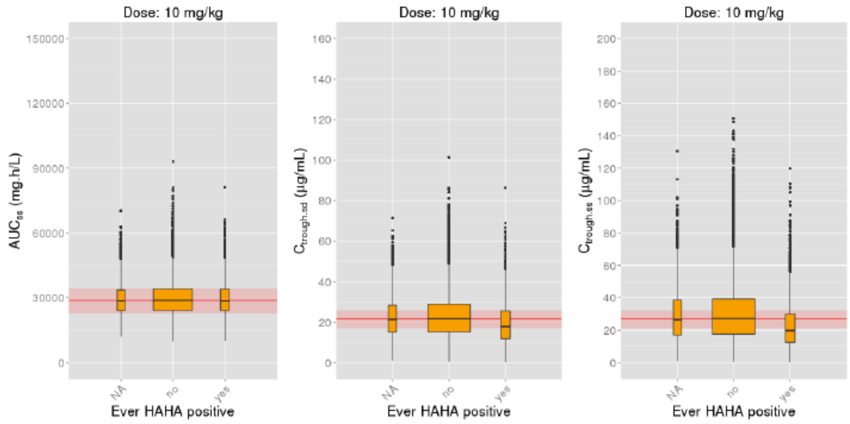
b N2 = The number of subjects with valid Baseline and at least one valid post-Baseline result.

c N3 = The number of subjects with at least one valid post-Baseline result and without positive Baseline results (including missing, not reported).

In study EMR100070-002, conducted in Japanese patients, samples for evaluation of HAHA response were available for 17 subjects from the dose-escalation cohorts.

In the dose-escalation cohorts, none of the 17 subjects were positive at baseline for HAHA. In the dose-escalation cohorts, a total of 1/17 (5.9%) subjects who were not positive for HAHA at baseline had subsequent positive HAHA results after treatment with avelumab; 1/5 (20.0%) subjects receiving 3.0 mg/kg dose and no HAHA positives at higher doses. The treatment-emergent response was persistent.

As part of the Population PK analysis, the influences of occurrence of an immunogenicity (HAHA) has been analysed and the results are showed below:



N=100,000 simulated datasets. Black horizontal lines are medians. Boxes are interquartile range. Solid red horizontal line represents median exposure. Red shaded area represents 80-120% range. Box widths represent the amount of observations in each category. Subjects were "Ever HAHA positive" if they had measurable antibodies at any point in the duration of the three studies. All other covariates set to population median or most common category. Dose set to 705 mg (10 mg/kg).

Filename: run124.hahaever.normal.png

Figure 14: Simulated exposure parameters by occurence of immunogenicity

The small observed changes in AUCss are consistent with that an influence of immunogenicity (ever HAHA) was only detected for V2 in the final population PK model.

2.4.4. Discussion on clinical pharmacology

Avelumab is expected to be distributed in the systemic circulation and to a lesser extent in the extracellular space. The volume of distribution at steady state was 4.72 L. The volume of distribution was low and it is in line with the expected for an IgG antibody i.e. mainly restrained to the systemic circulation. Consistent with a limited extravascular distribution, the volume of distribution of avelumab at steady state is small. As expected for an antibody, avelumab does not bind to plasma proteins in a specific manner.

The excretion and metabolism of avelumab has not been studied and no studies are required as an IgG antibody is degraded by proteolytic catabolism. Based on a population pharmacokinetic analysis from 1,629 patients, the value of total systemic clearance (CL) is 0.59 L/day. In the supplemental analysis, avelumab CL was found to decrease over time: the largest mean maximal reduction (% coefficient of variation [CV%]) from baseline value with different tumour types was approximately 32.1% (CV 36.2%). Steady-state concentrations of avelumab were reached after approximately 4 to 6 weeks (2 to 3 cycles) of repeated dosing at 10 mg/kg every 2 weeks, and systemic accumulation was approximately 1.25-fold. The elimination half-life ($t_{½}$) at the recommended dose is 6.1 days based on the population PK analysis.

The exposure of avelumab increased dose-proportionally in the dose range of 10 mg/kg to 20 mg/kg every 2 weeks. The more than proportionally increase of exposure for doses between 1 to 10 mg/kg is likely due to the presence of target mediated drug disposition (TMDD) at the lower dose levels. This is not considered to be clinically relevant. Based on non-compartmental analysis the exposure was dose proportional at the proposed therapeutic dose level 10 mg/kg and above.

The PopPK model appears reasonable for describing the PK of an IgG based mAb. A population pharmacokinetic analysis suggested no difference in the total systemic clearance of avelumab based on age, gender, race, PD-L1 status, tumour burden, renal impairment and mild or moderate hepatic impairment. Clearance was shown to correlate to body weight and serum albumin, which is similar compared to other mAbs. Total systemic clearance increases with body weight. Steady-state exposure was approximately uniform over a wide range of body weights (30 to 204 kg) for body weight normalised dosing. Also, modeling showed that clearance is time dependent but that an empirical function with time after first dose as the covariate was a better descriptor than tumour burden.

Body weight was identified to be a significant covariate for both CL and V1. Avelumab is intended to be administered with weight based dosing regimen at 10 mg/kg leading to that high-weight subjects are predicted to be slightly over-exposed relative to the general population and the opposite, slightly under-exposure for low-weight subjects. This is acceptable as the effect is limited and all clinical studies have been performed using weight-based dosing.

No influence of age on avelumab CL was observed in the range 20-91 years (n=1629 and geometric mean 63 years).

Race was identified to be a statistically significant covariate on CL in Black or African American race. However, the influence on exposure appear to be minor and not considered clinically meaningful, thus no dose adjustment is needed based on race.

Gender was identified to be a statistically significant covariate on both CL and V1 after accounting for weight, both being higher in male subjects. However, the influence on exposure appears to be minor and not considered clinically meaningful, thus no dose adjustment is needed based on gender.

No dedicated study in special population has been conducted. However, different demographic and pathophysiological covariates were evaluated in the population PK analysis to assess any influence on avelumab PK. Overall, no covariate effect in the special populations caused more than 20% change in exposure from the typical patient. No dose adjustment is needed for elderly patients (≥ 65 years) (see sections 5.1 and 5.2).

No formal study has been performed in renal or hepatic impaired patients, however it was evaluated in the PopPK analysis. This is acceptable as avelumab is an antibody and renal or hepatic impairment is not expected to influence the PK of avelumab.

To assess the influence of renal function on avelumab PK the estimated glomerular filtration rate (eGFR, mL/min/1.73 m2) was calculated according to the Modification of Diet in Renal Disease (MDRD) formula. Data included 671 patients with normal renal function (CRCL ≥90 mL/min), 623 with mild renal impairment (RI, CRCL ≥60 and <90 mL/min), 320 with moderate RI (≥30 and <60 mL/min) and 4 with severe RI CRCL (<30 mL/min). The PopPK analysis did not detect any relation between clearance of avelumab and renal function. No clinically important differences in the clearance of avelumab were found between patients with mild (glomerular filtration rate (GFR) 60 to 89 mL/min, Cockcroft-Gault Creatinine Clearance (CrCL); n=623), moderate (GFR 30 to 59 mL/min, n=320) and patients with normal (GFR ≥ 90 mL/min, n=671) renal function. Avelumab has not been studied in patients with severe renal impairment (GFR 15 to 29 mL/min). Of note only 4 patients were categorised as having severe RI and no firm conclusion can be drawn in this group. Therefore, no dose adjustment is needed for patients with mild or moderate renal impairment (see section 5.2). There are insufficient data in patients with severe renal impairment for dosing recommendations.

No clinically important differences in the clearance of avelumab were found between patients with mild hepatic impairment (bilirubin ≤ ULN and AST > ULN or bilirubin between 1 and 1.5 times ULN, n=217) and normal hepatic function (bilirubin and AST ≤ ULN, n=1,388) in a population PK analysis. Hepatic impairment was defined by National Cancer Institute (NCI) criteria of hepatic dysfunction. Avelumab has not been studied in patients with moderate hepatic impairment (bilirubin between 1.5 and 3 times ULN) or severe hepatic impairment (bilirubin > 3 times ULN). In patients with mild hepatic impairment (n=217, defined as (BILI≤ULN and AST>ULN) or (BILI>1*ULN and BILI≤1.5 *ULN and any AST) data indicated that no dose adjustment is needed for patients with mild hepatic impairment (see section 5.2). Very few subjects were available in moderate HI patients (n=4) and severe HI (n=1) and thus there are insufficient data in patients with moderate or severe hepatic impairment for dosing recommendations.

This medicinal product contains less than 1 mmol sodium (23 mg) per dose, i.e. essentially 'sodium-free'.

No interaction studies or studies with biomaterials have been conducted with avelumab. Avelumab is primarily metabolised through catabolic pathways, therefore, it is not expected that avelumab will have pharmacokinetic drug-drug interactions with other medicinal products.

The bioassays have in general been correctly validated. Several concerns were raised to account for possible underestimation of avelumab immunogenicity, with emphasis on assay cut-point determination (report TNJS13-170 and TNJS13-170A1, report TNJS15-062) and drug tolerance (report IP190 and IP373, report TNJS15-062). Finally, a solid tumour (ST) population cut point factor of 1.25 was identified and provided adequate assay sensitivity and drug tolerance level. With the updated cut point factor the number of subjects classified as ADA ever-positive and as ADA treatment-emergent were estimated to be 6.2% and 5.9%, respectively (data from the integrated safety summary, i.e. study EMR100070-001 and Study EMR100070-003 Part A; n=1738). There seemed to be a potential increased risk for infusion-related reactions (IRRs) in ADA ever-positive subjects versus ADA ever-negative subjects: the incidence of IRRs was about 40% and 25% in ADA ever-positive patients and ADA never-positive patients, respectively. It was also noted that a numerically higher percentage of ADA ever-positive subjects had serious TEAEs and TEAEs leading to permanent treatment discontinuation as compared to ADA never-positive subjects. But the interpretation of the data is limited by the low incidence and the confounding effect of increasing number of ever-positive subjects with time and the similarly increased risk for TEAEs. This data has been adequately described in section 4.8 of the SmPC. Immunogenicity and severe infusion-related reactions (grade ≥3) are also included in the RMP as an Important potential risk and Important identified risk, respectively.

The applicant's neutralising antibody (nAb) assay is considered inadequate, and the data on nAb positive subjects is therefore not reliable. As avelumab is a human immunoglobulin, it is expected that the propensity of patients to develop neutralising antibodies against avelumab is very low. Nevertheless, the applicant is strongly encouraged to improve their assay in terms of appropriate drug tolerance level compared to the avelumab C_{trough} values.

2.4.5. Conclusions on clinical pharmacology

The clinical pharmacology for avelumab has overall been adequately characterised.

The CHMP recommends the following measures to address the issues related to pharmacology:

- The current neutralizing antibody (nAb) assay is not considered adequate in terms of the assay's drug tolerance, i.e. there is a concern that the sensitivity of the assay cannot be ensured at the Ctrough levels of avelumab. The Applicant is therefore recommended to develop a method suitable to detect nAb in patients and, if possible, submit a re-analysis of the potential impact of nAb on pharmacokinetics, efficacy and safety.

The drug tolerance of the neutralizing antibody assay may be underestimated due to use of a polyclonal positive control. The applicant should evaluate feasibility to improve drug tolerance, including additional methodology for identification of a monoclonal positive control.

2.5. Clinical efficacy

2.5.1. Dose response study(ies)

Dose justification

The dose selected for the pivotal Study, EMR100070-003, was based on adequate safety and tolerability in Phase 1 Study EMR100070-001, a phase I study which enrolled adult subjects with metastatic or locally advanced solid tumors and expansion to selected indications (1490 enrolled as of 20 November 2015). The study included a dose escalation phase administered via iv infusion once every 2 weeks (dose escalation phase: 1.0 mg/kg (n = 4), 3.0 mg/kg (n =13), 10.0 mg/kg (n =15), 20.0 mg/kg (n =21); Treatment expansion phase: 10.0 mg/kg (n=1437)). Pharmacokinetic half-life was seen to increase with increasing doses indicating non-linear drug disposition. The 10 mg/kg dose once every 2 weeks achieved the high target occupancy (mean TO >90%) of PD-L1 in PBMC during the whole dose interval as determined from ex vivo studies. Based on the in vitro TO data and the observed trough serum avelumab levels, TO was predicted to reach or exceed 95% throughout the entire dosing interval for more subjects in 10 mg/kg dose group than those in 3 mg/kg dose group from the dose escalation cohorts of Study EMR100070-001.

2.5.2. Main study

EMR100070-003: A Phase II, open-label, multicenter trial to investigate the clinical activity and safety of avelumab (MSB0010718C) in subjects with Merkel cell carcinoma

Methods

Study Participants

Key inclusion criteria were the following:

- Histologically proven MCC
    a. Confirmation of the diagnosis by IHC (local laboratory testing)
    b. Metastatic disease (subjects with non-metastatic MCC that was only recurrent or unresectable were NOT eligible)
    c. At least 1 line of chemotherapy for metastatic MCC of the following chemotherapy regimens: cyclophosphamide, topotecan, doxorubicin, epirubicin, vincristine, carboplatin, cisplatin, etoposide in combination with carboplatin or cisplatin; must have progressed after the most

recent line of chemotherapy.

- Collection of biopsy material (fresh biopsy or archival material)
- ECOG PS 0-1
- Estimated life expectancy > 12 weeks
- Measurable disease by RECIST 1.1 (including skin lesions)
- Adequate hematological, renal and hepatic function

Key exclusion criteria were the following:

- Prior therapy with any antibody/drug targeting T-cell coregulatory proteins such as anti-PD-1, anti-PD-L1, or anti-CTLA-4 antibody
- Concurrent anticancer treatment
- Concurrent systemic therapy with steroids or other immunosuppressive agents, or use of any investigational drug within 28 days before the start of study treatment. Short-term administration of systemic steroids (that is, for allergic reactions or the management of immune-related adverse events) was allowed while on study
- Active CNS metastases (allowed if treated, fully recovered from treatment, not progressing for at least 2 months and not requiring steroids)
- Previous malignant disease within the last 5 years
- Prior organ transplantation, including allogeneic stem-cell transplantation
- Known history of HIV or known AIDS or hepatitis B or hepatitis C acute or chronic infection
- Active or history of any autoimmune disease (except for subjects with vitiligo) or immunodeficiencies that required treatment with systemic immunosuppressive drugs
- Clinically significant cardiovascular disease and all other significant diseases which might have impaired the subject’s tolerance of study treatment

Treatments

All patients received avelumab 10 mg/kg once every 2 weeks, as an IV infusion over approximately 1 hour.

Premedication with paracetamol and antihistamine, 30-60 minutes before avelumab infusion, was mandatory.

Subjects were treated until significant clinical deterioration, unacceptable toxicity, withdrawal of consent or any other criterion for withdrawal as per protocol. Treatment beyond radiological PD by RECIST 1.1 was allowed providing there was no significant clinical deterioration and if, in the opinion of investigator, the subject would benefit from continued treatment. Confirmation of PD by imaging was required, preferably 6 weeks (but no later) after progression has been diagnosed according to RECIST 1.1, before stopping treatment. Treatment during the confirmation period should continue as scheduled, despite a first observation of progression according to RECIST 1.1, until confirmation has been made, and further if there is no significant clinical deterioration.

Subjects who experienced a confirmed CR as reported by the Investigator could be treated for 6 to 12 months after confirmation, at the discretion of the Investigator, or beyond 12 months after discussion with the Sponsor. In case a subject with a confirmed CR relapsed after stopping treatment, but prior to the end of the study, one re-initiation of treatment was allowed, providing there were no prior toxicities leading to discontinuation of avelumab.

For the purpose of subject management, assessments were made by the Investigators, but the primary and secondary endpoint determinations were supported by tumour assessments performed by an Independent Endpoint Review Committee (IERC). The tumour imaging assessment was performed at baseline, at week 7, then every 6 weeks afterwards. Confirmation of response (CR, PR) was required (preferably at the regularly scheduled 6-week assessment interval, but no sooner than 5 weeks after the initial documentation of response).

Objectives

Primary objective:

The primary objective of the study is to assess the clinical activity of avelumab as determined by the objective response rate (ORR) according to RECIST 1.1 based on independent review (IERC) of tumour assessments in subjects with metastatic MCC after failing first-line chemotherapy.

Secondary objectives:

- To assess the duration of response (DOR) according to RECIST 1.1.
- To assess the progression-free survival time (PFS) according to RECIST 1.1.
- To assess the safety profile of avelumab in subjects with MCC.
- To assess overall survival (OS) time.
- To assess response status according to RECIST 1.1 at 6 and 12 months after start of study treatment.
- To characterize the population PK of avelumab in subjects with MCC by sparse sampling.
- To evaluate the immunogenicity of avelumab and to correlate it to exposure.

Exploratory objectives:

- To correlate the immunogenicity of avelumab with clinical results (ORR and AEs).
- To assess the immune-related best overall response (irBOR) and immune-related PFS (irPFS) using the modified immune-related response criteria (irRC), derived from RECIST 1.1.
- To compare time to progression (TTP) on last prior anticancer therapy to PFS time on treatment with avelumab.
- To evaluate tumour shrinkage in target lesions at each time point from Baseline.
- To evaluate changes in biomarkers in relation to disease responses to avelumab.
- To evaluate the association between PD-L1 expression and best overall response (BOR).
- To explore the benefits of avelumab as perceived by subjects with metastatic MCC.

Outcomes/endpoints

Primary endpoint:

- ORR according to RECIST 1.1 as determined by an IERC

  ORR definition: proportion of subjects with a confirmed BOR of CR or PR, based on independent tumor assessment according to RECIST 1.1

Secondary endpoints:

- DOR according to RECIST 1.1 as determined by an IERC

  DOR definition: the duration of response as determined by an independent tumor assessment was calculated for each subject with a confirmed response according to RECIST 1.1 as the time from first observation of response until first observation of documented disease progression or death when death occurred within 12 weeks of the last tumor assessment, whichever occurred first. For subjects with a confirmed response but neither documented disease progression nor death within 12 weeks after the last tumor assessment, as of the cut-off date for the analysis, the duration of response was censored at the date of the last tumor assessment. If death occurred more than 12 weeks after the last tumor assessment, the duration of response was also censored at the date of the last adequate tumor assessment.

- PFS according to RECIST 1.1 as determined by an IERC
- *PFS definition*: Time from first administration of study treatment until first observation of PD or death when death occurred within 12 weeks of the last tumour assessment or first administration of study treatment (whichever was later), based on independent tumour assessment according to RECIST 1.1. If death occurred after 12 weeks after the last tumour assessment, the PFS was also censored at the date of the last adequate tumour assessment.
- Occurrence, number and severity of treatment-related AEs according to NCI-CTCAE v 4.0
- OS
- Response status according to RECIST 1.1 at 6 and 12 months after start of study treatment
- Serum titers of anti-avelumab antibodies
- Population PK profile of avelumab (sparse sampling)

Exploratory endpoints:

- DOR and PFS per Investigator assessment
- irBOR and irPFS according to modified irRC per Investigator assessment
- TTP under the last prior anticancer drug therapy
- Tumour shrinkage in target lesions per time point from Baseline
- Expression of PD-L1 in tumour tissue
- changes in soluble factors (for example, cytokine profiles, soluble PD-1 and soluble PD-L1)
- changes in MCV-specific humoral responses
- changes in EQ-5D and FACT-M scores over the treatment period

- description of the effects of avelumab treatment as perceived and reported by subjects with metastatic MCC

Sample size

The planned total sample size was 84 subjects. The trial aims at demonstrating an ORR greater than 20% by means of an exact binomial test, based on the following assumptions: 1) an ORR of 35%; 2) an overall alpha = 0.025 (1-sided) for the test of the null hypothesis of an ORR ≤ 20%.

The planned enrollment was 84 patients, however, an additional 4 were enrolled since they were in the screening window at the time the decision was made to halt.

The study was designed to have an interim analysis for futility after 20 subjects were enrolled and observed for at least 3 months and an interim analysis for efficacy 6 months after 56 subjects were enrolled.

Applicant's justification of the assumption of a response rate of 35% to define the trial sample size: anti-PD-L1 monoclonal antibodies have demonstrated antitumor activity in tumours whose micro-environment express PD-L1 (data from International Association for Study of Lung Cancer annual meeting in 2013). Robust expression of PD-L1 in the immune-infiltrating cells of MCC has been reported in the literature (Lipson, 2013) and confirmed by the Sponsor. On these grounds, the expression of PD-L1 by the immune infiltrating cells as well as the expression of PD-L1 at the surface of MCC constitute a very strong rationale for the evaluation of avelumab in that disease. A response rate of approximately 20% has been reported in several tumour types where only a fraction of the tumours express PD-L1 at the surface of the tumours or at the surface of immune infiltrating cells.

Randomisation

This is a single arm study.

Blinding (masking)

This is an open-label study.

Statistical methods

A two-stage group sequential testing approach was applied for efficacy. The null hypothesis could be rejected if 20 subjects in the interim analysis after 56 subjects, or 26 subjects in the primary analysis after 88 subjects, showed a confirmed PR or CR according to RECIST 1.1. The corresponding nominal p-values of the exact binomial test were 0.0045 and 0.0211, respectively. The resulting overall probability of reaching a positive result in the interim or primary analysis under the null hypothesis assumption of an ORR ≤ 20% was ≤ 0.0223, as derived from the binomial distribution according to Jennison and Turnbull. The overall type I error rate is controlled at a level of 2.5% (one-sided).

Objective response rate was reported with corresponding two-sided Clopper-Pearson confidence intervals (CI). A repeated CI according to Jennison for the ORR (95.9% CI for the primary analysis) was calculated to account for the group sequential testing approach (i.e., interim analysis and primary analysis). Response status at 6 and 12 months was analysed by calculating the proportion of subjects in response at 6 (12) months among all subjects who started study treatment at least 6 (12) months prior to the cut-off date, respectively. Time-to event endpoints (duration of response, progression-free survival, and overall survival) were analysed using Kaplan-Meier methodology; median values were calculated with corresponding CI using the Brookmeyer-Crowley method. To compare PFS time on treatment with avelumab to time to progression (TTP) on the last prior anticancer therapy, the hazard ratio (including 95% CI) of avelumab treatment to the last prior anticancer therapy was calculated using a shared frailty model with a gamma distribution of the frailty term. In a post-hoc analysis, the 6-months DRR was estimated as the product of the ORR and the Kaplan-Meier estimate of 6-months durability of response. An asymptotic 95% confidence interval (CI) for the DRR was obtained by applying the standard formula for the variance of a product of independent random variables, using Greenwood's formula for the variance of the Kaplan-Meier estimate of 6-month durability.

Descriptive statistics for the absolute EQ-5D and FACT-M scores were summarized at baseline and at all scheduled requests to fill out questionnaires during the study. Changes in EQ-5D and FACT-M scores from baseline were summarized. The association between tumour size changes and HRQoL score changes from baseline was assessed by simple linear regression of time-matched values, with absolute change from baseline in HRQoL scores as the dependent variable and relative change in tumour size from baseline at the same time point as the predictor.

Since the study is not powered for any subgroup analysis, all the subgroup analysis are exploratory in nature.

The following sequence of analyses was planned:

- Interim futility analysis: was performed 3 months after start of study treatment of the 20th subject. Only the first 20 subjects at the clinical cut-off date were included in the primary endpoint and secondary efficacy endpoint analysis; and all subjects who received at least one dose of treatment up to the cut-off date were included in analysis of safety.
- Interim efficacy analysis: was performed 6 months after start of study treatment of the 56th subject. The analysis was performed on the ITT population and included subjects with minimum follow up of 6 weeks, 12 weeks and 24 weeks beyond day 1.
- The primary analysis: was performed 6 months after start of study treatment of the last subject enrolled in the trial. The primary analysis was performed on the ITT population.
- An exploratory analysis will be conducted 12 months after the accrual of the last subject.

The confirmatory statistical test for the primary efficacy endpoint analysis included procedures for controlling the overall type I error rate at level of 2.5% one-sided. All other statistical analyses performed on the secondary and other endpoints were exploratory. Two-sided 95% confidence intervals (CI) were used, if not otherwise specified.

Censoring rules applied in the analysis are presented in the table below:

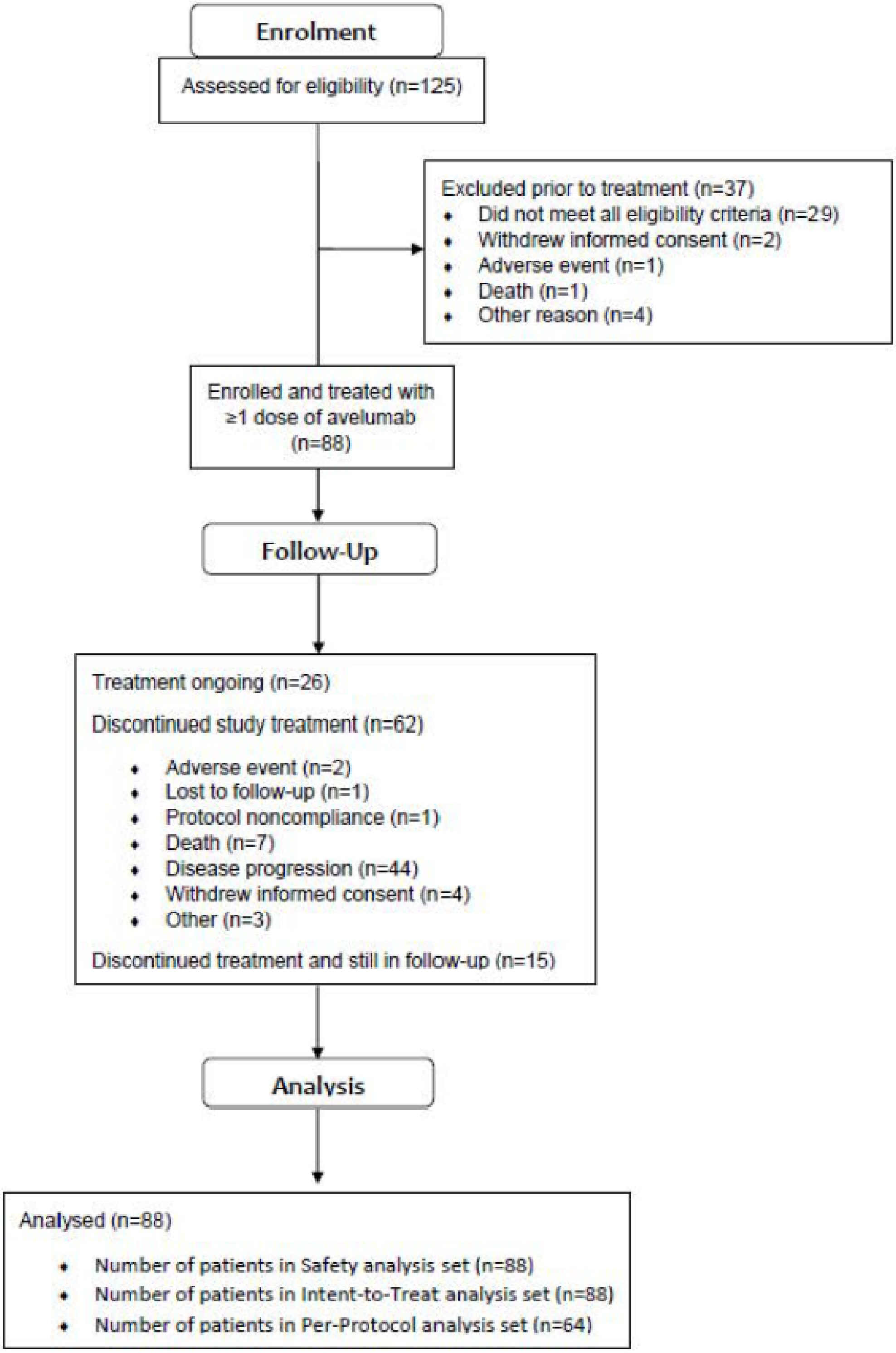
Endpoint	Censoring rules
DOR	For subjects with a confirmed response but neither documented disease progression nor death within 12 weeks after the last tumor assessment, as of the cut-off date for the analysis, the duration of response will be censored at the date of the last adequate tumor assessment.
PFS	For subjects with both the baseline tumor assessment and at least one adequate tumor assessment after start of treatment, the general censoring rules apply:

<table>
<tr>
<td></td>
<td>
<table>
<tr><th colspan="3"></th><th>Date of event / censoring</th><th>Censoring</th></tr>
<tr><td colspan="3">PD according to RECIST as assessed by the IERC</td><td>Date of PD</td><td>No</td></tr>
<tr><td rowspan="2">No PD</td><td rowspan="2">Death within 12 weeks after last tumor assessment or date of first study treatment (whichever is later)</td><td>Yes</td><td>Date of death</td><td>No</td></tr>
<tr><td>No</td><td>Date of last adequate tumor assessment</td><td>Yes</td></tr>
</table>
For subjects with no tumor assessment at baseline or no tumor response assessment post-baseline, the special censoring rules apply (and overrule the general censoring rules):
<table>
<tr><th colspan="3"></th><th>Date of event / censoring</th><th>Censoring</th></tr>
<tr><td rowspan="2">No tumor assessment at Baseline</td><td rowspan="2">Death within 12 weeks after first study treatment</td><td>Yes</td><td>Date of death</td><td>No</td></tr>
<tr><td>No</td><td>Date of first study treatment</td><td>Yes</td></tr>
<tr><td rowspan="2">No tumor response assessment post Baseline</td><td rowspan="2">Death within 12 weeks after first study treatment</td><td>Yes</td><td>Date of death</td><td>No</td></tr>
<tr><td>No</td><td>Date of first study treatment</td><td>Yes</td></tr>
</table>
</td>
</tr>
<tr>
<td>OS</td>
<td>For subjects who are still alive at the time of data analysis or who are lost to follow up, OS time will be censored at the last recorded date that the subject is known to be alive (date of last contact, last visit date, date of last trial treatment administration or date of last scan, whichever is the latest) as of the data cut-off date for the analysis.</td>
</tr>
</table>

PD=Progressive disease

Results

Participant flow



Recruitment

Study period was from 3rd July 2014 (consent date of first enrolled subject) to 3rd March 2016 (data cut-off for the primary analysis, corresponding to 6 months after start of study treatment of the last subject).

Estimated median duration of follow-up for PFS according to IERC assessment in the ITT Analysis Set (exploratory endpoint), calculated by applying reverse censoring rules, was 10.1 months (95% CI: 8.3, 11.1; range: 0.3, 18.8).

Conduct of the study

There were 3 SAP versions. The analyses were carried out according to the SAP Version 3.0 dated 31 March 2016. The changes from the planned analyses are summarized below:

- The on-treatment period was changed from + 29 days to + 30 days from first study drug administration.
- The planned sample size as specified in the clinical study protocol was 84 subjects. The actual enrolment was 88 subjects, all receiving at least 1 dose of study drug; therefore all 88 subjects are included in the primary analysis. Critical values for statistical testing in the primary analysis were adjusted
- Subgroup analysis according to the SAP was to include "number of prior systemic therapies for metastatic disease or locally advanced therapies", classified as 1 or ≥2; however, analyses were performed according to "Number of prior systemic therapies," classified as 1 or ≥2" and "number of prior systemic therapies for metastatic disease" classified as 1 or ≥2
- The subgroup for visceral metastases at Baseline was based on IERC assessment, as opposed to Investigator assessment, as specified in Section 10 of SAP v3.0. Furthermore, lesions recorded as having a site of 'SKIN', 'SOFTTISSUES' or 'EYE' were classified as skin lesions, and only lesions with a categorization of neither skin ('SKIN' / 'SOFTTISSUES' /'EYE') nor lymph node lesion were considered as visceral lesions

Protocol amendments 1, 2 and 2.1 were made before starting of enrolment in July 2014. Part B of the study, enrolling patients who had not received any prior systemic treatment for metastatic MCC, was added with Amendment 7.

Protocol deviations:

A total of 12/88 (13.6%) subjects in the ITT analysis set were reported with 1 or more important protocol deviation:

- 5 patients were enrolled although ineligible (1 subject had prior therapy with antibody targeting T-cell co-regulatory proteins [4-1BB agonist], 1 subject had a previous malignancy within 5 years, 2 subjects had lymphocytes below the required level at screening, 1 subject was found to have brain metastases)
- 5 subjects took prohibited medications (steroids used not to treat irAEs)
- 3 subjects did not receive premedication

Minor deviations were mainly related to laboratory assessment (e.g. missing tests), visit schedule (e.g. out of planned range) and study procedure (e.g. ECG not performed).

Baseline data

Baseline patients and tumor characteristics are presented in the tables below:

Table 14: Selected key demographic and baseline characteristics – ITT analysis set

Characteristic	Avelumab N=88 (100%) n (%)
Sex	
Male	65 (73.9)
Female	23 (26.1)
Race	
White	81 (92.0)
Black or African American	0
Asian	3 (3.4)
Not Collected at the Site	3 (3.4)
Unknown	1 (1.1)
Ethnicity	
Hispanic/Latino	
Yes	4 (4.5)
No	58 (65.9)
Missing	26 (29.5)
Japanese	
Yes	3 (3.4)
Missing	85 (96.6)[a]
Geographic region	
North America	51 (58.0)
Latin America	0
Western Europe	29 (33.0)
Eastern Europe	0
Australia	5 (5.7)
Asia	3 (3.4)
Missing	0
Age (Years)	
n	88 (100.0)
Mean ± standard deviation	69.7 ± 10.71
Median	72.5
Quartile 1; quartile 3	64.5; 77.0
Minimum; maximum	33; 88
Age Categories	
< 65 years	22 (25.0)
≥ 65 years	66 (75.0)
65 to < 75 years	35 (39.8)
75 to < 85 years	28 (31.8)
≥ 85 years	3 (3.4)
Eastern Cooperative Oncology Group Performance Status	
0	49 (55.7)
1	39 (44.3)

Source: Study EMR100070-003 CSR Table 15.1.3.1.

N/n: number of subjects.

[a] Only Asian subjects were asked whether they were Japanese.

Table 15: MCC disease history – ITT analysis set

Characteristic	Avelumab N=88 (100%) n (%)
Site of primary tumor	
Lymph nodes	12 (13.6)
Skin	67 (76.1)
Other	2 (2.3)
Missing	7 (8.0)
Visceral[a] metastases at Baseline per IERC	
Yes	47 (53.4)
No	41 (46.6)
Sum of target lesion diameter at Baseline per IERC, (mm)	
n (%)	77 (87.5)
Missing (%)	11 (12.5)
Mean ± standard deviation	101.0 ± 76.01
Median	79.0
Quartile 1; quartile 3	43.0; 138.0
Minimum; maximum	16; 404
Time since first metastatic disease (months)	
Median (range)	10.4 (1.5, 159.0)
Quartile 1; quartile 3	6.3, 17.2
Time since last progression of disease prior to study entry (months)	
n	84 (95.5)
Median (range)	1.3 (0.1, 11.6)
M stage at study entry, n (%)	
M1	88 (100.0)
Number of prior anti-cancer therapy regimens, n (%)	
1	52 (59.1)
2	26 (29.5)
3	7 (8.0)
≥4	3 (3.4)
Number of prior anti-cancer therapy lines for metastatic disease, n (%)	
1	57 (64.8)
2	27 (30.7)
3	3 (3.4)
≥4	1 (1.1)
PD-L1 tumor cell status at 1% cut-off, n (%)	
Positive	58 (65.9)
Negative	16 (18.2)
Not evaluable	14 (15.9)
PD-L1 tumor cell status at 5% cut-off, n (%)	
Positive	19 (21.6)

Characteristic	Avelumab N=88 (100%) n (%)
Negative	55 (62.5)
Not evaluable	14 (15.9)
MCV	
Positive	46 (52.3)
Negative	31 (35.2)
Not evaluable	11 (12.5)
Combination of PD-L1 (1%) and MCV	
PD-L1 + MCV +	36 (40.9)
PD-L1 + MCV -	19 (21.6)
PD-L1 - MCV +	9 (10.2)
PD-L1 - MCV -	7 (8.0)
Not evaluated	17 (19.3)

Source: Study EMR100070-003 CSR: Table 15.1.5.1, Table 15.1.5.3, Table 15.1.5.7, Table 15.2.1.21, Table 15.5.2.4.

IERC: Independent Endpoint Review Committee; MCV: Merkel cell polyomavirus (determined by immunohistochemistry); N/n: number of subjects; PD-L1: Programmed death ligand 1.

[a] Visceral defined as not skin and not lymph node; skin category included anything reported as skin, soft tissue and eyelid.

Table 16: Tumor PDL-1 expression status at baseline with 1% or 5% cut-offs (any staining intensity)

PD-L1 status (n, %)	≥ 1%	≥ 5%
Positive	58 (65.9%)	19 (21.6%)
Negative	16 (18.2%)	55 (62.5%)
Not evaluable	14 (15.9%)	14 (15.9%)

Numbers analysed

The primary analysis was conducted 6 months after the accrual of the last subject. An exploratory analysis was conducted 12 months after the accrual of the last subject.

Subject follow-up for progression and survival will continue until 1 year after the last subject receives the last dose of avelumab.

Table 17: Analysis sets

Disposition	Avelumab n (%)
Number of subjects in Screening analysis set[a]	125
Number of subjects in Safety analysis set[b]	88 (100.0)
Number of subjects in ITT analysis set[b]	88 (100.0)
Number of subjects in PP analysis set[c]	64 (72.7)

Efficacy analyses were based on the Intent-to-Treat (ITT) analysis set. The ITT population is including all 88 enrolled patients who received at least 1 dose of study treatment.

The PP population is made by 64 patients and it has been used for sensitivity analyses for efficacy endpoints. The Per-Protocol (PP) analysis set is a subset of the ITT analysis set and includes all ITT subjects who meet all of the following criteria:

• Measurable disease per RECIST 1.1 and IERC assessment

• Distant metastatic disease

• Histologically proven MCC with confirmation of the diagnosis by immuno-histochemistry detection of CK20 (or other appropriate cytokeratin expression such as pancytokeratin, AE1/AE3, or Cam5.2) according to the assessment documented in the "Disease History" eCRF page

• Have progressed after 1 line of chemotherapy that was administered for the treatment of distant metastatic MCC as defined in Section 5.3.1 Inclusion Criteria in the protocol

• Evaluable patients, defined as having at least one post-baseline tumor assessment with absence of non-assessable status

Outcomes and estimation

At the time of the data cut-off, 26 subjects (29.5%) were continuing on active treatment, 62 patients discontinued treatment; among them, 15 were still in follow-up at the cut-off date and 43 had died.

Primary endpoint

Confirmed best overall response (BOR)(ITT)

Results are presented below:

Table 18: Confirmed best overall response according to IERC assessment in subjects with previously treated mMCC - ITT analysis set (primary analysis) - EMR100070-003 Part A

	Avelumab N=88 (100%)
Best Overall Response, n(%)	
Complete Response	8 (9.1)
Partial Response	20 (22.7)
Stable Disease	9 (10.2)
Progressive Disease	32 (36.4)
Non-Complete Response/ Non-Progressive Disease	1 (1.1)
Not Evaluable	18 (20.5)
Objective Response Rate	
Response rate (CR+PR), n (%)	28 (31.8)
Repeated conf. interval (exact)(a)	(21.9,43.1)
95% conf. interval (exact)(b)	(22.3,42.6)
p-value (exact binomial test)(c)	0.0060

Updated ORR Results from the 12-months Follow-up Analysis of Subjects with Previously Treated mMCC from Study EMR100070-003 Part A

Table 19: Confirmed best overall response according to IERC assessment in subjects with previously treated mMCC - ITT analysis set (updated analysis) - EMR100070-003 Part A

	Avelumab N=88	
	6-month minimum follow-up analysis	12-month minimum follow-up analysis
Best Overall Response, n (%)		
Complete response	8 (9.1)	10 (11.4)
Partial response	20 (22.7)	19 (21.6)
Stable disease	9 (10.2)	9 (10.2)
Non-complete response / non-progressive disease	1 (1.1)	0 (0)
Progressive disease	32 (36.4)	32 (36.4)
Not evaluable	18 (20.5)	18 (20.5)
Objective Response Rate		
CR + PR (Response rate), n (%) [95% CI (exact)][a]	28 (31.8) (22.3, 42.6)	29 (33.0) (23.3, 43.8)

Source: EMR100070-003 CSR Addendum 2 Table 2

CI: confidence interval; CR: complete response; N/n: number of subjects; PR: partial response.

[a] 95% exact CI using the Clopper-Pearson method.

Secondary endpoints

DOR

Table 20: Duration of response according to IERC assessment – ITT Analysis Set

	Avelumab N=28 (100%)
Number of subjects without event (censored), n (%)	23 (82.1)
Number of subjects with an event, n (%)	5 (17.9)
Progressive disease, n (%)	5 (17.9)
Death, n (%)	0
Duration of response[a]	
Median (months)	NE
Minimum, maximum	2.8+, 17.5+
95% CI	(8.3, NE)
Proportion of duration of response, % (95% CI)[a, b]	
3 months	96 (77, 99)
6 months	92 (70, 98)
12 months	74 (47, 89)
Durable response rate (DRR)[c]	
6-month, % (95% CI)	29.1 (19.5, 38.8)

Source: Study EMR100070-003 CSR: Table 15.2.2.1 and Table 15.2.13.1.

CI: confidence interval; NE: Not estimable; N/n: number of subjects.

Note:

[a] Based on Kaplan-Meier estimates.

[b] The denominator is the number of subjects with a confirmed response of CR or PR according to IERC assessment.

[c] Post-hoc analysis based on the ORR and the KM estimate for 6-month durability.

Updated DoR Results from the 12-months Follow-up Analysis of Subjects with Previously Treated mMCC from Study EMR100070-003 Part A

In the 12-months minimum follow-up analysis, the longest response durations included 1 subject with a CR lasting 23.3+ months and 3 subjects with CR (n=2) or PR (n=1) lasting 18.0+ months, all of which were still ongoing at the last tumor assessment. In this more mature dataset, the median duration of response still has not been reached (95% CI: 18.0 months, not estimable), which provides evidence that responses to avelumab are durable (see Table 21 and see Figure 15 below with duration of response reported).

Table 21: Duration of response according to independent endpoint review committee assessment - ITT analysis set (updated analysis) - EMR100070-003 Part A and Part B

	Part A		Part B
	6-month minimum follow-up analysis (N=88)	**12-month minimum follow-up analysis (N=88)**	**≥13 weeks minimum follow-up analysis (N=16)**
Number of subjects with confirmed objective response, n (%)	28 (100)	29 (100)	10 (100)
Number of subjects without event (censored), n (%)	23 (82.1)	21 (72.4)	10 (100.0)
Number of subjects with an event, n (%)	5 (17.9)	8 (27.6)	0
Progressive disease, n (%)	5 (17.9)	8 (27.6)	0
Death, n (%)	0	0	0
Duration of response[a]			
Median (months)	NE	NE	NE
Minimum, maximum	2.8, 17.5+	2.8, 23.3+	1.2+, 6.5+
95% CI	(8.3, NE)	(18.0, NE)	(NE, NE)
Proportion of duration of response, % (95% CI)[b]			
3 months	96 (77, 99)	97 (78, 100)	100 (NE, NE)
6 months	92 (70, 98)	93 (74 , 98)	100 (NE, NE)
12 months	74 (47, 89)	74 (53 , 87)	Not determined

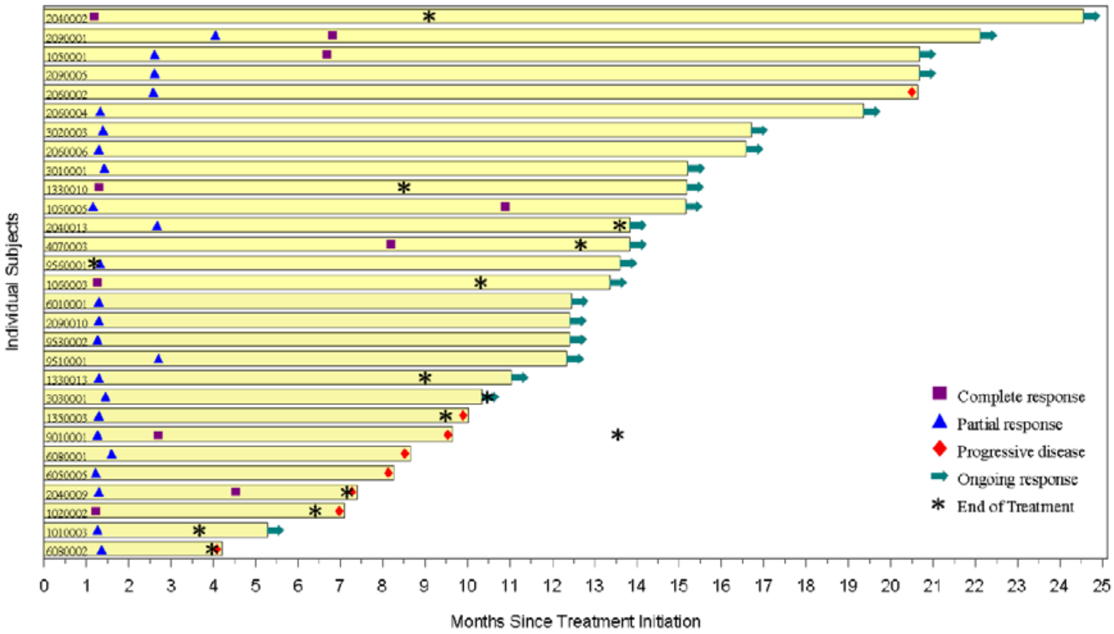
Source: Module 2.7.3 Table 5 and Part B Technical report 1 Table 15.2.2.1b

CI: confidence interval; N/n: number of subjects; NE: not estimable.

a Product-limit (Kaplan-Meier) estimates.

b Based on Kaplan-Meier estimates.

\+ Denotes a censored value (response was ongoing)



Source: Figure 15.2.2.20 (EMR100070-003 CSR Addendum 2)

Figure 15: Time to and duration of response by subject according to IERC - ITT analysis set (EMR100070-003 Part A)

Tumor shrinkage

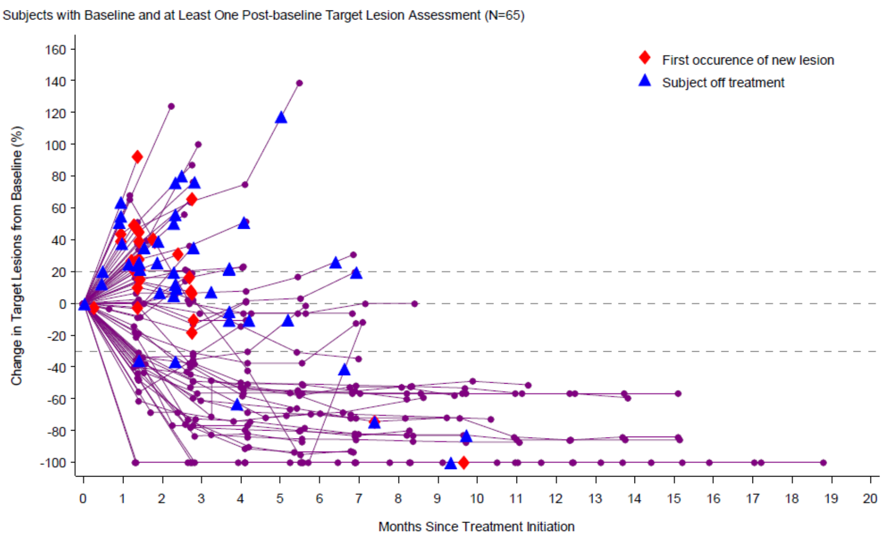


Figure 16: Percent change from baseline in target lesions vs. month according to IERC assessment - ITT analysis set

The target lesions decrease rapidly in the beginning of treatment (within the first 6 weeks) and few patients experienced a ≥ 20% increase from baseline in tumor dimensions, followed by a decrease.

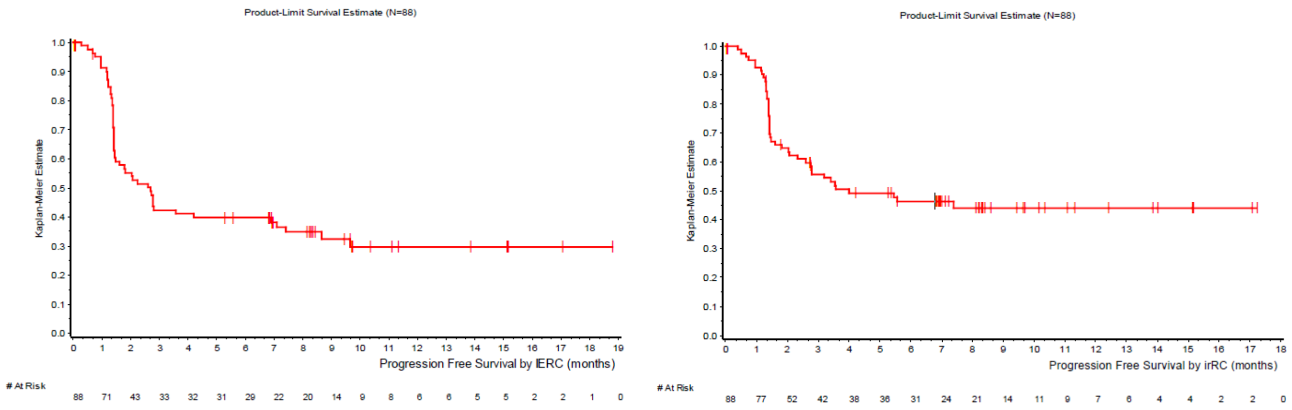
Progression-free survival

Table 22: Progression-Free Survival according to the IERC assessment, Part A - ITT - Study EMR100070-003

	PFS	irPFS
Number of subjects without event (censored), n (%)	36 (40.9)	44 (50.0)
Number of subjects with an event, n (%)	52 (59.1)	44 (50.0)
Progressive disease, n (%)	44 (50.0)	32 (36.4)
Death, n (%)	8 (9.1)	12 (13.6)
Progression-free survival time[a]		
Median (months)	2.7	4.0
Minimum, maximum	0.03, 18.8	0.03, 17.2
95% CI	(1.4, 6.9)	(2.3, -)
Number of subjects at risk / failed / progression-free rates, % up to (95% CI)[b]		
3 months	33 / 45 / 42 (31, 53)	42 / 36 / 56 (44, 66)
6 months	29 / 47 / 40 (29, 50)	31 / 43 / 46 (35, 57)
12 months	6 / 52 / 30 (19, 41)	7 / 44 / 44 (33, 55)

Figure 17: K-M estimates of PFS according to IERC assessment - ITT analysis set Part A 6 month mimimum follow up analysis - Study EMR100070-003

PFS **irPFS**



Updated PFS Results from the 12-months Follow-up Analysis of Subjects with Previously Treated mMCC from Study EMR100070-003 Part A

Table 23: PFS according to IERC assessment ITT analysis set - Study EMR100070-003

	Avelumab N=88	
	6-month minimum follow-up analysis	12-month minimum follow-up analysis
Number of subjects without event (censored), n (%)	36 (40.9)	33 (37.5)
Number of subjects with an event, n (%)	52 (59.1)	55 (62.5)
Progressive disease	44 (50.0)	47 (53.4)
Death	8 (9.1)	8 (9.1)
Progression-free survival time[a] (months)		
Median	2.7	2.7
Minimum, maximum	0.03, 18.8	0.03, 24.5
95% CI	(1.4, 6.9)	(1.4, 6.9)
Number of subjects at risk / failed / progression-free rates, % up to (95% CI)[b]		
3 months	33 / 45 / 42 (31, 53)	33 / 45 / 42 (31, 53)
6 months	29 / 47 / 40 (29, 50)	30 / 47 / 40 (29, 50)
12 months	6 / 52 / 30 (19, 41)	21 / 54 / 30 (21, 41)
15 months	Not determined[c]	11 / 54 / 30 (21, 41)

Source: Table 15.2.3.1 (12-month minimum follow-up analysis) and Study EMR100070-003 Clinical Study Report Table 20 (6-month minimum follow-up analysis).

CI: confidence interval; N/n: number of subjects.

a Product-limit (Kaplan-Meier) estimates.

b Based on Kaplan-Meier estimates.

c Not determined in the 6-month minimum follow-up analysis due to the low number of subjects having reached this time point.

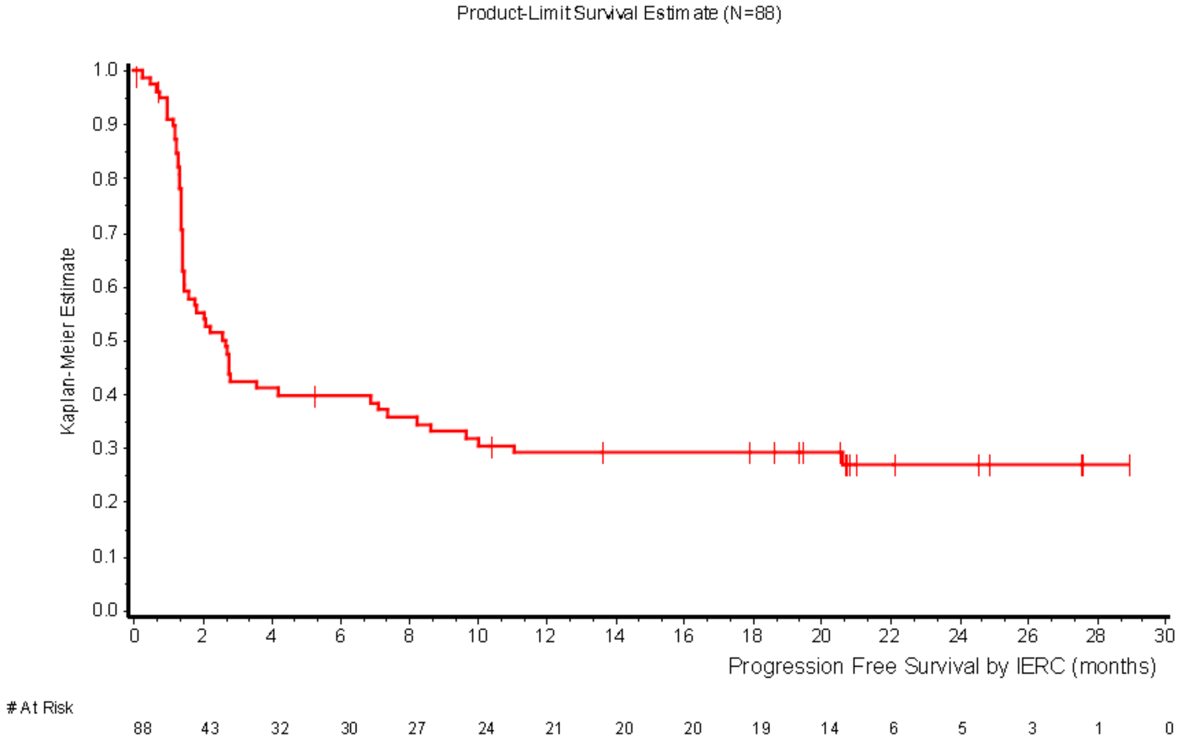


Figure 18: K-M estimates of PFS according IERC assessment in patients with previously treated mMCC ITT analysis set - Study EMR100070-003 Part A (18 months analysis)

Overall Survival (OS)

Table 24: Overall Survival, ITT analysis set - Study EMR100070-003 Part A

	Avelumab 6-month minimum follow-up analysis N=88 (100%)	Avelumab 12-month minimum follow-up analysis N=88 (100%)
Number of subjects without event (censored), n (%)	45 (51.1)	40 (45.5)
Number of subjects with an event (death), n (%)	43 (48.9)	48 (54.5)
Overall survival time[a]		
Median (months)	11.3	12.9
Minimum, maximum	0.4, 18.8	0.4, 24.7
95% CI	(7.5, 14.0)	(7.5, NE)
Survival rates, % (95% CI)[a]		
3 months	87 (78, 93)	87 (78, 93)
6 months	69 (58, 78)	70 (59, 78)
12 months	48 (35, 60)	52 (41, 62)
15 months	Not determined	44 (32 ,54)

Source: Study EMR100070-003 CSR Table 15.2.4.1; Study EMR100070-003 Addendum 2 Table 15.2.4.1 (12-month analysis set).

CI: confidence interval; N/n: number of subjects.

[a] Based on Kaplan-Meier estimates.

In the primary analysis, 49% of patients had died. The OS was 11.3 months (0.4-18.9). At 7.5 months the numbers at risk drop from 51 to 38, ITT (41 to 29 in PP).

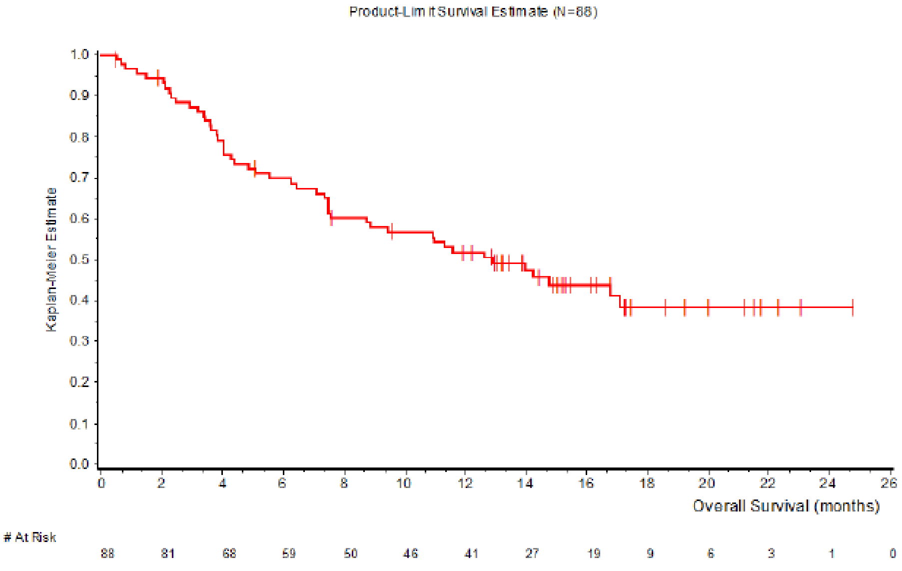


Figure 19: K-M estimates of overall survival ITT analysis set 12 month minimum follow-up analysis - Study EMR100070-003 Part A

The 12-months minimum follow-up analysis of overall survival (OS) reported an increased (compared to the 6-months minimum follow-up analysis) estimated median OS of 12.9 months (95% CI: 7.5, not estimable), with a 6-month OS rate of 70% (95% CI: 59, 78), a 12-month OS rate of 52% (95% CI: 41, 62), and a 15-month OS rate of 44% (95% CI: 32, 54).

Health-related QoL

a) EQ-5D VAS and index scores, ITT

Table 25: EQ-5D VAS and index scores, ITT analysis set – Study EMR100070-003

		Change in Score from Baseline							
	Baseline	**Week 7**	**Week 13**	**Week 19**	**Week 25**	**Week 31**	**Week 37**	**Week 43**	**EOT**
VAS[a], n	72	49	38	29	27	19	12	10	20
Mean (StDev)	64.9 (24.00)	0.4 (23.17)	-1.4 (16.22)	3.2 (19.66)	4.7 (23.22)	5.8 (16.15)	4.9 (14.07)	4.5 (16.27)	-0.6 (31.08)
Median	72.0	0.0	-1.0	1.0	1.0	5.0	3.5	2.0	-1.5
Index[b], n	72	49	38	29	27	19	12	10	20
Mean (StDev)	0.80 (0.155)	-0.04 (0.172)	-0.02 (0.105)	-0.02 (0.134)	-0.01 (0.068)	0.01 (0.109)	0.01 (0.089)	-0.01 (0.067)	-0.11 (0.100)
Median	0.83	0.00	-0.01	0.00	0.00	0.00	0.00	-0.01	-0.13

Source: Table 15.2.7.3.

EOT: End-of-Treatment visit; N/n: number of subjects; PR: partial response; StDev: standard deviation; VAS: Visual Analog Scale.

a EQ-5D Visual Analogue Scale ranges from 0 to 100 for self-rated health status. A higher score indicates better health status.

b EQ-5D Index score ranges from 0 to 1. A higher score indicates better health status.

Patient attrition can be observed from baseline to EOT. The results are relatively stable, with a slight tend towards lower scores at EOT.

b) Functional assessment of cancer therapy-melanoma (FACT-M)

Low score changes were observed patients beyond week 13; however, EOT results indicate a slight worsening, with a median score change for the FACT-M Total score of -9.3.

The health-related quality of life ED-5D and FACT-M do not contribute with definitive information regarding avelumab treatment. As for the semi-structured interviews, 10 patients responded at baseline as well as at weeks 13 and 17.

Ancillary analyses

Exploratory endpoints

OS by PD-L1 expression

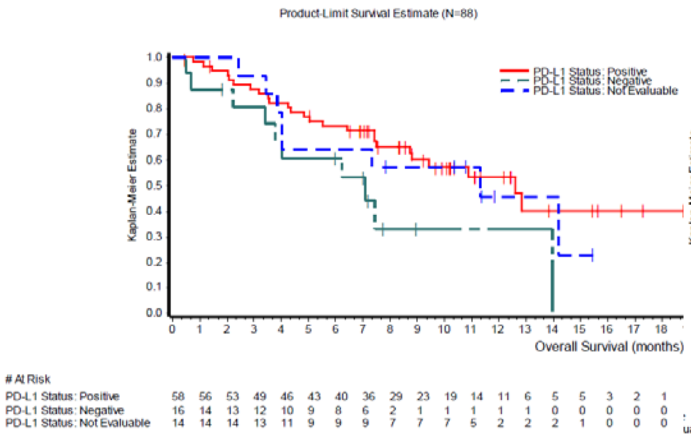


Figure 20: OS by PD-L1 expression at 1% cutoff, ITT

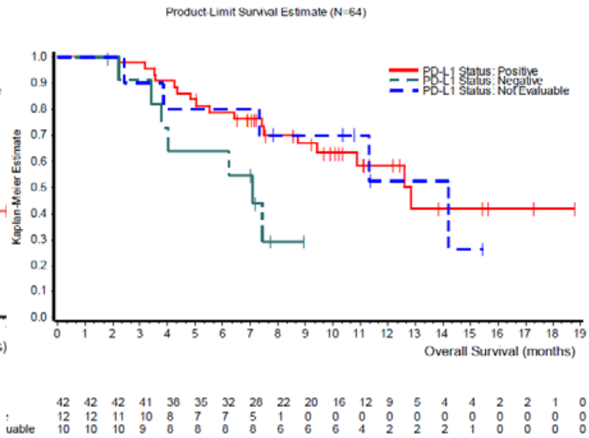


Figure 21: OS by PD-L1 expression at 1% cutoff, PP

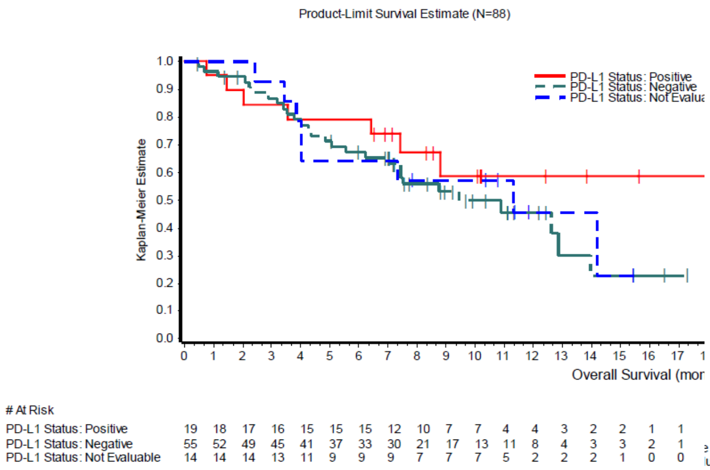


Figure 22: OS by PD-L1 expression at 5% cutoff, ITT

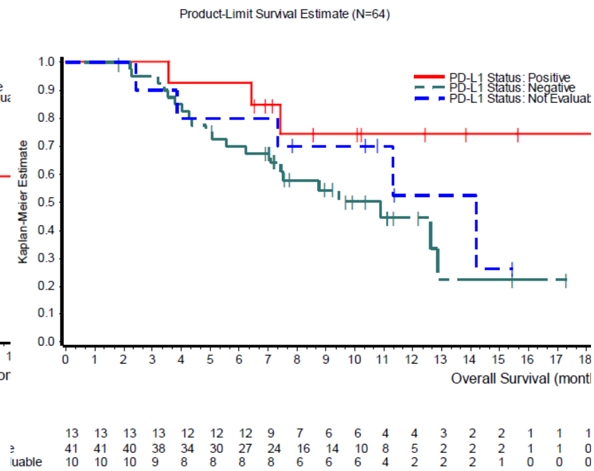


Figure 23: OS by PD-L1 expression at 5% cutoff, PP

Response and progression on the last prior therapy

Table 26: Best response on the last prior anti-cancer drug therapy for metastatic disease - ITT analysis set

	Avelumab N=88 (100%)
Best Overall Response, n (%)	
Complete Response	9 (10.2)
Partial Response	13 (14.8)
Stable Disease	20 (22.7)
Progressive Disease	39 (44.3)
Not Applicable	1 (1.1)
Not Assessable	0
Unknown	6 (6.8)
Objective Response Rate	
Response rate (CR+PR), n (%)	22 (25.0)
95% Conf. interval (exact) (a)	(16.4,35.4)

ORR by subgroup (selected analyses)

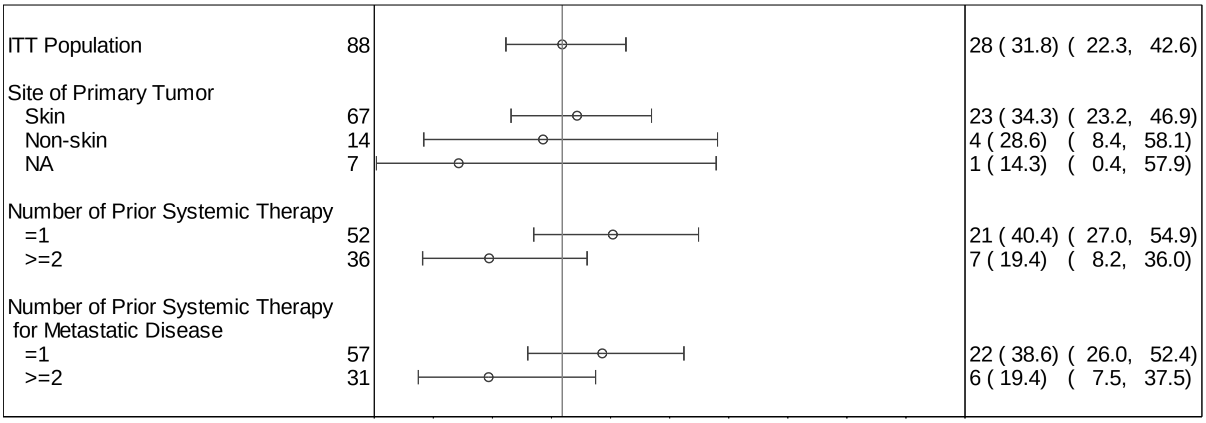
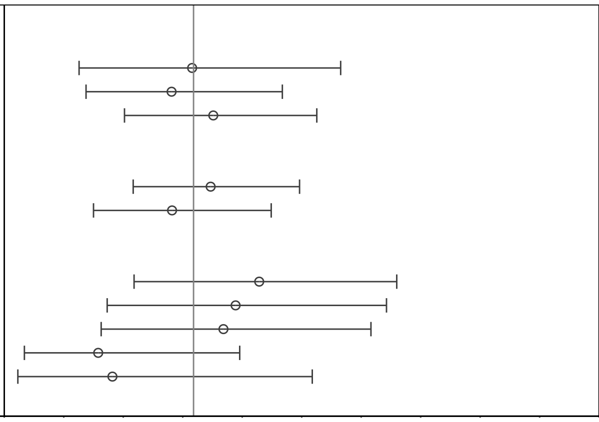
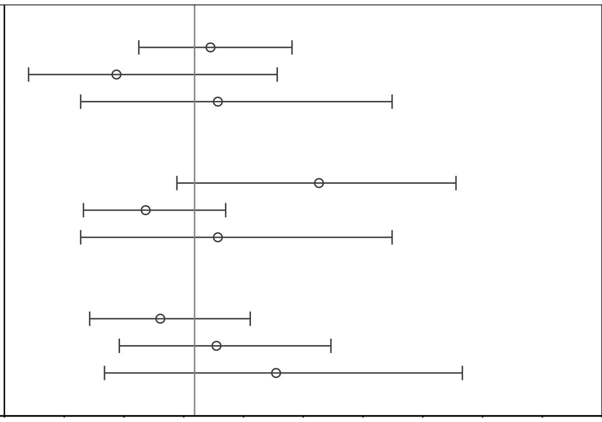


Figure 24: Forest plot of ORR subgroup analyses

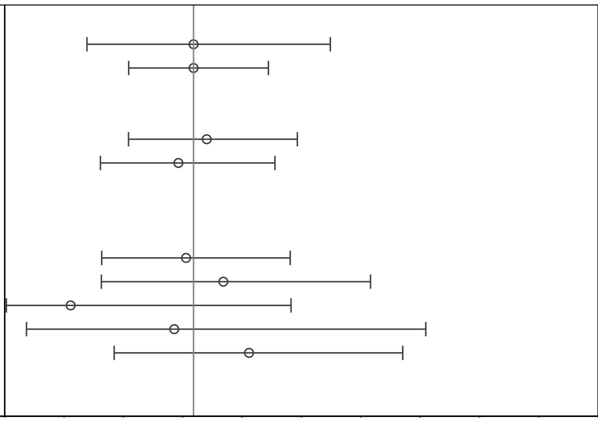
Time from initial diagnosis to study entry		
<= 1 year	19	6 (31.6) (12.6, 56.6)
> 1 year and <= 2 years	32	9 (28.1) (13.7, 46.7)
> 2 years	37	13 (35.1) (20.2, 52.5)
Baseline ECOG PS		
=0	49	17 (34.7) (21.7, 49.6)
=1	39	11 (28.2) (15.0, 44.9)
Disease Burden at Baseline SLD		
<= Q1	21	9 (42.9) (21.8, 66.0)
> Q1 and <= Median	18	7 (38.9) (17.3, 64.3)
> Median and <= Q3	19	7 (36.8) (16.3, 61.6)
> Q3	19	3 (15.8) (3.4, 39.6)
NA	11	2 (18.2) (2.3, 51.8)

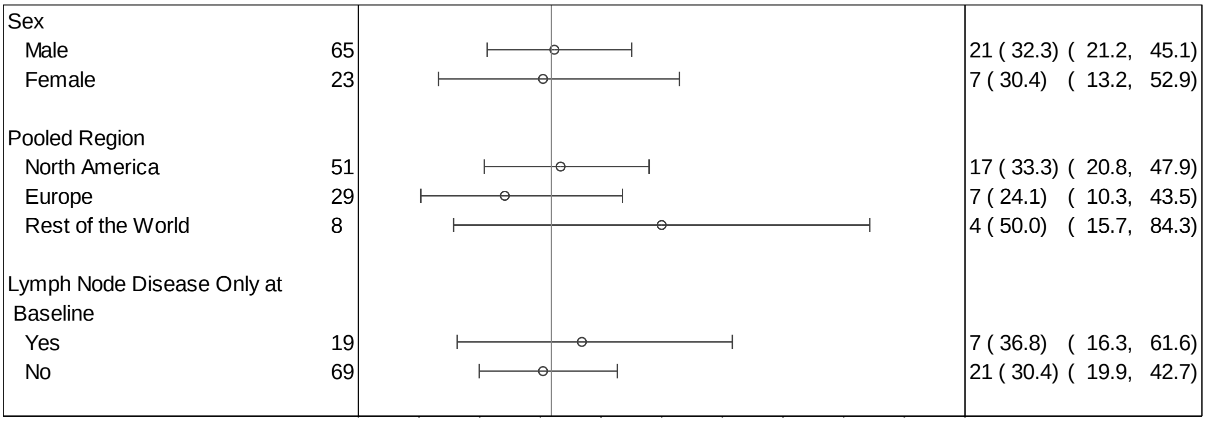


PD-L1 Expression at Cut-off of 1%		
Positive	58	20 (34.5) (22.5, 48.1)
Negative	16	3 (18.8) (4.0, 45.6)
Not Evaluable	14	5 (35.7) (12.8, 64.9)
PD-L1 Expression at Cut-off of 5%		
Positive	19	10 (52.6) (28.9, 75.6)
Negative	55	13 (23.6) (13.2, 37.0)
Not Evaluable	14	5 (35.7) (12.8, 64.9)
IHC-MCV Status		
Positive	46	12 (26.1) (14.3, 41.1)
Negative	31	11 (35.5) (19.2, 54.6)
NA	11	5 (45.5) (16.7, 76.6)



Age Group		
<65	22	7 (31.8) (13.9, 54.9)
>=65	66	21 (31.8) (20.9, 44.4)
Visceral Metastases at Baseline		
Present	47	16 (34.0) (20.9, 49.3)
Absent	41	12 (29.3) (16.1, 45.5)
Combination of MCV and PD-L1 Status at Baseline		
PD-L1+/MCV+	36	11 (30.6) (16.3, 48.1)
PD-L1+/MCV-	19	7 (36.8) (16.3, 61.6)
PD-L1-/MCV+	9	1 (11.1) (0.3, 48.2)
PD-L1-/MCV-	7	2 (28.6) (3.7, 71.0)
NA	17	7 (41.2) (18.4, 67.1)





<u>Durable response (defined as response for ≥6 months)</u>

Table 27: Durable response rate, proportion of subjects with at least 6 months durable response

Characteristic	ITT (N=88)	PP (N=64)
Objective Response Rate (%)	31.8	40.6
Proportion of duration of response at 6 month (a) (%)	91.6	90.8
Durable Response Rate (%) (95% confidence interval)	29.1 (19.5,38.8)	36.9 (24.8,48.9)

Seven patients had not yet reached the 6 months in response at the time of cutoff, so DRR may change. In the primary analysis, the DRR in the ITT population was 29%, while in the PP was 37%.

In the 12 months' analysis, 25 patients (28.4%) have been reported with a durable response as of the data cutoff. Two patients, still on study as of cut-off date had a DOR censored at a value < 6 months due to limited follow-up (possibly because subsequent tumor assessment was not evaluable).

<u>PD-L1 expression</u>

Predictive value of PD-L1 expression in terms of objective response rate (ORR) was evaluated at various cut-offs that were specified in the SAP. Four definitions for PD-L1 positivity were specified as follows:

- ≥ 1% positive tumour cell staining at any staining intensity
- ≥ 5% positive tumour cell staining at any staining intensity
- ≥ 25% positive tumour cell staining with at least 2+ staining intensity
- "PD-L1 hotspots" with ≥ 10% PD-L1 expressing immune cells

Results from study EMR100070-003 (Part A) 12-months minimum follow-up data are presented below:

Table 28: Summary of efficacy endpoints by PD-L1 subgroups - ITT analysis set

	Avelumab N=74[a]			
	PD-L1 at Cut-off of 1%		PD-L1 at Cut-off of 5%	
	Positive (n=58)	Negative (n=16)	Positive (n=19)	Negative (n=55)
BOR				
Confirmed ORR, % (95% CI)[b]	36.2 (24.0, 49.9)	18.8 (4.0, 45.6)	57.9 (33.5, 79.7)	23.6 (13.2,37.0)
Odds ratio (positive over negative) (95% CI)	2.46 (0.57,14.82)		4.44 (1.28,15.51)	
DOR	n=21	n=3	n=11	n=13
Median DOR (months) (95% CI)[c]	NE (18.0, NE)	NE (NE , NE)	NE (6.9, NE)	18.0 (18.0, NE)
Estimated Rate for 6-month DOR, % (95% CI)[d]	100 (NE , NE)	100 (NE , NE)	100 (NE, NE)	100 (NE , NE)
Estimated Rate for 12-month DOR, % (95% CI)[d]	80 (55, 92)	NE (NE, NE)	70 (33 , 89)	92 (54 , 99)
PFS				
Median PFS time (months) (95% CI)[c]	2.2 (1.4,9.7)	2.6 (1.0, NE)	10.0 (1.5, NE)	1.6 (1.4, 2.7)
6-month PFS rate, % (95% CI)[d]	43 (30, 56)	29 (9, 53)	63 (38, 80)	31 (19, 44)
12-month PFS rate, % (95% CI)[d]	34 (22, 47)	29 (9, 53)	47 (24, 67)	27 (15, 40)
15-month PFS rate, % (95% CI)[d]	34 (22 ,47)	NE (NE , NE)	47 (24 ,67)	27 (15 ,40)
Hazard ratio (positive over negative) (95% CI)	0.78 (0.38,1.57)		0.49 (0.24, 0.98)	
Overall survival				
Median overall survival time (months) (95% CI)[c]	14.8 (8.8, NE)	7.1 (3.4, NE)	NE (6.4, NE)	12.6 (7.1, 17.1)
6-month overall survival rate, % (95% CI)[d]	74 (60, 83)	61 (32, 80)	79 (53, 92)	68 (54 , 79)
12-month overall survival rate, % (95% CI)[d]	55 (41, 67)	40 (17, 63)	58 (33, 76)	50 (36, 63)
15-month overall survival rate, % (95% CI)[d]	49 (34, 61)	20 (2 ,54)	58 (33, 76)	40 (26, 53)
Hazard ratio (positive over negative) (95% CI)[d]	0.56 (0.27, 1.16)		0.65 (0.30,1.42)	

Source: Tables 15.2.1.10, 15.2.1.18, 15.2.2.8, 15.2.2.12, 15.2.3.8, 15.2.3.14, 15.2.4.8, 15.2.4.11, and 15.2.11.1 (12-month minimum follow-up analysis).

BOR: best overall response; CI: confidence interval; DOR: duration of response; IERC: Independent Endpoint Review Committee; N/n: number of subjects; NE: not evaluable; ORR: Objective Response Rate; PD-L1: Programmed death ligand 1; PFS: progression-free survival.
Note: For applicable endpoints, tumor assessment is according to the IERC.
a Please refer to source tables for data on subjects who were not evaluable for PD-L1.
b 95% exact confidence interval (CI) using the Clopper-Pearson method.
c Product-limit (Kaplan-Meier) estimates.
d Based on Kaplan-Meier estimates.

Tumour samples were evaluated for PD-L1 tumour cell expression, and for Merkel cell polyomavirus (MCV) using an investigational immunohistochemistry (IHC) assay. The tables below summarise the PD-L1 expression and MCV status of patients with metastatic MCC in study EMR100070-003 (Part A).

Table 29: ORR by combination of tumour PD-L1 expression (5%cutoff) and MCV-IHC status (minimum 12 month follow up)

Combination	ORR		Odds ratio[c] (95% CI)
	n/N1	% (95% CI)	
PD-L1+/MCV+	6/10	60.0 (26.2, 87.8)	6.00 (1.04, 36.13)
PD-L1+/MCV-	5/9	55.6 (21.2, 86.3)	5.00 (0.80, 31.47)
PD-L1-/MCV+	7/35	20.0 (8.4, 36.9)	
PD-L1-/MCV-	4/17	23.5 (6.8, 49.9)	1.23 (0.22, 5.91)

[c] Odds Ratio has been calculated as PD-L1+/MCV+ over PD-L1-/MCV+, PD-L1+/MCV- over PD-L1-/MCV+ and PD-L1-/MCV- over PD-L1-/MCV+.

Table 30: Objective response rates by MCV status in patients with metastatic MCC in study EMR100070-003 (Part A)

	Avelumab ORR (95% CI)
IHC-MCV tumour status	N=77[b]
Positive (n=46)	28.3% (16.0, 43.5)
Negative (n=31)	35.5% (19.2, 54.6)

IHC: Immunohistochemistry; MCV: Merkel cell polyomavirus; ORR: objective response rate

[a] Based on data from patients evaluable for PD-L1

[b] Based on data from patients evaluable for MCV by immunohistochemistry (IHC)

Table 31: Progression Free Survival and Overall Survival According to IERC Assessment by Combination of PD-L1 and IHC-MCV status – ITT Analysis Set (N=88)

PD-L1 and IHC-MCV Status at Baseline	PD-L1+ / MCV+	PD-L1+ / MCV-	PD-L1- / MCV+	PD-L1- / MCV-	NA[a]
Total subjects in subgroup, n (%)	**36 (100.0)**	**19 (100.0)**	**9 (100.0)**	**7 (100.0)**	**17 (100.0)**
Progression-free survival time[b]					
Number of subjects without event (censored), n (%)	13 (36.1)	7 (36.8)	3 (33.3)	3 (42.9)	7 (41.2)
Number of subjects with an event, n (%)	23 (63.9)	12 (63.2)	6 (66.7)	4 (57.1)	10 (58.8)
Progressive disease, n (%)	18 (50.0)	11 (57.9)	6 (66.7)	2 (28.6)	10 (58.8)
Death, n (%)	5 (13.9)	1 (5.3)	0	2 (28.6)	0
Median (months), 95% CI	2.1 (1.4, 8.6)	4.3 (1.3, NE)	1.4 (1.0, NE)	2.7 (0.5, NE)	4.2 (1.4, 20.6)
6-months PFS rate (%), 95% CI	36 (21 ,52)	50 (26 ,70)	25 (4 ,56)	33 (5 ,68)	33 (5 ,68)
12-months PFS rate (%), 95% CI	**30 (16 ,46)**	**33 (14 ,55)**	**25 (4 ,56)**	**33 (5 ,68)**	**31 (9 ,55)**
Overall survival time[b]					
Number of subjects without event (censored), n (%)	18 (50.0)	9 (47.4)	3 (33.3)	3 (42.9)	7 (41.2)
Number of subjects with an event (death), n (%)	18 (50.0)	10 (52.6)	6 (66.7)	4 (57.1)	10 (58.8)
Median (months), 95% CI	14.8 (5.1, NE)	17.1 (8.8, NE)	5.7 (2.2, NE)	7.1 (0.5, NE)	14.2 (4.0, NE)
6-months OS rate (%), 95% CI	63 (44 ,76)	89 (64 ,97)	50 (15 ,77)	71 (26 ,92)	71 (43 ,87)
12-months OS rate (%), 95% CI	**50 (32 ,66)**	**63 (38 ,80)**	**38 (9 ,67)**	**43 (10 ,73)**	**53 (28 ,73)**

Source: Refer to EMR100070-003 Addendum 2 (12-month analysis) Table 15.2.3.17 and Table 15.2.4.14
CI: confidence interval
[a] NA: not available due to not evaluable status for PD-L1 or MCV
[b] Product-limit (Kaplan-Meier) estimates
PD-L1 expression status is defined as cut-off value >=1% for positive and <1% for negative.
IHC-MCV status is Merkel cell polyomavirus status determined by immunohistochemistry method.
NE: not estimable

In conclusion, given the clinically meaningful response rates across the subgroups evaluated, the clinical utility of PD-L1 as a biomarker has not been established.

Summary of main study(ies)

The following tables summarise the efficacy results from the main studies supporting the present application. These summaries should be read in conjunction with the discussion on clinical efficacy as well as the benefit risk assessment (see later sections).

Title: Title: A Phase II, open-label, multicenter trial to investigate the clinical activity and safety of avelumab (MSB0010718C) in subjects with Merkel cell carcinoma (JAVELIN Merkel 200)	
Study identifier	EMR 100070-003
Design	Phase II, single arm, open-label, multicenter

	Duration of main phase: Duration of Run-in phase: Duration of Extension phase:		not applicable not applicable not applicable
Hypothesis	Exploratory. Overall alpha = 0.025 (1-sided) for the test of the null hypothesis of an ORR ≤ 20%.		
Treatments groups	Single arm		Avelumab 10 mg/kg IV every 2 weeks. Subject were treated until significant clinical deterioration, including treatment beyond radiological PD if the subject would benefit from continued treatment; unacceptable toxicity; withdrawal of consent; or any other protocol-specified criterion for withdrawal. 88 patients enrolled and treated (ITT population)
Endpoints and definitions	Primary endpoint	ORR (IERC assessed)	Proportion of subjects with a confirmed BOR of CR or PR, based on independent tumor assessment (IERC) according to RECIST 1.1
	Secondary endpoint	DOR (IERC assessed)	Calculated for each subject with a confirmed response (CR or PR) according to RECIST 1.1 as determined by IERC as the time from first observation of response until first observation of documented disease progression or death when death occurred within 12 weeks of the last tumor assessment, whichever occurred first. For subjects with a confirmed response but neither documented disease progression nor death within 12 weeks after the last tumor assessment, as of the cut-off date for the analysis, the duration of response was censored at the date of the last tumor assessment. If death occurred more than 12 weeks after the last tumor assessment, the duration of response was also censored at the date of the last adequate tumor assessment. Based on K-M estimates.
	Secondary endpoint	PFS (IERC assessed)	Time from first administration of study treatment until first observation of PD or death when death occurred within 12 weeks of the last tumor assessment or first administration of study treatment (whichever was later), based on independent tumor assessment (IERC) according to RECIST 1.1.
	Secondary endpoint	OS	Time from first administration of study treatment until the date of death.
Database lock	03 March 2016 (6 months after start of study treatment of the last subject)		

Results and Analysis		
Analysis description	**Primary Analysis**	
Analysis population and time point description	Intent to treat (88 patients) 6 months after start of study treatment of the last subject	
Descriptive statistics and estimate variability	Treatment group	Avelumab (Single arm)
	Number of subject	88
	BOR confirmed (IERC assessed) ORR (CR+PR) (%) (95.9% CI)	**28** (8+20) **(31.8%)** (21.9; 43.1)
Effect estimate per comparison	Not applicable (single arm study)	

Notes	-	
Analysis description	**Secondary Analyses**	
Analysis population and time point description	Intent to treat (88 patients) 6 months after start of study treatment of the last subject	
Descriptive statistics and estimate variability	Treatment group	Avelumab (Single arm)
	Number of subject	88
	DOR (IERC assessed) No of events (%) Median DOR (months) (95% CI)	 5 (17.9%) **Not Estimable** (8.3; NE)
	Durable response rate (DRR)[a] 6 months,% (95%CI)	**29.1** (19.5; 38.8)
	PFS (IERC assessed) No of PFS events (%) Median PFS (months) (95% CI)	 52 (59.1%) **2.7** (1.4; 6.9)
	Progression-free rate 6 months,% (95%CI)	**40** (29; 50)
	OS No of OS events (%) Median OS (months) (95% CI)	 43 (48.9%) **11.3** (7.5; 14.0)
Effect estimate per comparison	Not applicable (single arm study)	
Notes	[a] Durable response rate (DRR): estimated in a post-hoc analysis as the product of ORR and the Kaplan-Meier estimate of 6 months durability of response.	

Ongoing, multicenter, single-arm study designed in 2 parts to evaluate the efficacy and safety of avelumab in subjects with metastatic Merkel cell carcinoma (mMCC):

Part A tested avelumab in patients with mMCC previously treated with at least one line of chemotherapy and progressed after the most recent regimen.

Study identifier	EMR10070-003	
Design	open-label, single-arm study	
	duration of trial	ongoing

Treatments groups	mMCC proven histology		10 mg/kg q2w till until significant clinical deterioration; unacceptable toxicity; withdrawal. No significant clinical deterioration was defined as no new symptoms or worsening of existing symptoms, ECOG ≥3 lasting for >14 days and no investigator assessment that salvage therapy was necessary. Patients with a confirmed CR could continue to be treated with avelumab for a min, of 6 months and a max. of 12 months.
Endpoints and definitions	Primary endpoint	ORR	
	Secondary endpoints	DOR, PFS, OS, response at 6 and 12 months	
	Exploratory endpoints	-To assess immune-related best overall response (irBOR) and immune-related PFS (irPFS) using the modified immune-related response criteria (irRC) -To compare TTP on last prior anticancer therapy to PFS time on treatment with avelumab -To evaluate tumor shrinkage in target lesions at each time point from baseline -To evaluate changes in biomarkers in relation to disease responses to avelumab (MCV status by IHC and real-time PCR); expression and localization of pre-existing CD8+ T cells in tumor tissue; cytokine profiles over time -To evaluate the association between PD-L1 expression (IHC) and BOR -To explore the benefits of avelumab as perceived by subjects with metastatic MCC	
Database lock	March 2016		

Results and Analysis

Analysis description	**Primary Analysis**	
Analysis population and time point description	ITT, March 2016 (at 6 months): n=88 PP, sensitivity analysis, n=64 ITT September 2016 (12 months): n=88	
Descriptive statistics and estimate variability	ORR (%) 95% CI	31.8, 95% CI: (22.3, 42.6) (ITT), 40.6, 95% CI: (28.5, 53.6) (PP) *Note: 12 months' analysis, cutoff Sept.2016: ORR 33%, 95% CI: (23.3, 43.8)*

	DOR (mo)	Median -, 95% CI (8.3, -), range = (2.8-17.5+) March 2016 cutoff: of the 28 patients with a confirmed BOR, 23 were still having a response and were censored in the primary analysis of DOR. 19/23 patients had DOR≥6 mo. *Note: 12 months' analysis, cutoff Sept.2016. The 6-months' DRR was 30.6% (95% CI 20.9-40.3).* *A proportion of 93% (95% CI: 74, 98) of patients with confirmed response were estimated to have a DOR of at least 6 months by KM method. DOR data is maturing, mDOR has not been reached (the lower bound of the 95% CI is 18 mo).* *There seems to be prolonged benefit after discontinuation of treatment.*
	TTR (weeks), ITT	7.7±3 *12-months analysis: 8.6±6.1*
	PFS (mo)	2.7, 95% CI 1.4-6.9 range: (0.03-19) *12-months analysis: 2.7, 95% CI 1.4-6.9* *range: (0.03-24.5)*
	irPFS (mo)	4 95% CI 2.3, - range: (0.03-17.2) *12-months analysis:* *4 95% CI 2.3, -* *range: (0.03-22.1)*
	OS (mo)	11.3, 95% CI 7.5, 14 range: (0.4-18.8) *12-months analysis:* *12.9, 95% CI 7.5, -* *range: (0.4-24.7)*Patients with PD-L1<1% have shorter OS
	Response status at 6 and 12 mo (ITT)	At 6 mo: 27 patients in response At 12 mo: 6 patients in response *12-months analysis:* *Response status at 12-months: 21 subjects in response*
	PFS vs TTP on last prior therapy	HR 0.7, 95% CI: 0.5-1.0 *12-months analysis:* *HR 0.65, 95% CI: 0.46-0.92*
	PFS duration on follow-up (mo)	10.1, 95% CI: 8.3, 11.1 *12-months analysis:* *15.2 mo, 95% CI: 13.8, 17.3*
	Tumor shrinkage	Rapid: first 6 w. The majority of 29 pat with ≥30% decrease in target lesions had responses with duration ≥6 months
	HRQoL	ED-5D and FACT-M do not contribute with definitive information regarding avelumab treatment.
Effect estimate per comparison	N/A, historical control	

Analysis performed across trials (pooled analyses and meta-analysis)

Clinical studies in special populations

Table 32: Proportion of patients by age in the different clinical studies for avelumab

	Age 65-74 (Older subjects number /total number)	Age 75-84 (Older subjects number /total number)	Age 85+ (Older subjects number /total number)
EMR100070-001 Dose Escalation	10/53 (18.9%)	3/53 (5.7%)	0/53 (0%)
EMR100070-001 Expansion	431/1437 (30.0%)	184/1437 (12.8%)	24/1437 (1.7%)
EMR100070-002 Dose Escalation	7/17 (41.2%)	0/17 (0%)	0/17 (0%)
EMR100070-002 Expansion	12/34 (35.3%)	2/34 (5.9)	0/34 (0%)
EMR100070-003	35/88 (39.8%)	28/88 (31.8%)	3/88 (3.4%)
Pooled	495/1629 (30.4%)	217/1629 (13.3%)	27/1629 (1.7%)

Supportive study(ies)

Study EMR100070-003 Part B: Results of Treatment-naïve Patients with metastatic Merkel cell carcinoma (mMCC)

For Part B, an interim analysis of efficacy was conducted with 39 patients who received at least one dose. Of those, 30 (77%) were males, the median age was 75 years (range: 47 years to 88 years), 33 (85%) patients were Caucasian, and 31 (79%) patients and 8 (21%) patients had an ECOG performance status 0 and 1, respectively. Twenty-nine patients had at least 13 weeks of follow-up at the time of the data cut-off (24 March 2017). Results are shown in Table 33.

Table 33: Best overall response (BOR) according to IERC assessment in treatment-naive subjects with mMCC - Full analysis set - Study EMR100070-003 Part B

	Avelumab ≥6-month follow-up N=14 (100%)	Avelumab ≥13-week follow-up N=29 (100%)
Best Overall Response, n (%)		
Complete response	4 (28.6)	4 (13.8)
Partial response	6 (42.9)	14 (48.3)
Stable disease	1 (7.1)	3 (10.3)
Progressive disease	2 (14.3)	7 (24.1)
Not evaluable[a]	1 (7.1)	1 (3.4)
Objective Response Rate		
Response rate (CR + PR), n (%)	10 (71.4)	18 (62.1)
95% CI	41.9, 91.6	42.3, 79.3

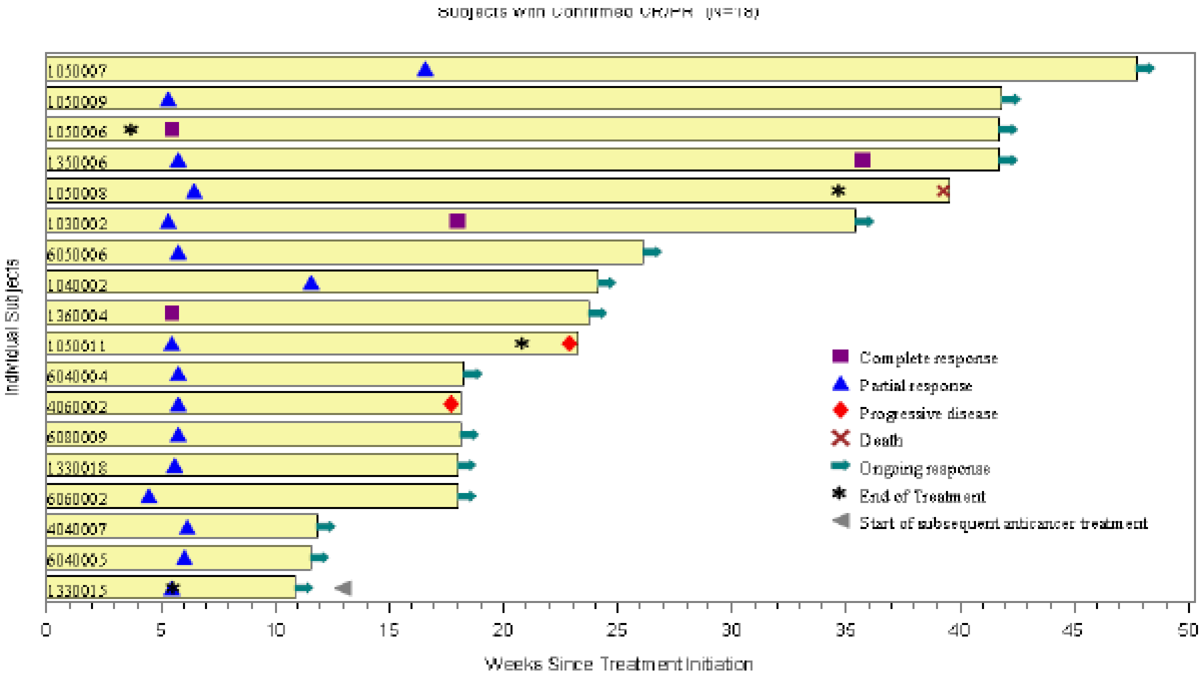
Source: Table 15.2.1.1.1a, Table 15.2.1.1.1b.

CI: confidence interval; CR: complete response; PR: partial response.

[a] One subject was not evaluable for BOR and experienced adverse events, unrelated to study drug, resulting in death due to disease progression per the Investigator.

Among the 39 subjects included in the interim analysis of Part B of study EMR100070-003, there are 29 subjects who were followed for a minimum of 13 weeks. Among these 29 treatment naïve subjects, an objective response rate (ORR) of 62.1% (95% CI: 42.3, 79.3) was reported. Per the IERC, there were 18 subjects with confirmed responses, including 4 subjects with a complete response (CR) and 14 subjects with a partial response (PR).

Among the 29 patients followed for a minimum of 13 weeks, 14 confirmed responses were ongoing at the time of the data cut-off, with a median duration that could not be estimated (95% CI for DOR: (4.0, -), min 1.2+, max 8.3+). Among treatment-naïve subjects in Part B, there were 6 responses with a duration of at least 6 months as of the cut-off in March 2017, 5 of which were ongoing at the cut-off date (see Figure 25).



Source: Figure 15.2.2.20b.
CR: complete response; PR: partial response.

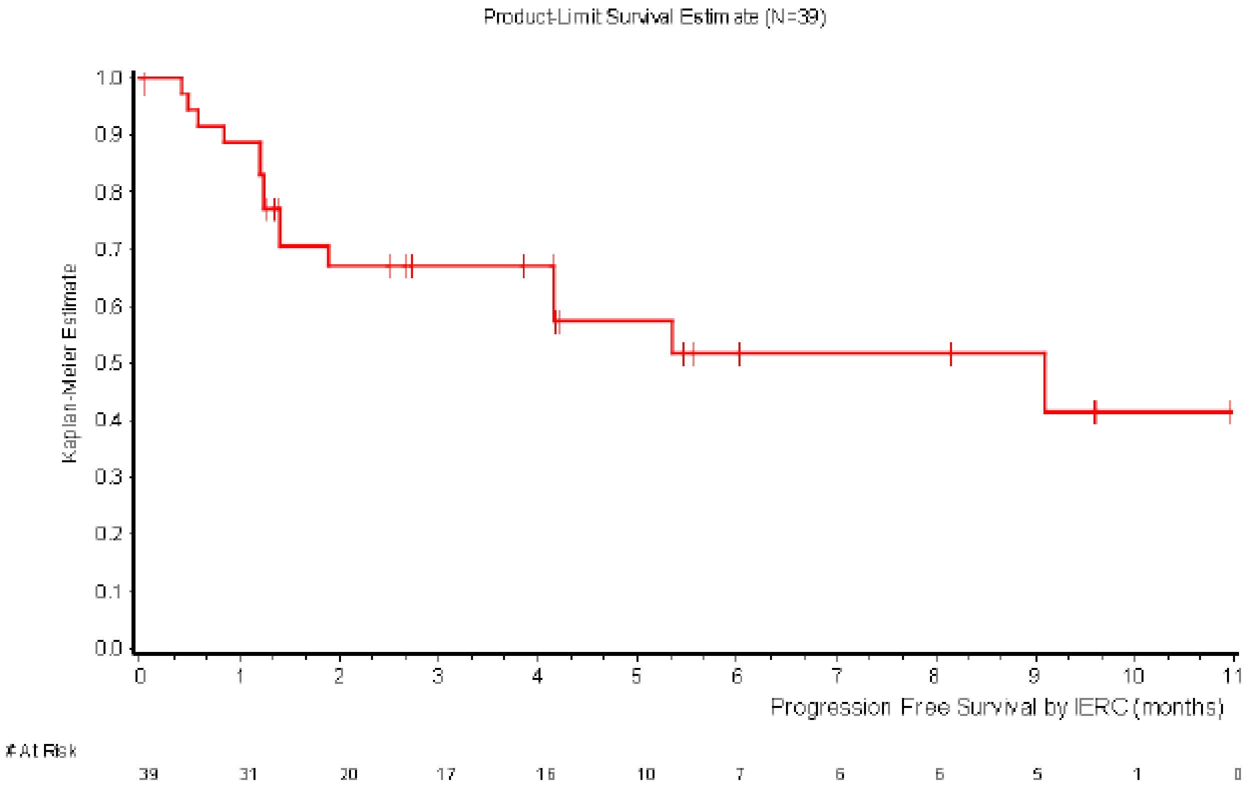
Figure 25: Time to and duration of response by subject according to IERC assessment - Full analysis set with minimum 13 weeks follow up - Study EMR100070-003 Part B

For PFS, at the last datacut, there were 15 (38.5%) subjects with a PFS event and 24 subjects (61.5 %) with no reported PFS event (censored). The estimated median PFS was 9.1 months (95% CI: 1.9 months, NE).

Table 34: PFS according to IERC assessment (Full Analysis Set)

	Avelumab N=39 (100%)
Number of subjects without event (censored), n (%)	24 (61.5)
Number of subjects with an event, n (%)	15 (38.5)
Progressive disease, n (%)	12 (30.8)
Death, n (%)	3 (7.7)
Progression-free survival time[a] (months)	
Median (95% CI)	9.1 (1.9, NE)
Minimum, maximum	0.03, 11.0
Progression-free survival rates, % (95% CI)[a]	
3 months	67 (48, 80)
6 months	52 (31, 69)

Source: Table 15.2.3.1.
CI: confidence interval; NE: not estimable; N/n: number of subjects.
[a] Based on Kaplan-Meier estimates.



Source: Figure 15.2.3.22.
IERC: Independent Endpoint Review Committee.

Figure 26: K-M estimates of PFS according to IERC assessment (Full Analysis Set)

Study 100070-Obs001: Observational study of treatment outcomes following chemotherapy

Methods

Investigators/Study Centers:

Part A: US Oncology Network (USON) outpatient medical oncology practices across the United States (US) (19 states)

Part B: MCC Registry with 56 clinical sites (Germany - 53, Austria - 2, and Switzerland - 1)

Study Period (years):

Part A: November 2004 – June 2015

Part B: November 2004 – December 2015

Objectives:

Part A: The primary objective was to assess objective response rate (ORR) based on best overall response (BOR) to the index (second line [2L] or later) chemotherapy treatment; BOR used

Response Evaluation Criteria in Solid Tumors (RECIST) 1.1 as a guide. The secondary objectives included the following:

- To assess duration of response (DOR), progression-free survival (PFS), overall survival (OS), time to treatment discontinuation (TTD), and durable response rate (DRR) to the index (2L or later) chemotherapy treatment
- To assess the above objectives in the first line (1L) chemotherapy treatment populations (1L cohorts) based on the 1L treatment.
- Part B: The primary objective was to assess ORR based on BOR to the index (2L or later) chemotherapy treatment; BOR was described based on data provided by the physician based on best clinical judgment. In Part B, the secondary objectives were identical to those in Part A, with 2 key differences.
- BOR was described based on data provided by the physician based on best clinical judgment rather than RECIST criteria
- As only patients treated with 2L chemotherapy were available in Part B, 1L treatment outcomes would be presented among those patients that received 2L chemotherapy treatment, not among patients that may have received 1L chemotherapy but not 2L.

For both Part A and Part B, all the objectives were evaluated among immunocompetent patients as the primary analysis as well as all patients meeting the study criteria (all qualified patients).

This was a retrospective, observational, descriptive study.

Number of Subjects:

Part A: Twenty qualified patients, including 14 immunocompetent patients, were identified who received 2L or later chemotherapy. A total of 67 qualified patients, including 51 immunocompetent patients, were identified who received 1L chemotherapy.

Part B: Overall, 34 qualified patients, including 29 immunocompetent patients, with distant mMCC were identified who received 2L or later chemotherapy. In addition, 32 patients, including

28 immunocompetent patients, were identified who received 1L chemotherapy among the patients that had received 2L or later chemotherapy.

<u>Study participants</u>

Main Criteria for Inclusion:

Part A

1. Male and female patients aged > 18 years at index date (initiation of 2L or later systemic chemotherapy) or 1L start date

2. Documented as diagnosed with distant mMCC on or any time before 30 September 2014

3. Diagnosed with mMCC disease on or any time before 30 September 2014

- Detection of a numbered line of therapy or reference to metastatic in line of therapy description
- Stage IV at initial diagnosis or current status
- Tumor node metastasis with M=1, indicating metastatic
- Detection of a documented location of metastatic disease (e.g., bone, lung)
- Searching for current or past recorded status of MCC disease with reference to metastases

4. Had evidence of 1L or later systemic chemotherapy in the metastatic setting between 01 November 2004 and 30 September 2014

5. For index line (2L or later): Confirmed patients received at least 1 line of systemic chemotherapy (including at least 1 agent from Table 3 of the protocol) for the treatment of distant mMCC prior to the index date and that they progressed after the most recent line of chemotherapy administered prior to the index date. For 1L therapy: Confirmed patients received 1L therapy for treatment of distant metastatic disease

6. First visit to the USON occurred 30 or more days prior to the index date to review prior medical history to rule-out initiation of index line of therapy prior to the patient identification period. The prior medical history period ended the day prior to the index date

7. During the study period, observed with either ≥ 1 visit within the USON in addition to the index date visit or a documented record of death, to approximate continuity of care

Part B

1. Male and female patients aged ≥ 18 years

2. Diagnosed with distant mMCC

3. Patients received any 2L or later systemic chemotherapy.

Main Criteria for Exclusion:

Part A

1. During the observed prior medical history, patients with evidence of treatment with any antibody/drug targeting T-cell co-regulatory proteins

2. During the study period, patients enrolled in any clinical trial within the USON (patients on only observational trials were retained if applicable)

3. Any time in the prior 3 years up to and including the index date, patients observed with any solid tumor, with the exception of basal or squamous cell carcinoma of the skin, bladder carcinoma in situ, or cervical carcinoma in situ. (Note: this was a change to the protocol from 5 years to 3 years after availability of first results.)

4. Patients whose 1L was observed prior to their mMCC diagnosis date (represented confounding data)

5. Patients whose index date was observed prior to their mMCC diagnosis date (represented confounding data).

Part B

Patients observed with any solid tumor at any time in the prior 3 years to study start, with the exception of basal or squamous cell carcinoma of the skin, bladder carcinoma in situ, or cervical carcinoma in situ (other malignant tumors).

Duration of Study:

Part A: The study observation period was 01 November 2004 through 30 September 2014, and patients were followed through 30 June 2015.

Part B: The study observation period was 01 November 2004 through 15 September 2015, and patients were followed through December 2015.

Criteria for Evaluation:

Efficacy: The primary endpoint was the patients' BOR to their line of systemic chemotherapy.

Other efficacy endpoints included DOR, durable response, TTD, PFS, and OS. All endpoints were calculated for patients with 2L or later chemotherapy and for the 1L chemotherapy populations.

The primary analysis population was the immunocompetent patients. All analyses were also conducted for all qualified patients.

Safety: Safety was not assessed in this study.

Statistical Methods:

Descriptive statistics were provided. Results were reported in aggregate. The number and percent of patients along with descriptive statistics (mean, standard deviation [SD], median, 25th and 75^{th} percentiles, and number of missing values) were reported for continuous data. Categorical variables (e.g., age groups, race) were reported with the number and percent of patients. Counts of missing observations were provided. Descriptive analyses were conducted for demographic characteristics, selected clinical characteristics, treatment, and outcome.

Outcome analyses were performed among all immunocompetent patients as well as in the overall study population. Efficacy endpoints were analyzed in the 1L chemotherapy treated population.

For the primary analysis, ORR described the proportion of immunocompetent patients with CR or PR among all immunocompetent patients fulfilling the inclusion and exclusion criteria. The

Kaplan-Meier method was used for time to event endpoints.

Outcomes:

Table 35: Demographic and baseline characteristics for patients with second line or later chemotherapy

Parameter	Part A All Qualified (N = 20)	Part A Immunocompetent (N = 14)	Part B All Qualified (N = 34)	Part B Immunocompetent (N = 29)
Gender, n (%)				
Male	14 (70.0)	11 (78.6)	22 (64.7)	18 (62.1)
Female	6 (30.0)	3 (21.4)	12 (35.3)	11 (37.9)
Race, n (%)				
Caucasian	9 (45.0)	8 (57.1)		
ND	11 (55.0)	6 (42.9)	34 (100)	29 (100)
Age at index in years				
average (SD)	73.3 (10.4)	72.5 (11.7)	64.4 (11.3)	64.4 (11.6)
median	73.5	75.2	67.5	67.0
(25th, 75th percentiles)	(66.0, 81.1)	(63.6, 81.1)	(61.0, 72.0)	(61.0, 73.0)
ECOG, n (%)			N/A	N/A
0	1 (5.0)	0		
1	16 (80.0)	13 (92.9)		
2	1 (5.0)	0		
Unknown	2 (10.0)	1 (7.1)		
Metastatic status[a]				
Visceral metastases	14 (70.0)	10 (71.4)	20 (58.8)[b]	16 (55.2)[b]
Not noted or missing	4 (20.0)	2 (14.3)	7 (20.6)	6 (20.7)
Non visceral metastases (bone / lymph node only)	2 (10.0)	2 (14.3)	7 (20.6)[c]	7 (24.1)[c]

ECOG = Eastern Cooperative Oncology Group, N/A = not available, ND = not documented, and SD = standard deviation

[a] For Part A, based on documented location of target lesions. If a patient did not have a target lesion identified or site not noted, it was classified under not noted. Non-visceral = bone or lymph node metastases only (no other sites); visceral = any other metastatic site other than bone or lymph node only.

[b] Visceral metastases and/or elevated lactate dehydrogenase according to classification of malignant melanoma.

[c] Distant soft tissue and lymph node metastases according to classification of malignant melanoma.

Table 36: Demographic and baseline characteristics for patients with first line chemotherapy (Part A only)

Parameter	All Qualified (N = 67)	Immunocompetent (N = 51)
Gender, n (%)		
Male	53 (79.1)	43 (84.3)
Female	14 (20.9)	8 (15.7)
Race, n (%)		
Caucasian	43 (64.2)	34 (66.7)
Other or ND	24 (35.8)	17 (33.3)
Age at index in years		
average (SD)	74.4 (10.5)	75.8 (10.7)
median	75.8	78.1
(25th, 75th percentiles)	(67.1, 82.3)	(67.9, 83.7)
ECOG, n (%)		
0	14 (20.9)	11 (21.6)
1	32 (47.8)	25 (49.0)
2 or 3[a]	8 (9.0)	5 (5.9)
Unknown	13 (19.4)	10 (19.6)
Metastatic status,[a] n (%)		
Visceral metastases	46 (68.65)	34 (66.67)
Non visceral metastases (bone/lymph node only)	14 (20.90)	10 (19.61)
Not noted	7 (10.45)	7 (13.72)

ECOG = Eastern Cooperative Oncology Group, ND = not documented, and SD = standard deviation

[a] Based on documented location of target lesions. If a patient did not have a target lesion identified or site not noted, it was classified under not noted. Non-visceral = bone or lymph node metastases only (no other sites); visceral = any other metastatic site other than bone or lymph node only.

Efficacy Results:

In both Part A and Part B, no immunocompetent or qualified patient achieved a CR on 2L or later chemotherapy (see table below).

Table 37: Best Overall response and objective response rate for patients with second line or later chemotherapy

	Part A		Part B	
Response	All Qualified (N = 20) n (%)	Immunocompetent (N = 14) n (%)	All Qualified (N = 34) n (%)	Immunocompetent (N = 29) n (%)
Best Overall Response				
Complete response	0	0	0	0
Partial response	4 (20.0)	4 (28.6)	3 (8.8)	3 (10.3)
Stable disease	2 (10.0)	2 (14.3)	3 (8.8)	3 (10.3)
Progressive disease	8 (40.0)	5 (35.7)	28 (82.4)	23 (79.3)
Not evaluable	6 (30.0)	3 (21.4)	0	0
Objective Response Rate[a,b] (95% CI)	20.0 (5.7 – 43.7)	28.6 (8.4 – 58.1)	8.8 (1.9 - 23.7)	10.3 (2.2 - 27.4)

CI = confidence interval

[a] Objective response rate was calculated as patients with complete response or partial response out of the total number of qualified patients.

[b] Clopper-Pearson exact binomial confidence intervals were provided.

Note: The immunocompetent group was the primary analysis population.

Table 38: Best overall response and objective response rate for patients with first line chemotherapy (Part A only)

Response	All Qualified (N = 67) n (%)	Immunocompetent (N = 51) n (%)
Best Overall Response		
Complete response	10 (14.9)	7 (13.7)
Partial response	11 (16.4)	8 (15.7)
Stable disease	1 (1.5)	1 (2.0)
Progressive disease	31 (46.3)	21 (41.2)
Not evaluable	14 (20.9)	14 (27.5)
Objective Response Rate[a,b] (95% CI)	31.3 (20.6 - 43.8)	29.4 (17.5 - 43.8)

CI = confidence interval, CR = complete response, and PR = partial response
[a] Objective response rate was calculated as patients with CR or PR out of the total number of qualified patients.
[b] Clopper-Pearson exact binomial confidence intervals were provided.
Note: The immunocompetent group was the primary analysis population.

For both Part A and Part B, none of the patients had DOR longer than 3 months, and the median DOR was less than 2 months (see table below). For both Part A and Part B, OS was between 4 to 6 months.

Table 39: Other efficacy analyses for patients with second line or later chemotherapy

Response	Part A		Part B	
	All Qualified (N = 20)	Immunocompetent (N = 14)	All Qualified (N = 34)	Immunocompetent (N = 29)
No. with CR or PR	4	4	3	3
DRR[a] (95% CI)	0.0 (0.0-16.8)	0.0 (0.0 - 23.2)	0.0 (0.0 – 10.3)	0.0 (0.0 – 11.9)
DOR[b] median (95% CI)	1.7 months (0.5 - 3.0)	1.7 months (0.5 - 3.0)	1.9 months (1.3 – 2.1)	1.9 months (1.3 – 2.1)
TTD[b] median (95% CI)	1.5 months (0.3 - 2.5)	1.8 months (0.3 – 3.3)	2.7 months (2.5-2.9)	2.8 months (2.5-4.3)
PFS[b] median (95% CI)	2.1 months (1.0 - 3.2)	2.2 months (1.2 – 3.5)	3.0 months (2.6 – 3.1)	3.0 months (2.5 – 3.2)
PFS rate[b] (95% CI) 6 months 12 months	 0 0	 0 0	 2.9 (0.2 - 13.0) 0	 3.4 (0.3 - 14.9) 0
OS[b] median (95% CI)	4.4 months (2.2 - 6.2)	4.3 months (2.1 – 6.2)	5.3 months (4.3 – 5.8)	5.3 months (4.3 – 6.0)
OS[b] rate (95% CI) 6 months 12 months	 30.2 (11.6 - 51.4) 0	 26.8 (7.3 - 51.5) 0	 26.4 (13.1 - 41.8) 0	 27.5 (13.0 - 44.2) 0

CI = confidence interval, CR = complete response, DOR = duration of response, DRR = durable response rate, N/A = not available, OS = overall survival, PFS = progression-free survival, PR = partial response, and TTD = time to treatment discontinuation
[a] DRR was calculated as patients with CR or PR lasting ≥ 6 months out of the total number of qualified patients. Clopper-Pearson exact binomial confidence intervals are provided.
[b] Kaplan-Meier estimate.
Note: The immunocompetent group was the primary analysis population.

The BOR for immunocompetent patients with 1L chemotherapy was CR in 7 patients (13.7%) and PR in 8 patients (15.7%) (see table below). The BOR for all qualified patients with 1L chemotherapy was CR in 10 patients (14.9%) and PR in 11 patients (16.4%).

Table 40: Other efficacy analyses for patients with first line chemotherapy (Part A only)

Response	All Qualified (N = 67)	Immunocompetent (N = 51)
No. with CR or PR	21	15
DRR[a,b] (95% CI)	14.9 (7.4 - 25.7)	15.7 (7.0 - 28.6)
DOR median (95% CI)	5.7 months (2.6 - 8.7)[c]	6.7 months (1.2 – 10.5)
TTD median (95% CI)	2.5 months (2.2 - 3.2)	2.4 months (2.2 – 2.9)
PFS median (95% CI)	4.6 months (3.0 - 7.0)[d]	4.6 months (2.8 – 7.7)[e]
PFS rate[f] (95% CI)		
6 months	44.8 (32.7-56.2)	47.1 (33.0-59.9)
12 months	21.8 (12.7-32.4)	24.8 (13.8-37.4)
18 months	16.3 (8.4-26.5)	17.3 (8.1-29.5)
24 months	10.2 (4.0-19.7)	8.7 (2.4-20.0)
OS median (95% CI)	10.2 months (7.4 - 15.2)[g]	10.5 months (7.2 – 10.2)[h]
OS rate (95% CI)		
6 months	70.1 (57.5-79.5)	66.7 (52.0-77.8)
12 months	44.0 (31.5-55.8)	45.3 (31.0-58.6)
18 months	28.7 (17.7-40.7)	30.2 (17.5-44.0)
24 months	24.5 (14.1-36.4)	24.4 (12.7-38.3)

CI = confidence interval, CR = complete response, DOR = duration of response, DRR = durable response rate, OS = overall survival, PFS = progression-free survival, PR = partial response, and TTD = time to treatment discontinuation

[a] DRR was calculated as patients with CR or PR lasting ≥ 6 months (180 days) out of qualified patients. There were 21 patients with CR or PR, and 10 of these had a response lasting ≥ 6 months. Fifteen responders were immunocompetent, and of these patients, 8 had a response lasting ≥ 6 months. Seven patients had PFS > 20 months that were driving the PFS rate at 24 months.

[b] Clopper-Pearson exact binomial confidence intervals are provided.

[c] One patient had data greater than the study end date, and was censored at the study end date of 30 June 2015 with a duration of 42.8 months.

[d] Four patients had data greater than the study end date, and were censored at the study end date of 30 June 2015.

[e] Two patients had data greater than the study end date, and were censored at the study end date of 30 June 2015.

[f] Kaplan-Meier estimates.

[g] Seven patients had overall survival data greater than the study end date, and were censored at the study end date of 30 June 2015.

[h] Four patients had data greater than the study end date, and were censored at the study end date of 30 June 2015.

Note: The immunocompetent group was the primary analysis population.

2.5.3. Discussion on clinical efficacy

Design and conduct of clinical studies

The applicant submitted the results of a single-arm phase II study EMR100070-003 Part A, in 2[nd] + line metastatic Merkel cell carcinoma (mMCC) patients who progressed after the most recent regimen, as well as interim results from study EMR100070-003 Part B to include treatment-naïve mMCC patients.

The design of the study was discussed at the SAWP and the advice was generally followed. The study design as a single arm study with patients that had been pre-treated with chemotherapy as standard of care was considered appropriate and acceptable. It had also been agreed that a randomised clinical trial was not feasible as the confirmatory trial for the conditional MA since this is an orphan condition and pembrolizumab (anti-PD-1) had shown efficacy in mMCC. Hence, a single arm study in naive patients was considered appropriate as the basis for the fulfilment of the conditional MA. The inclusion-exclusion criteria are considered acceptable as they reflect the patient population with metastatic disease. The primary endpoint (ORR) as well as secondary endpoint (DoR, PFS and OS) were also considered acceptable considering that MCC is a rare tumour and there are few patients that can be recruited in order to appropriately power a randomised controlled trial with the conventional endpoints of PFS and OS. Patients with active or a history of central nervous system (CNS) metastasis; active or a history of autoimmune disease; a history of other malignancies within the last 5 years; organ transplant; conditions requiring therapeutic immune suppression or active infection with HIV, or hepatitis B or C were excluded (SmPC section 4.4 and 5.1). Therefore, patients with autoimmune disease, HIV, Hepatitis B or C and with organ transplants have included in the RMP as missing information. As many patients that are immunocompromised have a higher risk for mMCC, efficacy data in mMCC patients that are immunocompromised will be collected during the conduct of the PASS study as well as in early access programmes.

As the study did not have a control arm, Study 100070OBS-001 was a multicenter, multi-country, retrospective, observational study that was performed to collect response information available for both 1L therapy and 2L+ therapy. The retrospective observational study was intended to address a lack of information on reliable historical data and also the lack of a comparator arm in the Phase II trial EMR100070-003. The purpose of the observational study was to generate data on observed clinical outcomes of patients with metastatic MCC who have progressed after one line of chemotherapy in current clinical practice. The cohort was similar to the patient population in EMR100070-003 study. Part A of the study was conducted in the US focusing on patients in community oncology settings to reflect real-world care in the US, which was primarily in an academic setting in the US. The observational study Part B was conducted in Europe, in the German-speaking countries of Germany, Austria and Switzerland (DACH) among patients in academic centers that are part of a common MCC Registry for the DACH. In addition, the data was supplemented with recent literature citing response rates and duration of response for subjects with distant metastatic MCC (Stage IV) in the 1L and 2L chemotherapy disease settings. Taking into account the caveats with registries and observational studies, the data can only be considered as supportive as there were divergences observed in terms of objective response rates in the registry study and in published clinical experience in first line treatment. Combined with an ultra-orphan disease with no approved therapy or consensus guideline on the most appropriate chemotherapy, the observational results cannot be used as intended by the advice, which required good quality comparative controls and a compelling difference between a comparative analysis of the the clinical effect observed in the chemotherapy group vs avelumab treated patients that would be indisputably positive. The limited data showed marked geographic differences reflective of the lack of consensus regarding therapy. Nevertheless, baseline characteristics and demographics for both Part A and Part B showed that the majority of patients recruited were male with an average age of 73 years and 64 years old respectively. The majority of patients had an ECOG 0-1 for Part A whereas no information was given for Past B. Most patients also had visceral disease for both Part A and B. There were fewer patients recruited with stage IV for Part A than for Part B. The majority of patients with 1L chemotherapy were Stage I, II, or III at initial MCC diagnosis. The mean (SD) duration of 1L chemotherapy was 90.8 (75.6) days and 91.8 (69.8) days in immunocompetent and qualified patients, respectively.

The avelumab dose was 10 mg/kg q2w and was based on studies demonstrating optimal receptor occupancy in the dose escalation studies. Administration of Bavencio should continue according to the recommended schedule until disease progression or unacceptable toxicity. Patients with radiological disease progression not associated with significant clinical deterioration, defined as no new or worsening symptoms, no change in performance status for greater than two weeks, and no need for salvage therapy, could continue treatment.

Efficacy data and additional analyses

Patients who have progressed after receiving at least 1 line of previous chemotherapy for the treatment of mMCC

Updated results with 12 month minimum follow-up data from the **Part A cohort** of 88 subjects with treatment experienced mMCC subjects, demonstrated an ORR of 33% (95% CI: 23.3, 43.8) with CR in 10/88 and PR in 19/88 individuals compared to 31.8% (95% CI: 22.3, 42.6) previously reported with the 6-months minimum follow-up (primary) analysis.

The durable response rate defined as duration of response 6 months or more was 31% and PFS rate at 12 months was 30% (95% CI 21; 41%). Median DOR was not reached (95%CI 18.0, NE). In the 12-months minimum follow-up analysis, the longest response durations included 1 subject with a CR lasting 23.3+ months and 3 subjects with CR (n=2) or PR (n=1) lasting 18.0+ months, all of which were still ongoing at the last tumour assessment. The estimated median PFS time was 2.7 months (95% CI: 1.4, 6.9), with a 6-month PFS rate of 40% (95% CI: 29, 50), and 12-month and 15-month PFS rates of 30% (95% CI: 21, 41) each. Although ORR is not very impressive, the duration of response is considered clinically relevant advantage over chemotherapy. The duration of response with avelumab therapy in 2L+ is favourable when placed in context with chemotherapy. Specifically, subjects treated with avelumab in the 2L+ setting, had a median duration of response that was not yet reached and a lower bound of the 95% CI of 18.0 months, while the median duration of response for 1L chemotherapy was 6.7 months with an upper bound of 10.5 months for the 95% CI in Part A of Study 100070-Obs001 (1L) and a median duration of response as 2.8 months in Iyer et al. This difference in median duration of response between 2L+ avelumab treatment and chemotherapy in the 1L setting, suggests that avelumab treatment in subjects with mMCC may lead to durable responses regardless of line of therapy.

In the updated analysis conducted 12 months after the accrual of the last subject, there were 3 additional PFS events (all progression of disease), compared to the primary analysis at 6-months minimum follow-up (52 events [59.1%] vs. 55 events [62.5%]). An apparent plateau of the Kaplan-Meier PFS curve, observed in the 6-months analysis, is maintained in the updated 12-months analysis, possibly reflecting the proportion of subjects with durable responses.

Compared to chemotherapy in 2L+, while the median PFS is numerically similar (2.7 months), long-term PFS rates exceed results with 2L chemotherapy. Indeed, PFS rates with avelumab 2L+ was 40%, 30% and 30% at 6, 12 and 15 months respectively, while it was between 0 and 13% at 6 months, and 0% at 12 and 15 months with chemotherapy.

Compared to chemotherapy in 1L, PFS rates at 6, 12 and 15 months are overall similar to Obs-001 study results. Literature 1L data from Iyer 2016 showed PFS rates lower compared to the Obs-001 study. Median OS with avelumab 2L+ is 12.9 months (95% CI: 7.5, NE), similar to mOS of chemotherapy 1L (10.5 months [95%CI 7.2, 10.2] in immunocompetent patients Part A Obs-001). Overall, the long-term benefit with avelumab in pre-treated patients appeared not inferior to the outcome of chemotherapy in treatment-naïve patients.

There is a lack of data on patients with active central nervous system (CNS) metastasis, active or with a history of any autoimmune disease, a history of other malignancies within the last 5 years, organ transplant, conditions requiring therapeutic immune suppression or active infection with HIV or hepatitis B or C, as these patients were excluded from the clinical trial (this is reflected in sections 4.4 and 5.1 of the SmPC). Hence, in the absence of clinical efficacy data, avelumab should be used with caution in these populations after careful consideration of the potential risk-benefit on an individual basis.

Patients naïve for systemic chemotherapy with mMCC

In **Part B**, meaningful data are only available in a small number of patients (35) with limited follow-up (≥6 weeks). The unconfirmed ORR was 60%. Altogether 39 patients were enrolled thereof 29 with a minimum of 13 weeks follow-up. The overall confirmed ORR was 62% (95% CI 42; 79) and the proportion of patients with at least 6 months durability of response was 83% (95% CI 46; 96). The study is ongoing with planned enrolment of 112 subjects. The interim data from Part B provides initial evidence of durable response with 14 of 18 responders with at least 13 weeks of follow-up still reporting ongoing responses at the time of database cut-off. As further replication for the data showing in Part A, there was, in Part B, a rapid onset of responses with 16 of 18 confirmed responses reported at the first visit occurring approximately 6 weeks after initiation of study drug treatment in the analysis set of subjects with at least 13 weeks of follow-up. This rapid onset of response in treatment naïve subjects is similar to the timing of response to avelumab observed in treatment experienced subjects from Part A. Finally, among these responders, 6 subjects reported to have duration of response ⩾ 6 months, of which 5 were continuing at the time of the data cutoff. Confirmed ORR in the 14 patients followed for a minimum of 6 months was 71.4% (95%CI 41.9-91.6) with 10 confirmed responses (4 CR and 6 PR).

Among the 29 patients followed for a minimum of 13 weeks, 14 confirmed responses were ongoing at the time of the data cut-off, with the median and 95% CI DOR that could not be estimated (95% CI for DOR: (4.0, -), min 1.2+, max 8.3+). Median duration of response from the interim analysis for 1L avelumab was not yet estimable.

The European Medicines Agency has waived the obligation to submit the results of studies with Bavencio in all subsets of the paediatric population for the treatment of Merkel cell carcinoma (see section 4.2 for information on paediatric use).

PD-L1 expression and MCV positivity

Of the total 88 patients, 74 (84%) had tumour samples evaluable for PD-L1 expression. Cut-off used for tumour PD-L1 expression were ≥ 1% and ≥ 5% positive tumour cell IHC staining at any staining intensity (only 4 samples were positive with a cut-off 25%). The updated 12-months follow up analysis confirmed that the PD-L1 positive tumours subgroup achieved overall better ORR and DRR compared to PD-L1 negative tumour, with a trend toward improved outcome with higher cut-off positivity. Median DOR was not reached in all subgroups with the exception of PD-L1 negative at 5% cut-off (median DOR 18.0 months, 95% CI [18, NE]). The subgroup analysis of PFS and OS favoured PD-L1 positive subjects with expression ≥ 1% or ≥ 5%. No conclusion can be drawn on patients with PD-L1 and MCV positivity status due to too few patients in each subgroup. In the framework of a rare disease and the lack of an appropriate cut-off for PD-L1, it is difficult to implement a selection of patients for avelumab therapy based on biomarker in current clinical practice. The clinical utility of PD-L1 as a predictive biomarker in MCC has not been established. The CHMP recommends taking the opportunity to further study the predictive value of PD-L1 and MCV in the ongoing Part B of Study 003 in first line mMCC, if possible, while acknowledging the difficulty of obtaining tumour samples from patients in this rare disease.

Study 100070OBS-001

For both Part A and Part B, the majority of patients recruited were male with an average age of 73 years and 64 years old respectively. The majority of patients had a ECOG 0-1 for Part A whereas no information was given for Past B. Most patients also had visceral disease for both Part A and B. For patients being treated with 2nd line therapy, ORR was 20.0% (95%CI 5.7, 43.7) and 28.6% (95%CI 8.4, 58.1) for Part A and 8.8% (95%CI 1.9, 23.7) and 10.3% (95%CI 2.2, 27.4) for Part B, for all qualified and immunocompetent patients, respectively. DOR was found to be similar in the two group, the qualified population for Part A had a DOR of 1.7 months (95%CI 0.5, 0.3) and for Part B was 1.9 months (95%CI

1.3, 2.1). In this study, there were no durable responses (> 6 months) to 2L or later chemotherapy, there were no complete responders, and the median DOR was < 2 months. In the qualified population, median PFS was 2.1 months (95%CI1.0, 3.2) for Part A and 3.0 months (95%CI 2.6, 3.1) for Part B and median OS was found to be 4.4 month (95%CI 2.2, 6.2) and 5.3 month (95%CI 4.3, 5.8). Similar results were found for immunocompetent patients.

For patients treated with 1st line chemotherapy, therapy, ORR was 31.3% (95%CI 20.6, 43.8) and 29.4% (95%CI 17.5, 43.8) for Part A only, for all qualified and immunocompetent patients, respectively. DOR was found to be similar between the all qualified and immunocompetent group for Part A with a DOR of 5.7 months (95%CI 2.6,8.7) and 6.7 months (95%CI 1.2, 10.5). In both groups, median PFS was 4.6 months (95%CI 3.0, 7.0) and 4.6 months (95%CI 2.8, 7.7) and median OS was found to be 10.2 month (95%CI 7.4, 15.2) and 10.5 month (95%CI 7.2, 10.2).

Additional efficacy data needed in the context of a conditional MA

The CHMP were of the opinion that the preliminary efficacy data presented from Study EMR100070-003 Part B indicated a clinical benefit to mMCC patients that have not been treated with chemotherapy. Therefore, the CHMP agreed that the conditional marketing authorisation requested by the applicant could be granted to an indication covering naïve as well as patients pre-treated with chemotherapy given that, although the data in 1st line was promising, the data is limited as few patients have reached the 6 month milestone and further confirmatory data from additional treated patients would be needed to confirm the effect size of the benefit. As further data is necessary to confirm the efficacy in the indication proposed on the basis of the Part B of the study which is ongoing and includes treatment-naïve mMCC patients (none enrolled prior to the cut-off date for the primary analysis of Part A, March 2016), the CHMP imposes the specific obligation to submit the final study results by January 2020.

2.5.4. Conclusions on the clinical efficacy

The results in 2nd line treatment show an ORR of 33% with some patients having durable responses resulting in an apparent PFS plateau. The long duration of response and high durable response rate observed after 6-month minimum follow-up is further substantiated by the 12-months minimum follow-up analysis. The median OS time in the updated results exceeded 1 year. Therefore, the clinical benefit in 2nd line treatment is considered clinically meaningful and the magnitude of the effect is significant compared to chemotherapy.

The data provided in 1st line, although preliminary, showed activity of avelumab in mMCC in terms of response rate, and evidence of a similar durable response from the small subset of patients with a longer follow-up compared to 2nd line treatment.

Taking into account the intrinsic limitation of single arm studies, the rarity of the disease and the challenges to compare the results with data from historical controls and in the literature, the currently available data are deemed to support the efficacy of avelumab in both pre-treated and chemotherapy-naïve patients.

This medicinal product has been authorised under a so-called ‘conditional approval’ scheme. This means that further evidence on this medicinal product is awaited. The European Medicines Agency will review new information on this medicinal product at least every year and this SmPC will be updated as necessary.

The CHMP considers the following measures necessary to address the missing efficacy data for 1st line treatment in the context of a conditional MA:

- In order to confirm the efficacy for chemotherapy-naïve treated patients, the MAH should submit the final results of study EMR 100070-003 – Part B. The final results of the study should be submitted by 30th January 2020.

The CHMP considers that additional supportive data on efficacy will be provided also from the following post-authorisation safety study:

- PASS: German real-world cohort study should be submitted as additional PhV activity to address the missing information of safety and efficacy in immune compromised patients.

2.6. Clinical safety

The safety review was based on the primary analysis of the pivotal Phase II EMR100070-003 (003) study conducted in patients with MCC (data cut-off date 03 March 2016). In addition, safety data with data cut-off date 20 November 2015 was included from the Phase I EMR100070-001 (001) study in patients with advanced solid tumours treated with 10 mg/kg q 2 weeks, which at the data cut-off date included 1452 patients with the following tumour types: NSCLC (23 %), ovarian cancer (16 %), gastric and gastroesophageal cancer (13 %), metastatic breast cancer (12 %), urothelial carcinoma (11 %) and head/neck cancer (10 %). A pooled safety set with study EMR-003 has also been compiled. Updated safety data for both studies were also taken into account (data cut off: 09 Jun 2016) yielding a total patient population of 1738 subjects.

Patient exposure

Table 41: Summary of Drug Exposure - Safety Analysis Set

Characteristic	Statistics	001 (N=1452)	003 (N=88)	Total (N=1540)
Treatment Duration (weeks)	N	1452	88	1540
	Missing	0	0	0
	Mean ±SD	15.27 ±15.72	23.15 ±18.92	15.72 ±16.02
	Median	11.9	17.0	11.9
	Q1; Q3	6.0; 18.0	7.4; 37.0	6.0; 18.0
	Min; Max	2.0; 107.9	2.0; 76.0	2.0; 107.9
Number of Avelumab Administrations	N	1452	88	1540
	Missing	0	0	0
	Mean ±SD	7.4 ±7.5	11.1 ±9.2	7.6 ±7.7
	Median	5	7	6
	Q1; Q3	3; 9	3; 18	3; 9
	Min; Max	1; 51	1; 35	1; 51
Cumulative Dose (mg/kg)	N	1452	88	1540
	Missing	0	0	0
	Mean ±SD	73.70 ±75.36	110.80 ±92.19	75.82 ±76.87
	Median	50.2	70.0	56.9
	Q1; Q3	30.0; 90.0	30.2; 180.1	30.0; 90.0
	Min; Max	3.0; 510.1	9.3; 351.0	3.0; 510.1
Dose Intensity (mg/kg/cycle)	N	1452	88	1540
	Missing	0	0	0
	Mean ±SD	9.71 ±0.86	9.53 ±0.90	9.70 ±0.86
	Median	10.0	9.9	10.0
	Q1; Q3	9.8; 10.0	9.3; 10.0	9.8; 10.0
	Min; Max	2.2; 11.2	5.0; 10.5	2.2; 11.2
Relative Dose Intensity (%)	N	1452	88	1540
	Missing	0	0	0
	Mean ±SD	97.1 ±8.6	95.3 ±9.0	97.0 ±8.6
	Median	100	99	100
	Q1; Q3	98; 100	93; 100	98; 100
	Min; Max	22; 112	50; 105	22; 112
Relative Dose Intensity (%) categories, N (%)	n (missing)	1452 (0)	88 (0)	1540 (0)
	≤ 80	67 (4.6)	6 (6.8)	73 (4.7)
	> 80 - ≤ 90	100 (6.9)	10 (11.4)	110 (7.1)
	> 90	1285 (88.5)	72 (81.8)	1357 (88.1)

Source: SCS Table 12.5.1

Notes:

1. Treatment duration (weeks) = (last dose date - first dose date + 14)/7.
2. Cumulative dose (mg/kg) = sum of all actual (non-zero) dose (mg) received/weight (kg) during the study.
3. Dose intensity (mg/kg/2 weeks) = cumulative dose (mg/kg)/(treatment duration (weeks)/2).
4. Relative dose intensity=cumulative dose intensity(mg/kg/2weeks)/planned dose level(mg/kg/2weeks)*100.
5. A cycle has a planned length of 2 weeks.

Table 42: Summary of Dose Reductions and Dose Delays - Safety Analysis Set

Characteristic	Statistics	001 (N=1452)	003 (N=88)	Total (N=1540)
At least one Dose Reduction, N (%)	0	1405 (96.8)	88 (100.0)	1493 (96.9)
	1	40 (2.8)	0 (0.0)	40 (2.6)
	2	6 (0.4)	0 (0.0)	6 (0.4)
	3	1 (0.1)	0 (0.0)	1 (0.1)
	≥ 4	0 (0.0)	0 (0.0)	0 (0.0)
No Dose Delay, N (%)[a]	No delay	1151 (79.3)	49 (55.7)	1200 (77.9)
	0 day	786 (54.1)	23 (26.1)	809 (52.5)
	1 day	278 (19.1)	19 (21.6)	297 (19.3)
	2 days	87 (6.0)	7 (8.0)	94 (6.1)
Subjects with at least one Dose Delay, N (%)		301 (20.7)	39 (44.3)	340 (22.1)
	Worst dose delay			
	3-6 days delay	63 (4.3)	10 (11.4)	73 (4.7)
	3 days	39 (2.7)	8 (9.1)	47 (3.1)
	4 days	18 (1.2)	2 (2.3)	20 (1.3)
	5 days	4 (0.3)	0 (0.0)	4 (0.3)
	6 days	2 (0.1)	0 (0.0)	2 (0.1)
	≥ 7 days delay	238 (16.4)	29 (33.0)	267 (17.3)
	7-13 days	61 (4.2)	7 (8.0)	68 (4.4)
	14-20 days	151 (10.4)	18 (20.5)	169 (11.0)
	21-27 days	5 (0.3)	1 (1.1)	6 (0.4)
	≥ 28 days	21 (1.4)	3 (3.4)	24 (1.6)

Source: SCS Table 12.5.1

[a] For completeness, the number of subjects with no delay (0 days), 1 or 2 days deviation from the planned 14-day infusion schedule are presented but are not considered as a delay.

Adverse events

Table 43: Most common treatment emergent adverse events (at least 3% in the total group in SUR column) by System Organ Class and Preferred Term - All subjects

Body System or Organ Class Preferred term	001 SCS (N=1452) n (%)	 SUR (N=1650) n (%)	 Delta (%)	003 SCS (N=88) n (%)	 SUR (N=88) n (%)	 Delta (%)	Total SCS (N=1540) n (%)	 SUR (N=1738) n (%)	 Delta (%)
Number of Subjects With At Least One Event	1359 (93.6)	1611 (97.6)	4.0	86 (97.7)	86 (97.7)	0.0	1445 (93.8)	1697 (97.6)	3.8
Blood and lymphatic system disorders	254 (17.5)	321 (19.5)	2.0	20 (22.7)	20 (22.7)	0.0	274 (17.8)	341 (19.6)	1.8
Anaemia	188 (12.9)	246 (14.9)	2.0	13 (14.8)	13 (14.8)	0.0	201 (13.1)	259 (14.9)	1.8
Endocrine disorders	100 (6.9)	144 (8.7)	1.8	5 (5.7)	6 (6.8)	1.1	105 (6.8)	150 (8.6)	1.8
Hypothyroidism	66 (4.5)	106 (6.4)	1.9	4 (4.5)	5 (5.7)	1.2	70 (4.5)	111 (6.4)	1.9
Gastrointestinal disorders	850 (58.5)	1064 (64.5)	6.0	51 (58.0)	53 (60.2)	2.2	901 (58.5)	1117 (64.3)	5.8
Nausea	326 (22.5)	418 (25.3)	2.8	18 (20.5)	19 (21.6)	1.1	344 (22.3)	437 (25.1)	2.8
Diarrhoea	242 (16.7)	309 (18.7)	2.0	20 (22.7)	20 (22.7)	0.0	262 (17.0)	329 (18.9)	1.9
Constipation	248 (17.1)	305 (18.5)	1.4	15 (17.0)	15 (17.0)	0.0	263 (17.1)	320 (18.4)	1.3
Vomiting	214 (14.7)	270 (16.4)	1.7	10 (11.4)	11 (12.5)	1.1	224 (14.5)	281 (16.2)	1.7
Abdominal pain	184 (12.7)	239 (14.5)	1.8	11 (12.5)	11 (12.5)	0.0	195 (12.7)	250 (14.4)	1.7
Abdominal pain upper	65 (4.5)	90 (5.5)	1.0	3 (3.4)	3 (3.4)	0.0	68 (4.4)	93 (5.4)	1.0
Dysphagia	51 (3.5)	76 (4.6)	1.1	1 (1.1)	1 (1.1)	0.0	52 (3.4)	77 (4.4)	1.0
Ascites	46 (3.2)	64 (3.9)	0.7	1 (1.1)	1 (1.1)	0.0	47 (3.1)	65 (3.7)	0.6
Dry mouth	50 (3.4)	64 (3.9)	0.5	3 (3.4)	3 (3.4)	0.0	53 (3.4)	67 (3.9)	0.5
Abdominal distension	54 (3.7)	65 (3.9)	0.2	0 (0.0)	0 (0.0)	0.0	54 (3.5)	65 (3.7)	0.2
Dyspepsia	43 (3.0)	64 (3.9)	0.9	1 (1.1)	1 (1.1)	0.0	44 (2.9)	65 (3.7)	0.8
Gastrooesophagealreflux disease	45 (3.1)	52 (3.2)	0.1	2 (2.3)	2 (2.3)	0.0	47 (3.1)	54 (3.1)	0.0
General disorders and administration site conditions	870 (59.9)	1090 (66.1)	6.2	61 (69.3)	61 (69.3)	0.0	931 (60.5)	1151 (66.2)	5.7
Fatigue	418 (28.8)	530 (32.1)	3.3	33 (37.5)	33 (37.5)	0.0	451 (29.3)	563 (32.4)	3.1
Pyrexia	181 (12.5)	231 (14.0)	1.5	6 (6.8)	6 (6.8)	0.0	187 (12.1)	237 (13.6)	1.5
Oedema peripheral	131 (9.0)	189 (11.5)	2.5	16 (18.2)	17 (19.3)	1.1	147 (9.5)	206 (11.9)	2.4
Disease progression	125 (8.6)	175 (10.6)	2.0	4 (4.5)	4 (4.5)	0.0	129 (8.4)	179 (10.3)	1.9
Chills	133 (9.2)	162 (9.8)	0.6	6 (6.8)	7 (8.0)	1.2	139 (9.0)	169 (9.7)	0.7
Asthenia	104 (7.2)	140 (8.5)	1.3	11 (12.5)	11 (12.5)	0.0	115 (7.5)	151 (8.7)	1.2
Non-cardiac chest pain	70 (4.8)	91 (5.5)	0.7	1 (1.1)	1 (1.1)	0.0	71 (4.6)	92 (5.3)	0.7
Influenza like illness	65 (4.5)	78 (4.7)	0.2	2 (2.3)	2 (2.3)	0.0	67 (4.4)	80 (4.6)	0.2
Infections and infestations	432 (29.8)	609 (36.9)	7.1	33 (37.5)	34 (38.6)	1.1	465 (30.2)	643 (37.0)	6.8
Urinary tract infection	115 (7.9)	163 (9.9)	2.0	4 (4.5)	4 (4.5)	0.0	119 (7.7)	167 (9.6)	1.9

Body System or Organ Class Preferred term	001 SCS (N=1452) n (%)	 SUR (N=1650) n (%)	 Delta (%)	003 SCS (N=88) n (%)	 SUR (N=88) n (%)	 Delta (%)	Total SCS (N=1540) n (%)	 SUR (N=1738) n (%)	 Delta (%)
Upper respiratory tract infection	53 (3.7)	98 (5.9)	2.2	5 (5.7)	5 (5.7)	0.0	58 (3.8)	103 (5.9)	2.1
Pneumonia	59 (4.1)	76 (4.6)	0.5	3 (3.4)	3 (3.4)	0.0	62 (4.0)	79 (4.5)	0.5
Injury, poisoning and procedural complications	342 (23.6)	438 (26.5)	2.9	21 (23.9)	21 (23.9)	0.0	363 (23.6)	459 (26.4)	2.8
Infusion related reaction[a]	246 (16.9)	284 (17.2)	0.3	13 (14.8)	13 (14.8)	0.0	259 (16.8)	297 (17.1)	0.3
Fall	36 (2.5)	58 (3.5)	1.0	3 (3.4)	3 (3.4)	0.0	39 (2.5)	61 (3.5)	1.0
Investigations	529 (36.4)	680 (41.2)	4.8	36 (40.9)	39 (44.3)	3.4	565 (36.7)	719 (41.4)	4.7
Weight decreased	225 (15.5)	275 (16.7)	1.2	12 (13.6)	13 (14.8)	1.2	237 (15.4)	288 (16.6)	1.2
Aspartate aminotransferase increased	74 (5.1)	95 (5.8)	0.7	5 (5.7)	6 (6.8)	1.1	79 (5.1)	101 (5.8)	0.7
Alanine aminotransferase increased	64 (4.4)	77 (4.7)	0.3	6 (6.8)	6 (6.8)	0.0	70 (4.5)	83 (4.8)	0.3
Blood creatinine increased	50 (3.4)	72 (4.4)	1.0	5 (5.7)	5 (5.7)	0.0	55 (3.6)	77 (4.4)	0.8
Weight increased	58 (4.0)	68 (4.1)	0.1	3 (3.4)	3 (3.4)	0.0	61 (4.0)	71 (4.1)	0.1
Blood alkaline phosphatase increased	53 (3.7)	66 (4.0)	0.3	0 (0.0)	1 (1.1)	1.1	53 (3.4)	67 (3.9)	0.5
Metabolism and nutrition disorders	531 (36.6)	697 (42.2)	5.6	30 (34.1)	31 (35.2)	1.1	561 (36.4)	728 (41.9)	5.5
Decreased appetite	232 (16.0)	302 (18.3)	2.3	17 (19.3)	18 (20.5)	1.2	249 (16.2)	320 (18.4)	2.2
Dehydration	86 (5.9)	118 (7.2)	1.3	4 (4.5)	4 (4.5)	0.0	90 (5.8)	122 (7.0)	1.2
Hyponatraemia	76 (5.2)	110 (6.7)	1.5	5 (5.7)	5 (5.7)	0.0	81 (5.3)	115 (6.6)	1.3
Hypokalaemia	83 (5.7)	109 (6.6)	0.9	3 (3.4)	3 (3.4)	0.0	86 (5.6)	112 (6.4)	0.8
Hypomagnesaemia	61 (4.2)	83 (5.0)	0.8	2 (2.3)	2 (2.3)	0.0	63 (4.1)	85 (4.9)	0.8
Hypoalbuminaemia	50 (3.4)	75 (4.5)	1.1	3 (3.4)	3 (3.4)	0.0	53 (3.4)	78 (4.5)	1.1
Hypophosphataemia	42 (2.9)	60 (3.6)	0.7	2 (2.3)	2 (2.3)	0.0	44 (2.9)	62 (3.6)	0.7
Hyperglycaemia	47 (3.2)	58 (3.5)	0.3	1 (1.1)	1 (1.1)	0.0	48 (3.1)	59 (3.4)	0.3
Musculoskeletal and connective tissue disorders	523 (36.0)	664 (40.2)	4.2	43 (48.9)	45 (51.1)	2.2	566 (36.8)	709 (40.8)	4.0
Back pain	154 (10.6)	196 (11.9)	1.3	9 (10.2)	9 (10.2)	0.0	163 (10.6)	205 (11.8)	1.2
Arthralgia	135 (9.3)	166 (10.1)	0.8	14 (15.9)	14 (15.9)	0.0	149 (9.7)	180 (10.4)	0.7
Pain in extremity	69 (4.8)	98 (5.9)	1.1	13 (14.8)	14 (15.9)	1.1	82 (5.3)	112 (6.4)	1.1
Musculoskeletal pain	74 (5.1)	97 (5.9)	0.8	2 (2.3)	2 (2.3)	0.0	76 (4.9)	99 (5.7)	0.8
Myalgia	71 (4.9)	90 (5.5)	0.6	4 (4.5)	4 (4.5)	0.0	75 (4.9)	94 (5.4)	0.5
Musculoskeletal chest pain	53 (3.7)	72 (4.4)	0.7	0 (0.0)	0 (0.0)	0.0	53 (3.4)	72 (4.1)	0.7
Muscle spasms	41 (2.8)	51 (3.1)	0.3	6 (6.8)	6 (6.8)	0.0	47 (3.1)	57 (3.3)	0.2
Neck pain	44 (3.0)	50 (3.0)	0.0	1 (1.1)	2 (2.3)	1.2	45 (2.9)	52 (3.0)	0.1
Nervous system disorders	382 (26.3)	489 (29.6)	3.3	31 (35.2)	32 (36.4)	1.2	413 (26.8)	521 (30.0)	3.2
Headache	123 (8.5)	151 (9.2)	0.7	8 (9.1)	9 (10.2)	1.1	131 (8.5)	160 (9.2)	0.7
Dizziness	82 (5.6)	117 (7.1)	1.5	11 (12.5)	12 (13.6)	1.1	93 (6.0)	129 (7.4)	1.4

Body System or Organ Class Preferred term	001 SCS (N=1452) n (%)	 SUR (N=1650) n (%)	 Delta (%)	003 SCS (N=88) n (%)	 SUR (N=88) n (%)	 Delta (%)	Total SCS (N=1540) n (%)	 SUR (N=1738) n (%)	 Delta (%)
Psychiatric disorders	205 (14.1)	295 (17.9)	3.8	20 (22.7)	21 (23.9)	1.2	225 (14.6)	316 (18.2)	3.6
Insomnia	77 (5.3)	112 (6.8)	1.5	6 (6.8)	6 (6.8)	0.0	83 (5.4)	118 (6.8)	1.4
Anxiety	73 (5.0)	105 (6.4)	1.4	5 (5.7)	5 (5.7)	0.0	78 (5.1)	110 (6.3)	1.2
Respiratory, thoracic and mediastinal disorders	565 (38.9)	733 (44.4)	5.5	30 (34.1)	34 (38.6)	4.5	595 (38.6)	767 (44.1)	5.5
Cough	168 (11.6)	224 (13.6)	2.0	15 (17.0)	16 (18.2)	1.2	183 (11.9)	240 (13.8)	1.9
Dyspnoea	163 (11.2)	221 (13.4)	2.2	5 (5.7)	8 (9.1)	3.4	168 (10.9)	229 (13.2)	2.3
Dyspnoea exertional	99 (6.8)	109 (6.6)	-0.2	3 (3.4)	3 (3.4)	0.0	102 (6.6)	112 (6.4)	-0.2
Pleural effusion	66 (4.5)	79 (4.8)	0.3	3 (3.4)	3 (3.4)	0.0	69 (4.5)	82 (4.7)	0.2
Productive cough	41 (2.8)	61 (3.7)	0.9	1 (1.1)	1 (1.1)	0.0	42 (2.7)	62 (3.6)	0.9
Oropharyngeal pain	41 (2.8)	54 (3.3)	0.5	3 (3.4)	3 (3.4)	0.0	44 (2.9)	57 (3.3)	0.4
Haemoptysis	39 (2.7)	52 (3.2)	0.5	0 (0.0)	0 (0.0)	0.0	39 (2.5)	52 (3.0)	0.5
Skin and subcutaneous tissue disorders	334 (23.0)	451 (27.3)	4.3	35 (39.8)	35 (39.8)	0.0	369 (24.0)	486 (28.0)	4.0
Pruritus	86 (5.9)	119 (7.2)	1.3	8 (9.1)	9 (10.2)	1.1	94 (6.1)	128 (7.4)	1.3
Rash	77 (5.3)	112 (6.8)	1.5	10 (11.4)	12 (13.6)	2.2	87 (5.6)	124 (7.1)	1.5
Dry skin	33 (2.3)	61 (3.7)	1.4	4 (4.5)	5 (5.7)	1.2	37 (2.4)	66 (3.8)	1.4
Rash maculo-papular	42 (2.9)	48 (2.9)	0.0	5 (5.7)	5 (5.7)	0.0	47 (3.1)	53 (3.0)	-0.1
Vascular disorders	227 (15.6)	313 (19.0)	3.4	23 (26.1)	24 (27.3)	1.2	250 (16.2)	337 (19.4)	3.2
Hypertension	103 (7.1)	155 (9.4)	2.3	11 (12.5)	11 (12.5)	0.0	114 (7.4)	166 (9.6)	2.2
Hypotension	44 (3.0)	57 (3.5)	0.5	4 (4.5)	4 (4.5)	0.0	48 (3.1)	61 (3.5)	0.4

Source: SUR Table 12.6.20.2.1
Events are coded in the latest MedDRA versions available at SCS and SUR cutoff times respectively
Delta is calculated as difference in displayed percentage between SCS and SUR
[a] Numbers (percentages) refer to the preferred term (PT) of infusion related reaction only, and differ from the numbers (percentages) for the composite of PTs reported elsewhere in this document.
Note: Reasons for negative Delta: the number of subjects with an AE stayed the same but the number of overall subjects increased or the number of subjects with an AE changed in the database by investigator's re-assessment leading to a lower number of subjects with that specific AE.

Immune related adverse events

Table 44: Immune-Related AEs – All Categories and All Grades cut-off 9th June 2016

	001			003			Total		
	SCS (N=1452) n (%)	SUR (N=1650) n (%)	Delta (%)	SCS (N=88) n (%)	SUR (N=88) n (%)	Delta (%)	SCS (N=1540) n (%)	SUR (N=1738) n (%)	Delta (%)
Number of Subjects with at least one Event	165 (11.4)	232 (14.1)	2.7	14 (15.9)	15 (17.0)	1.1	179 (11.6)	247 (14.2)	2.6
Hypothyroidism	55 (3.8)	84 (5.1)	1.3	4 (4.5)	4 (4.5)	0.0	59 (3.8)	88 (5.1)	1.3
Rash	20 (1.4)	35 (2.1)	0.7	3 (3.4)	5 (5.7)	2.3	23 (1.5)	40 (2.3)	0.8
Pruritus	22 (1.5)	25 (1.5)	0.0	1 (1.1)	1 (1.1)	0.0	23 (1.5)	26 (1.5)	0.0
Rash maculo-papular	20 (1.4)	19 (1.2)	-0.2	1 (1.1)	1 (1.1)	0.0	21 (1.4)	20 (1.2)	-0.2
Diarrhoea	11 (0.8)	19 (1.2)	0.4	2 (2.3)	2 (2.3)	0.0	13 (0.8)	21 (1.2)	0.4
Pneumonitis	12 (0.8)	21 (1.3)	0.5	0 (0.0)	0 (0.0)	0.0	12 (0.8)	21 (1.2)	0.4
Aspartate aminotransferase increased	9 (0.6)	9 (0.5)	-0.1	0 (0.0)	1 (1.1)	1.1	9 (0.6)	10 (0.6)	0.0
Alanine aminotransferase increased	9 (0.6)	8 (0.5)	-0.1	0 (0.0)	1 (1.1)	1.1	9 (0.6)	9 (0.5)	-0.1
Hyperthyroidism	7 (0.5)	6 (0.4)	-0.1	2 (2.3)	1 (1.1)	-1.2	9 (0.6)	7 (0.4)	-0.2
Adrenal insufficiency	6 (0.4)	8 (0.5)	0.1	0 (0.0)	0 (0.0)	0.0	6 (0.4)	8 (0.5)	0.1
Rash pruritic	5 (0.3)	7 (0.4)	0.1	0 (0.0)	0 (0.0)	0.0	5 (0.3)	7 (0.4)	0.1
Erythema	4 (0.3)	4 (0.2)	-0.1	1 (1.1)	1 (1.1)	0.0	5 (0.3)	5 (0.3)	0.0
Rash generalised	5 (0.3)	5 (0.3)	0.0	0 (0.0)	0 (0.0)	0.0	5 (0.3)	5 (0.3)	0.0
Autoimmune hepatitis	4 (0.3)	5 (0.3)	0.0	0 (0.0)	0 (0.0)	0.0	4 (0.3)	5 (0.3)	0.0
Myositis	4 (0.3)	4 (0.2)	-0.1	0 (0.0)	0 (0.0)	0.0	4 (0.3)	4 (0.2)	-0.1
Autoimmune disorder	2 (0.1)	5 (0.3)	0.2	0 (0.0)	0 (0.0)	0.0	2 (0.1)	5 (0.3)	0.2
Rash erythematous	3 (0.2)	4 (0.2)	0.0	0 (0.0)	0 (0.0)	0.0	3 (0.2)	4 (0.2)	0.0

Source: SUR Table 12.7.6.1.0
SCS: Summary of Clinical Safety; SUR: Safety Update Reporting
Delta is calculated as difference in displayed percentage between SCS and SUR.
[a] PTs of Blood creatinephosphokinase increased, hyperglycaemia and diabetes mellitus was newly added with this SUR (updated irAE definition for analysis, Version 2).
Note: Reasons for negative Delta: the number of subjects with an AE stayed the same but the number of overall subjects increased or the number of subjects with an AE changed in the database by Investigator's re-assessment leading to a lower number of subjects with that specific AE.

Table 45: Immune-Related AEs – All Categories Grade≥ 3 cut-off 9th June 2016

	001			003			Total		
	SCS (N=1452) n (%)	SUR (N=1650) n (%)	Delta (%)	SCS (N=88) n (%)	SUR (N=88) n (%)	Delta (%)	SCS (N=1540) n (%)	SUR (N=1738) n (%)	Delta (%)
Number of Subjects with at least one Event	23 (1.6)	36 (2.2)	0.6	1 (1.1)	3 (3.4)	2.3	24 (1.6)	39 (2.2)	0.6
Pneumonitis	4 (0.3)	7 (0.4)	0.1	0 (0.0)	0 (0.0)	0.0	4 (0.3)	7 (0.4)	0.1
Autoimmune hepatitis	4 (0.3)	4 (0.2)	-0.1	0 (0.0)	0 (0.0)	0.0	4 (0.3)	4 (0.2)	-0.1
Alanine aminotransferase increased	2 (0.1)	3 (0.2)	0.1	0 (0.0)	1 (1.1)	1.1	2 (0.1)	4 (0.2)	0.1
Aspartate aminotransferase increased	3 (0.2)	3 (0.2)	0.0	0 (0.0)	0 (0.0)	0.0	3 (0.2)	3 (0.2)	0.0
Autoimmune disorder	2 (0.1)	3 (0.2)	0.1	0 (0.0)	0 (0.0)	0.0	2 (0.1)	3 (0.2)	0.1
Colitis	2 (0.1)	3 (0.2)	0.1	0 (0.0)	0 (0.0)	0.0	2 (0.1)	3 (0.2)	0.1
Diarrhoea	1 (0.1)	4 (0.2)	0.1	0 (0.0)	0 (0.0)	0.0	1 (0.1)	4 (0.2)	0.1
Hypothyroidism	1 (0.1)	2 (0.1)	0.0	0 (0.0)	1 (1.1)	1.1	1 (0.1)	3 (0.2)	0.1

Table 46: Summary Statistics of Time to First Onset of Any Immune-related Adverse Events (Any Grade) Up to 90 Days Extended Safety Period – Overall cut-off 9th June 2016

	Statistics	001 (N=1650)	003 (N=88)	Total (N=1738)
Time to first onset of event (weeks)	n (%)	235 (14.2)	15 (17.0)	250 (14.4)
	Mean	14.65	11.02	14.43
	Median	11.86	9.14	11.71
	Q1; Q3	5.29; 18.29	5.57; 18.43	5.43; 18.29
	Min; Max	0.1; 101.1	0.1; 26.1	0.1; 101.1

Source: SUR Table 12.7.1.6.0
Statistics do not account for competing events and provide a simple description of the observations in the trial.
Includes events until end of expanded safety period (90 days after last treatment).

Immune-related Adverse Events by severity

In the overall population, 14.2% of the patients experienced immune-related AEs (irAEs) (17% in Study 003), the most common events being hypothyroidism and rash, along with an overall low incidence of Grade ≥ 3 irAEs (2.2%). Grade 4 irAEs were reported only in Study 001. There were 3 subjects with irAE with fatal outcomes (acute hepatic failure, pneumonitis, hepatic failure and autoimmune hepatitis), none of them new in the updated safety analysis. However, new irAEs have been observed (type I diabetes mellitus, myositis), which have been added to the SmPC. It is noted that 5 subjects (1 subject in SCS) experienced immune-related psoriasis, 4 cases were grade 2 and 1 case Grade 3. Three cases were considered related to avelumab by the Investigator. In addition, one case of Grade 3 systemic inflammatory response syndrome (SIRS) have been reported among the immune-related AEs and considered related to avelumab by investigator. There were 2 events recorded as immune-related rheumatoid arthritis. However, review of both subjects with rheumatoid arthritis by a rheumatologist led to the conclusion that the formal criteria for diagnosing rheumatoid arthritis were not met, as in both subjects, no information about joint stiffness, symmetry of involved joints, and the specific joints involved was available. This is noted.

Table 47: Summary of immune-related adverse events by disease subcategory

	EMR100070-001				EMR100070-003			
	SCS N = 1452		SUR N = 1650		SCS N = 88		SUR N = 88	
	Any Grade (%)	Grade ≥ 3 (%)	Any Grade (%)	Grade ≥ 3 (%)	Any Grade (%)	Grade ≥ 3 (%)	Any Grade (%)	Grade ≥ 3 (%)
All irAEs	11.4	1.6	14.1	2.2	15.9	1.1	17.0	3.4
Immune-related pneumonitis	0.8	0.3	1.3	0.4	0	0	0	0
Immune-related colitis	0.9	0.2	1.5	0.4	2.3	0	2.3	0
Immune-related hepatitis	0.9	0.6	0.8	0.7	1.1	1.1	2.3	2.3
Immune-related endocrinopathies	4.8	0.1	6.1	0.3	5.7	0	5.7	1.1
Thyroid disorder	4.5	0.1	5.6	0.1	5.7	0	5.7	1.1
Hypothyroidism events[a]	3.9	0.1	5.2	0.1	4.5	0	4.5	1.1
Hyperthyroidism events[a]	0.5	0	0.4	0	2.3	0	1.1	0
Thyroiditis events[a]	0.3	0	0.2	0	0	0	0	0
Adrenal insufficiency	0.4	0.1	0.5	0.1	0	0	0	0
Type I diabetes mellitus	0	0	0.1	0.1	0	0	0	0
Pituitary dysfunction	0.1	0	0.1	0	0	0	0	0
Hypogonadism	0	0	0	0	0	0	0	0
Immune-related nephritis and renal dysfunction	0.1	0	0	0	1.1	0	1.1	0
Immune-related rash (including pruritis)	4.3	0.1	5.0	0.1	5.7	0	8.0	0
All other immune-related AEs	0.8	0.3	1.2	0.5	0	0	0	0
Myositis	0	0	0.5	0.3	0	0	0	0
Myocarditis[b]	0	0	0	0	0	0	0	0
Other immune-mediated AEs	0.8	0.3	0.6	0.2	0	0	0	0

Source: SUR Tables 12.7.6.1.0, 12.7.6.1.1, 12.7.6.1.2, 12.7.6.1.3, 12.7.6.1.4, 12.7.6.1.5, 12.7.6.1.6.0, 12.7.6.1.6.1.0, 12.7.6.1.6.1.1, 12.7.6.1.6.1.2, 12.7.6.1.6.1.3, 12.7.6.1.6.2, 12.7.6.1.6.3, 12.7.6.1.6.4, 12.7.6.1.6.5, 12.7.6.1.7.0, 12.7.6.1.7.1, 12.7.6.1.7.2, 12.7.6.2.0, 12.7.6.2.1, 12.7.6.2.2, 12.7.6.2.3, 12.7.6.2.4, 12.7.6.2.5, 12.7.6.2.6.0, 12.7.6.2.6.1.0, 12.7.6.2.6.1.1, 12.7.6.2.6.1.2, 12.7.6.2.6.1.3, 12.7.6.2.6.2, 12.7.6.2.6.3, 12.7.6.2.6.4, 12.7.6.2.6.5, 12.7.6.2.7.0, 12.7.6.2.7.1, 12.7.6.2.7.2

irAE: immune-related adverse event; SCS: Summary of Clinical Safety; SUR: Safety Update Reporting

[a] Summary terms were not included in initial submission. Numbers were regenerated post-hoc based on the data cutoff of the SCS.

[b] Myocarditis was classified as an identified risk based on 1 event reported in the 20 mg/kg cohort of Study EMR100070-001 and 1 report of myocarditis reported in Study B9991002 (avelumab in combination with axitinib).

Infusion-related Reactions (IRR)

Table 48: Overview of Infusion-related Reactions

	001 (N=1452)	003 (N=88)	Total (N=1540)
	n (%)	n (%)	n (%)
Number of Subjects with at least one IRR (All Grades)	352 (24.2)	19 (21.6)	371 (24.1)
Infusion related reaction	245 (16.9)	13 (14.8)	258 (16.8)
Chills	74 (5.1)	2 (2.3)	76 (4.9)
Pyrexia	50 (3.4)	2 (2.3)	52 (3.4)
Back pain	7 (0.5)	1 (1.1)	8 (0.5)
Dyspnoea	6 (0.4)	0 (0.0)	6 (0.4)
Hypotension	5 (0.3)	1 (1.1)	6 (0.4)
Drug hypersensitivity	4 (0.3)	1 (1.1)	5 (0.3)
Abdominal pain	4 (0.3)	0 (0.0)	4 (0.3)
Flushing	4 (0.3)	0 (0.0)	4 (0.3)
Hypersensitivity	2 (0.1)	1 (1.1)	3 (0.2)
Anaphylactic reaction	1 (0.1)	0 (0.0)	1 (0.1)
Type I hypersensitivity	1 (0.1)	0 (0.0)	1 (0.1)
Number of Subjects with at least one IRR Grade ≥ 4	3 (0.2)	0 (0.0)	3 (0.2)
Infusion related reaction	3 (0.2)	0 (0.0)	3 (0.2)
Number of Subjects with at least one IRR Grade ≥ 3	14 (1.0)	0 (0.0)	14 (0.9)
Infusion related reaction	10 (0.7)	0 (0.0)	10 (0.6)
Dyspnoea	2 (0.1)	0 (0.0)	2 (0.1)
Anaphylactic reaction	1 (0.1)	0 (0.0)	1 (0.1)
Chills	1 (0.1)	0 (0.0)	1 (0.1)
Number of Subjects with at least one treatment-related IRR (All Grades)	336 (23.1)	19 (21.6)	355 (23.1)
Number of Subjects with at least one treatment-related IRR Grade ≥ 3	14 (1.0)	0 (0.0)	14 (1.0)
Number of Subjects with at least one Serious IRR (All Grades)	18 (1.2)	1 (1.1)	19 (1.2)
Time to First Onset of IRRs – (All Grades) Median (Range) (weeks)	0.14 (0.1; 26.3)	0.14 (0.1; 2.1)	0.14 (0.1; 26.3)
Number of Subjects with at least one Interruption Due to IRRs (All Grades)	109 (7.5)	8 (9.1)	117 (7.6)
Number of Subjects with at least one Infusion Rate Reduction Due to IRR (All Grades)	96 (6.6)	9 (10.2)	105 (6.8)
Number of Subjects Who Discontinued Permanently Due to IRR (All Grades)	27 (1.9)	0 (0.0)	27 (1.8)

Source: SCS Tables 12.7.2.1.0, 12.7.2.2.0, 12.7.2.3.0, 12.7.2.4.0, 12.7.2.5.0, 12.7.2.6, 12.7.4.1, and 12.7.4.7

Table 49: Time to Onset of First Infusion-related Reactions Related to Number of Infusions - updated

Number of Subjects with IRRs at Infusion	001 (N=1650)		003 (N=88)		Total (N=1738)	
	At risk	n (%)	At risk	n (%)	At risk	n (%)
Infusion 1	1650	333 (20.2)	88	16 (18.2)	1738	349 (20.1)
Infusion 2	1240	58 (4.7)	66	3 (4.5)	1306	61 (4.7)
Infusion 3	1084	17 (1.6)	60	0 (0.0)	1144	17 (1.5)
Infusion 4	885	6 (0.7)	52	0 (0.0)	937	6 (0.6)
Infusion 5 or later	794	6 (0.8)	47	0 (0.0)	841	6 (0.7)

Source: SUR Table 12.7.2.10
IRR: infusion-related reaction

Table 50: Infusion-Related Reactions and Premedication – Expansion Cohorts Safety Analysis Set – Study EMR100070-001

	Without Premedication[a] Expansion Cohorts (N=142) n (%)	With Premedication[a] Expansion Cohorts (N=1395) n (%)
Number of subjects with ≥ 1 event[b]	25 (17.6)	229 (16.4)
Grade 1	4 (2.8)	52 (3.7)
Grade 2	18 (12.7)	169 (12.1)
Grade 3	2 (1.4)	6 (0.4)
Grade 4	1 (0.7)	2 (0.1)
Grade 5	0	0
Missing	0	0

Source: Refer to Study EMR100070-001 Clinical Study Report Table 15.3.1.15w.

[a] Prior to a letter sent to Investigators, IRBs, and IECs on 29 January 2014, except for subjects who had previously experienced an infusion-related reaction, premedication with an antihistamine and with paracetamol was optional. Upon receipt of the letter, premedication as outlined became mandatory. Premedication was formalized in the protocol with Protocol Amendment 7 (Protocol Version 8.0; dated 30 July 2014).

[b] The same subject may have received study drug with and without premedication.

Among patients who had IRRs (439), 63 (14.4%) experienced one or more subsequent IRR (46 one further episode only, 14 two further episodes and 3 three or more further episodes of IRR).

Adverse Drug Reactions

Based on guidance documents, criteria were established and clinical review was conducted to identify relevant adverse drug reactions (ADRs) for inclusion in Table 2 of section 4.8 of the SmPC:

- In the absence of control groups in studies EMR 100070-001 and EMR 100070-003, all Preferred Terms of AEs in these studies regardless of causality assessment by Investigators reported at a frequency of ≥ 10% of subjects in the pooled safety dataset of 1738 subjects treated with avelumab in various solid tumours were included.

- In addition, all Preferred Terms of events that, following medical evaluation, met the pre-specified criteria for classification of immune-related adverse events or an infusion-related reaction with a timely relationship as per the definitions provided in updated Module 2.7.4, section 2.1.8 were included. Note: All IRR diagnoses (irrespective of incidence) and any IRR symptoms occurring in ≥ 3% subjects were to be included in the product information.

As per D120 List of Question comment on the SmPC, the ADR table in section 4.8 was revised to reflect the pooled safety dataset of studies EMR 100070-001 and EMR 100070-003 (data cut-off date: 09Jun2016). As data from controlled studies are not available, a table including all ADRs and matching Table 2 of section 4.8 of the SmPC by severity is included below (see updated SmPC, Module 1.3.1, section 4.8, Table 2). This table also provides frequencies of respective events assessed as related by investigator.

Table 51: Expected adverse reactions in patients treated with avelumab in clinical studies

Frequency	MedDRA SOC and PT	Avelumab (N = 1738)		
		All Grades n (%)	Grade ≥ 3 n (%)	Serious n (%)
Blood and lymphatic system disorder				
Very common	Anaemia	259 (14.9)	104 (6.0)	24 (1.4)
Common	Lymphopenia	27 (1.6)	14 (0.8)	0
Uncommon	Thrombocytopenia	12 (0.7)	4 (0.2)	2 (0.1)
Uncommon	Eosinophilia§	0	0	0
Immune system disorders				
Uncommon	Drug hypersensitivity	8 (0.5)	0	0
Uncommon	Hypersensitivity	6 (0.3)	0	0

Frequency	MedDRA SOC and PT	Avelumab (N = 1738)		
		All Grades n (%)	Grade ≥ 3 n (%)	Serious n (%)
Uncommon	Anaphylactic reaction	2 (0.1)	2 (0.1)	2 (0.1)
Uncommon	Type I hypersensitivity	1 (0.1)	0	1 (0.1)
Endocrine disorders				
Common	Hypothyroidism*	88 (5.1)	3 (0.2)	4 (0.2)
Uncommon	Adrenal insufficiency*	8 (0.5)	1 (0.1)	2 (0.1)
Uncommon	Hyperthyroidism*	7 (0.4)	0	1 (0.1)
Uncommon	Thyroiditis*	2 (0.1)	0	1 (0.1)
Uncommon	Autoimmune thyroiditis*	2 (0.1)	0	0
Uncommon	Autoimmune hypothyroidism*	2 (0.1)	0	0
Uncommon	Adrenocortical insufficiency acute*	1 (0.1)	0	1 (0.1)
Uncommon	Hypopituitarism*	1 (0.1)	0	0
Metabolism and nutrition disorders				
Very common	Decreased appetite	320 (18.4)	19 (1.1)	4 (0.2)
Uncommon	Diabetes mellitus*	1 (0.1)	1 (0.1)	1 (0.1)
Uncommon	Type 1 diabetes mellitus	1 (0.1)	0	0
Nervous system disorders				
Common	Headache	160 (9.2)	7 (0.4)	3 (0.2)
Common	Dizziness	129 (7.4)	2 (0.1)	2 (0.1)
Common	Neuropathy peripheral	22 (1.3)	1 (0.1)	1 (0.1)
Uncommon	Guillian-Barré syndrome*	1 (0.1)	1 (0.1)	1 (0.1)
Eye disorder				
Uncommon	Uveitis*	1 (0.1)	0	0
Cardiac disorder				
Rare	Myocarditis	1 (0.1)	0	1 (0.1)
Vascular disorders				
Common	Hypertension	166 (9.6)	75 (4.3)	2 (0.1)
Common	Hypotension	61 (3.5)	17 (1.0)	12 (0.7)
Uncommon	Flushing	15 (0.9)	0	1 (0.1)
Respiratory, thoracic and mediastinal disorders				
Very common	Cough	240 (13.8)	2 (0.1)	1 (0.1)
Very common	Dyspnea	229 (13.2)	68 (3.9)	48 (2.8)
Common	Pneumonitis*	21 (1.2)	7 (0.4)	10 (0.6)
Gastrointestinal disorders				
Very common	Nausea	437 (25.1)	27 (1.6)	22 (1.3)
Very common	Diarrhea	329 (18.9)	22 (1.3)	12 (0.7)
Very common	Constipation	320 (18.4)	17 (1.0)	12 (0.7)
Very common	Vomiting	281 (16.2)	31 (1.8)	27 (1.6)
Very common	Abdominal pain	250 (14.4)	52 (3.0)	41 (2.4)
Common	Dry mouth	67 (3.9)	0	0
Uncommon	Ileus	9 (0.5)	5 (0.3)	6 (0.3)
Uncommon	Colitis*	5 (0.3)	4 (0.2)	4 (0.2)
Uncommon	Autoimmune colitis*	1 (0.1)	0	0
Uncommon	Enterocolitis*	1 (0.1)	0	0
Hepatobiliary disorders				
Uncommon	Autoimmune hepatitis*	5 (0.3)	4 (0.2)	3 (0.2)
Uncommon	Acute hepatic failure*	1 (0.1)	1 (0.1)	1 (0.1)
Uncommon	Hepatic failure*	1 (0.1)	1 (0.1)	1 (0.1)
Uncommon	Hepatitis*	1 (0.1)	1 (0.1)	1 (0.1)
Skin and subcutaneous tissue disorders				
Common	Dry skin	66 (3.8)	0	0
Common	Rash*	40 (2.3)	1 (0.1)	0
Common	Pruritus*	26 (1.5)	0	0
Common	Rash maculo-papular*	20 (1.2)	0	0
Uncommon	Eczema	15 (0.9)	0	0
Uncommon	Dermatitis	9 (0.5)	0	0
Uncommon	Rash pruritic*	7 (0.4)	0	0
Uncommon	Erythema*	5 (0.3)	0	0
Uncommon	Rash generalised*	5 (0.3)	0	1 (0.1)
Uncommon	Psoriasis*	5 (0.3)	1 (0.1)	1 (0.1)

Frequency	MedDRA SOC and PT	Avelumab (N = 1738)		
		All Grades n (%)	Grade ≥ 3 n (%)	Serious n (%)
Uncommon	Rash erythematous*	4 (0.2)	0	0
Uncommon	Rash macular*	3 (0.2)	0	0
Uncommon	Rash papular*	2 (0.1)	0	0
Uncommon	Dermatitis exfoliative*	1 (0.1)	0	0
Uncommon	Erythema multiforme*	1 (0.1)	0	0
Uncommon	Pemphigoid*	1 (0.1)	0	0
Uncommon	Pruritus generalised*	1 (0.1)	0	0
Musculoskeletal and connective tissue disorders				
Very common	Back pain	205 (11.8)	24 (1.4)	14 (0.8)
Very common	Arthralgia	180 (10.4)	18 (1.0)	4 (0.2)
Common	Myalgia	94 (5.4)	1 (0.1)	1 (0.1)
Uncommon	Myositis*	5 (0.3)	2 (0.1)	2 (0.1)
Renal and urinary disorders				
Uncommon	Tubulointerstitial nephritis*	1 (0.1)	0	1 (0.1)
General disorders and administration site conditions				
Very common	Fatigue	563 (32.4)	51 (2.9)	10 (0.6)
Very common	Pyrexia	237 (13.6)	5 (0.3)	22 (1.3)
Very common	Oedema peripheral	206 (11.9)	8 (0.5)	3 (0.2)
Common	Chills	169 (9.7)	1 (0.1)	1 (0.1)
Common	Asthenia	151 (8.7)	29 (1.7)	18 (1.0)
Common	Influenza like illness	80 (4.6)	1 (0.1)	2 (0.1)
Uncommon	Systemic inflammatory response syndrome*	1 (0.1)	1 (0.1)	2 (0.1)
Investigations				
Very common	Weight decreased	288 (16.6)	12 (0.7)	2 (0.1)
Common	Blood creatinine increased	77 (4.4)	3 (0.2)	0
Common	Blood alkaline phosphatase increased	67 (3.9)	24 (1.4)	0
Common	Gamma-glutamyltransferase increased	51 (2.9)	32 (1.8)	1 (0.1)
Common	Lipase increased	45 (2.6)	29 (1.7)	2 (0.1)
Common	Amylase increased	33 (1.9)	10 (0.6)	1 (0.1)
Uncommon	Aspartate aminotransferase (AST) increased*	10 (0.6)	3 (0.2)	1 (0.1)
Uncommon	Alanine aminotransferase (ALT) increased*	9 (0.5)	4 (0.2)	0
Uncommon	Blood creatine phosphokinase increased*	5 (0.3)	3 (0.2)	2 (0.1)
Uncommon	Transaminases increased*	2 (0.1)	2 (0.1)	2 (0.1)
Injury, Poisoning and Procedural Complications				
Very common	Infusion related reaction	297 (17.1)	10 (0.6)	15 (0.9)

MedDRA: Medical Dictionary for Regulatory Activities; n: number; PT: preferred term; SOC: System Organ and Class.
* Immune-related adverse reaction based on medical review
§Reaction only observed from study EMR 100070-003 (part B) after the data cut-off of the pooled analysis, hence frequency as provided in the SmPC estimated.

In a safety update (data cut-off date 09 June 2016), distribution of events of Grades ≥ 3 in the combined **001** + **003** study population were presented for a total of 1738 patients. Among Grade 3 events (N=650, 37 %), anaemia was the most frequently reported (5.8%), followed by hypertension (4.3%), hyponatremia (3.7%) and dyspnoea (3.5%). Other TEAEs with an incidence of ≥ 2 % included abdominal pain (2.9%), fatigue (2.9%), and pneumonia (2.0%). Grade 4 events constituted 7.5 % (N=130) with sepsis was the most frequently reported (1.3%), followed by lipase increased (0.6%) and respiratory failure (0.6%). Other TEAEs with at least 5 subjects reporting included hypokalaemia and dyspnoea (each 0.5%), acute kidney injury (0.3%), and hyponatremia (0.3%). Grade 5 events (13 %) were in the vast majority due to disease progression (8.4%) and 0.5% died due to respiratory failure. Other fatal events that occurred in at least 5 subjects included pneumonia (0.3%) and sepsis (0.3%).

Serious adverse event/deaths/other significant events

Table 52: Most Common Serious TEAEs (at Least 2 Subjects in the Total Group) by SOC and PT – All Subjects

Body System or Organ Class Preferred term	001 Subjects (N=1452) n (%)	003 Subjects (N=88) n (%)	Total Subjects (N=1540) n (%)
Number of Subjects With At Least One Event	578 (39.8)	36 (40.9)	614 (39.9)
Blood and lymphatic system disorders	25 (1.7)	5 (5.7)	30 (1.9)
Anaemia	17 (1.2)	3 (3.4)	20 (1.3)
Cardiac disorders	31 (2.1)	3 (3.4)	34 (2.2)
Atrial fibrillation	8 (0.6)	0 (0.0)	8 (0.5)
Cardiac tamponade	4 (0.3)	0 (0.0)	4 (0.3)
Myocardial infarction	3 (0.2)	0 (0.0)	3 (0.2)
Endocrine disorders	13 (0.9)	0 (0.0)	13 (0.8)
Adrenal insufficiency	6 (0.4)	0 (0.0)	6 (0.4)
Hypothyroidism	3 (0.2)	0 (0.0)	3 (0.2)
Gastrointestinal disorders	147 (10.1)	6 (6.8)	153 (9.9)
Abdominal pain	33 (2.3)	2 (2.3)	35 (2.3)
Vomiting	26 (1.8)	0 (0.0)	26 (1.7)
Nausea	22 (1.5)	0 (0.0)	22 (1.4)
Ascites	17 (1.2)	0 (0.0)	17 (1.1)
Small intestinal obstruction	17 (1.2)	0 (0.0)	17 (1.1)
Dysphagia	11 (0.8)	0 (0.0)	11 (0.7)
Gastrointestinal haemorrhage	9 (0.6)	1 (1.1)	10 (0.6)
Constipation	9 (0.6)	0 (0.0)	9 (0.6)
Diarrhoea	8 (0.6)	0 (0.0)	8 (0.5)
General disorders and administration site conditions	174 (12.0)	11 (12.5)	185 (12.0)
Disease progression	120 (8.3)	4 (4.5)	124 (8.1)
Pyrexia	18 (1.2)	0 (0.0)	18 (1.2)
Asthenia	15 (1.0)	2 (2.3)	17 (1.1)
Hepatobiliary disorders	16 (1.1)	2 (2.3)	18 (1.2)
Hepatic failure	3 (0.2)	1 (1.1)	4 (0.3)
Autoimmune hepatitis	3 (0.2)	0 (0.0)	3 (0.2)
Immune system disorders	6 (0.4)	0 (0.0)	6 (0.4)
Autoimmune disorder	3 (0.2)	0 (0.0)	3 (0.2)
Infections and infestations	107 (7.4)	7 (8.0)	114 (7.4)
Pneumonia	32 (2.2)	1 (1.1)	33 (2.1)
Sepsis	18 (1.2)	0 (0.0)	18 (1.2)
Cellulitis	7 (0.5)	2 (2.3)	9 (0.6)
Urinary tract infection	8 (0.6)	1 (1.1)	9 (0.6)
Injury, poisoning and procedural complications	40 (2.8)	1 (1.1)	41 (2.7)
Infusion related reaction	13 (0.9)	1 (1.1)	14 (0.9)
Investigations	22 (1.5)	1 (1.1)	23 (1.5)
Blood bilirubin increased	5 (0.3)	0 (0.0)	5 (0.3)
Transaminases increased	4 (0.3)	1 (1.1)	5 (0.3)
Metabolism and nutrition disorders	41 (2.8)	2 (2.3)	43 (2.8)
Musculoskeletal and connective tissue disorders	34 (2.3)	4 (4.5)	38 (2.5)
Back pain	11 (0.8)	0 (0.0)	11 (0.7)
Arthralgia	4 (0.3)	0 (0.0)	4 (0.3)
Neoplasms benign, malignant and unspecified (incl cysts and polyps)	34 (2.3)	5 (5.7)	39 (2.5)
Tumour pain	5 (0.3)	0 (0.0)	5 (0.3)
Nervous system disorders	42 (2.9)	2 (2.3)	44 (2.9)
Cerebrovascular accident	4 (0.3)	0 (0.0)	4 (0.3)
Seizure	4 (0.3)	0 (0.0)	4 (0.3)

Psychiatric disorders	11 (0.8)	3 (3.4)	14 (0.9)
Renal and urinary disorders	19 (1.3)	5 (5.7)	24 (1.6)
Acute kidney injury	12 (0.8)	4 (4.5)	16 (1.0)
Respiratory, thoracic and mediastinal disorders	152 (10.5)	3 (3.4)	155 (10.1)
Dyspnoea	42 (2.9)	1 (1.1)	43 (2.8)
Pleural effusion	33 (2.3)	1 (1.1)	34 (2.2)
Respiratory failure	17 (1.2)	0 (0.0)	17 (1.1)
Vascular disorders	29 (2.0)	2 (2.3)	31 (2.0)
Hypotension	8 (0.6)	0 (0.0)	8 (0.5)
Deep vein thrombosis	6 (0.4)	1 (1.1)	7 (0.5)
Embolism	4 (0.3)	0 (0.0)	4 (0.3)
Jugular vein thrombosis	3 (0.2)	0 (0.0)	3 (0.2)

Source: SCS Table 12.6.8.1

About 40 % (similar in the respective studies) experienced a serious TEAE. The most frequently reported by PT was disease progression in both studies (4.5% in study 003 and 8 % in study 001). In the 003 study serious TEAEs reported in > 1 subject each were acute kidney injury, anaemia, abdominal pain, asthenia, cellulitis, and general physical health deterioration. In study 001, other serious TEAEs reported > 2% of subjects each were dyspnoea, abdominal pain, pleural effusion, and pneumonia.

Relatedness to study drug was considered in about 6 % of the patients in study 003 (enterocolitis, infusion related reaction, transaminases increased, chondrocalcinosis, synovitis and tubule-interstitial nephritis in one patient each). In study 001, a similar proportion was reported.

Deaths

Table 53: Deaths by Primary Reason – All Subjects

	001		003		Total	
	SCS (N=1452) n (%)	SUR (N=1650) n (%)	SCS (N=88) n (%)	SUR (N=88) n (%)	SCS (N=1540) n (%)	SUR (N=1738) n (%)
Number of Subjects Who Died	545 (37.5)	867 (52.5)	43 (48.9)	44 (50.0)	588 (38.2)	911 (52.4)
Primary Reason for Death:						
Disease Progression	421 (29.0)	703 (42.6)	40 (45.5)	41 (46.6)	461 (29.9)	744 (42.8)
Adverse Event Related to Study Treatment	6 (0.4)	4 (0.2)	0	0	6 (0.4)	4 (0.2)
Adverse Event not Related to Study Treatment	39 (2.7)	59 (3.6)	0	0	39 (2.5)	59 (3.4)
Other	15 (1.0)	17 (1.0)	0	0	15 (1.0)	17 (1.0)
Unknown	61 (4.2)	80 (4.8)	3 (3.4)	3 (3.4)	64 (4.2)	83 (4.8)
Missing	3 (0.2)	4 (0.2)	0	0	3 (0.2)	4 (0.2)
Number of Deaths from date of first dose to date of last dose +30 days	159 (11.0)	216 (13.1)	12 (13.6)	12 (13.6)	171 (11.1)	228 (13.1)
Primary Reason for Death:						
Disease Progression	121 (8.3)	163 (9.9)	11 (12.5)	11 (12.5)	132 (8.6)	174 (10.0)
Adverse Event Related to Study Treatment	6 (0.4)	4 (0.2)	0	0	6 (0.4)	4 (0.2)
Adverse Event not Related to Study Treatment	23 (1.6)	41 (2.5)	0	0	23 (1.5)	41 (2.4)
Other	5 (0.3)	4 (0.2)	0	0	5 (0.3)	4 (0.2)
Unknown	4 (0.3)	4 (0.2)	1 (1.1)	1 (1.1)	5 (0.3)	5 (0.3)
Missing	0	0	0	0	0	0
Number of Subjects Who Died Within 60 days from First Treatment Administration	149 (10.3)	185 (11.2)	5 (5.7)	5 (5.7)	154 (10.0)	190 (10.9)
Primary Reason for Death:						
Disease Progression	110 (7.6)	142 (8.6)	5 (5.7)	5 (5.7)	115 (7.5)	147 (8.5)
Adverse Event Related to Study Treatment	6 (0.4)	4 (0.2)	0	0	6 (0.4)	4 (0.2)
Adverse Event not Related to Study Treatment	24 (1.7)	30 (1.8)	0	0	24 (1.6)	30 (1.7)
Other	6 (0.4)	4 (0.2)	0	0	6 (0.4)	4 (0.2)
Unknown	3 (0.2)	5 (0.3)	0	0	3 (0.2)	5 (0.3)
Missing	0	0	0	0	0	0

Source: SUR Table 12.6.14.1 and SCS Table 12.6.14.1

In the 003 study about half of the patients died (49 %) with the vast majority due to disease progression (46 %). There were no patients who died due to AE as the primary reason although it is to be noted whilst 8 patients (9%) were reported with TEAEs with a fatal outcome, the primary cause of death was considered due to PD.

In the larger 001 study, few AE-related fatalities were reported (3 % in total regardless of related or not).

Immune related adverse events

Table 54: Summary of irAEs - Study EMR100070- 001 and -003

	Study 001 (N=1452)					**Study 003** (N=88)				
	N (%)			**Median TTO Weeks (range)**	**Median Time to resolution Days**	**N (%)**			**Median TTO Weeks (range)**	**Median Time to resolution Days**
	Any	**Grade 3/4**	**Grade 5**			**Any**	**Grade 3/4**	**Grade 5**		
IrAEs	164 (11.4)			9 (0.1; 70.3)		14 (15.9)			9 (0.1; 26.1)	
Ir Pneumonitis	12 (0.8)	3 (0.2)	1 (0.1)	6 (0.1; 23.6)	21 (4; 99)	0	0	0	-	
Ir Colitis	13 (0.9)	3 (0.2)	0	5 (0.3; 49.9)	18 (2; 299+)	2 (2.3)	0	0	7 (4.1; 9.4)	28.0 (5; 51)
Ir Hepatitis	13 (0.9)	7 (0.48)	1 (0.1)	5.00 (1.1; 42.3)	50 (1; 83)	1 (1.1)	1 (1.1)	0	2 (2.1; 2.1)	77 (77; 77)
Ir Endocrinopathie	70 (4.8)					5 (5.7)				
Ir Thyroid Disorders	66 (4.5)	1 (0.1)	0	12 (2.0; 55.7)	NE (1; 535+)	5 (5.7)	0	0	10 (8.9; 26.1)	105 (42; 372+)
Ir Adrenal Insufficiency	6 (0.4)	1 (0.1)	0	11 (0.1; 32.9)	NE (2; 170+)	0	0	0	-	-
Ir Rash	63 (4.3)	2 (0.1)	0	6 (0.1; 67.1)	57.0 (1; 562+)	5 (5.7)	0	0	9 (0.1; 22.3)	150 (15; 218+)

Abbreviations: IrAEs = Immune-Related Adverse Events, TTO= Time to Onset

Laboratory findings

Haematology

Table 55: Haematology - Summary by Worst on Treatment NCI-CTCAE Toxicity Grade

Parameter	Worst On-Treatment Grade	001 (N=1452) N (%)	003 (N=88) N (%)	Total (N=1540) N (%)
Hemoglobin (g/L) High	Any Grade ≥ 1	14 (1.0)	1 (1.1)	15 (1.0)
	Any Grade ≥ 3	0	0	0
Hemoglobin (g/L) Low	Any Grade ≥ 1	1112 (76.6)	77 (87.5)	1189 (77.2)
	Any Grade ≥ 3	62 (4.3)	9 (10.2)	71 (4.6)
Platelet (10^9/L) Low	Any Grade ≥ 1	245 (16.9)	30 (34.1)	275 (17.9)
	Any Grade ≥ 3	15 (1.0)	1 (1.1)	16 (1.0)
Leukocytes (10^9/L) High	Any Grade ≥ 1	2 (0.1)	0	2 (0.1)
	Any Grade ≥ 3	2 (0.1)	0	2 (0.1)
Leukocytes (10^9/L) Low	Any Grade ≥ 1	260 (17.9)	29 (33.0)	289 (18.8)
	Any Grade ≥ 3	5 (0.3)	1 (1.1)	6 (0.4)
Lymphocytes (10^9/L) High	Any Grade ≥ 1	44 (3.0)	0 (0.0)	44 (2.9)
	Any Grade ≥ 3	2 (0.1)	0 (0.0)	2 (0.1)
Lymphocytes (10^9/L) Low	Any Grade ≥ 1	829 (57.1)	60 (68.2)	889 (57.7)
	Any Grade ≥ 3	182 (12.5)	18 (20.5)	200 (13.0)
Neutrophils (10^9/L) Low	Any Grade ≥ 1	150 (10.3)	5 (5.7)	155 (10.1)
	Any Grade ≥ 3	12 (0.8)	1 (1.1)	13 (0.8)

Source: SCS Tables 12.8.1.11.1 and 12.8.1.11.2

Table 56: Haematology - Shift from Baseline to Highest NCI-CTCAE Grade on Treatment (Pooled Safety Set)

		Worst NCI-CTC Grade at Treatment						
Study	NCI-CTC Grade at Baseline	Grade 0	Grade 1	Grade 2	Grade 3	Grade 4	Missing	Total
Anemia								
Total N=1540	Grade 0	245 (15.9)	253 (16.4)	44 (2.9)	3 (0.2)	0 (0.0)	13 (0.8)	558 (36.2)
	Grade 1	10 (0.6)	451 (29.3)	232 (15.1)	28 (1.8)	0 (0.0)	39 (2.5)	760 (49.4)
	Grade 2	0 (0.0)	12 (0.8)	122 (7.9)	37 (2.4)	0 (0.0)	14 (0.9)	185 (12.0)
	Grade 3	0 (0.0)	0 (0.0)	0 (0.0)	3 (0.2)	0 (0.0)	0 (0.0)	3 (0.2)
	Grade 4	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
	Missing	1 (0.1)	3 (0.2)	1 (0.1)	0 (0.0)	0 (0.0)	26 (1.7)	31 (2.0)
	Total	256 (16.6)	719 (46.7)	399 (25.9)	71 (4.6)	0 (0.0)	92 (6.0)	1537 (99.8)
Lymphocyte count decreased								
Total N=1540	Grade 0	474 (30.8)	155 (10.1)	165 (10.7)	32 (2.1)	1 (0.1)	30 (1.9)	857 (55.6)
	Grade 1	7 (0.5)	105 (6.8)	112 (7.3)	44 (2.9)	2 (0.1)	19 (1.2)	289 (18.8)
	Grade 2	7 (0.5)	16 (1.0)	123 (8.0)	94 (6.1)	8 (0.5)	15 (1.0)	263 (17.1)
	Grade 3	0 (0.0)	3 (0.2)	3 (0.2)	17 (1.1)	1 (0.1)	3 (0.2)	27 (1.8)
	Grade 4	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	1 (0.1)
	Missing	2 (0.1)	3 (0.2)	4 (0.3)	0 (0.0)	0 (0.0)	84 (5.5)	93 (6.0)
	Total	490 (31.8)	282 (18.3)	407 (26.4)	187 (12.1)	13 (0.8)	151 (9.8)	1530 (99.4)

Source: SCS Table 12.8.1.12.2
Note: No Grade 1 events were reported.

Biochemistry

Table 57: Summary of Liver Function Tests by Worst on Treatment NCICTCAE Toxicity Grade

Parameter	Worst On-Treatment Grade	001 (N=1452) N (%)	003 (N=88) N (%)	Total (N=1540) N (%)
Total bilirubin (μmol/L)	Any Grade ≥ 1	114 (7.9)	6 (6.8)	120 (7.8)
	Any Grade ≥ 3	29 (2.0)	1 (1.1)	30 (1.9)
Aspartate aminotransferase (IU/L)	Any Grade ≥ 1	446 (30.7)	40 (45.5)	486 (31.6)
	Any Grade ≥ 3	55 (3.8)	1 (1.1)	56 (3.6)
Alanine aminotransferase (IU/L)	Any Grade ≥ 1	332 (22.9)	21 (23.9)	353 (22.9)
	Any Grade ≥ 3	33 (2.3)	3 (3,4)	36 (2.3)

Source: SCS Table 12.8.2.24.1

Table 58: Biochemistry - Shift from Baseline to Highest NCI-CTCAE Grade on Treatment (Pooled Safety Set)

Study	NCI-CTC Grade at Baseline	Worst NCI-CTC Grade at Treatment						
		Grade 0	Grade 1	Grade 2	Grade 3	Grade 4	Missing	Total
Alanine aminotransferase increased								
Total N=1540	Grade 0	1071 (69.5)	201 (13.1)	22 (1.4)	24 (1.6)	2 (0.1)	55 (3.6)	1375 (89.3)
	Grade 1	36 (2.3)	82 (5.3)	9 (0.6)	9 (0.6)	1 (0.1)	8 (0.5)	145 (9.4)
	Grade 2	0 (0.0)	2 (0.1)	1 (0.1)	0 (0.0)	0 (0.0)	0 (0.0)	3 0.2)
	Grade 3	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
	Grade 4	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
	Missing	5 (0.3)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	7 (0.5)	12 (0.8)
	Total	1112 (72.2)	285 (18.5)	32 (2.1)	33 (2.1)	3 (0.2)	70 (4.5)	1535 (99.7)
Aspartate aminotransferase increased								
Total N=1540	Grade 0	953 (61.9)	275 (17.9)	22 (1.4)	27 (1.8)	0 (0.0)	56 (3.6)	1333 (86.6)
	Grade 1	19 (1.2)	100 (6.5)	30 (1.9)	21 (1.4)	2 (0.1)	7 (0.5)	179 (11.6)
	Grade 2	0 (0.0)	2 (0.1)	1 (0.1)	5 (0.3)	0 (0.0)	1 (0.1)	9 (0.6)
	Grade 3	1 (0.1)	0 (0.0)	0 (0.0)	1 (0.1)	0 (0.0)	0 (0.0)	2 (0.1)
	Grade 4	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
	Missing	3 (0.2)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	9 (0.6)	12 (0.8)
	Total	976 (63.4)	377 (24.5)	53 (3.4)	54 (3.5)	2 (0.1)	73 (4.7)	535 (99.7)
Blood bilirubin increased								
Total N=1540	Grade 0	1323 (85.9)	56 (3.6)	25 (1.6)	19 (1.2)	8 (0.5)	63 (4.1)	1494 (97.0)
	Grade 1	5 (0.3)	4 (0.3)	2 (0.1)	2 (0.1)	0 (0.0)	2 (0.1)	15 (1.0)
	Grade 2	0 (0.0)	1 (0.1)	2 (0.1)	1 (0.1)	0 (0.0)	0 (0.0)	4 (0.3)
	Grade 3	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 0.0)	0 (0.0)
	Grade 4	1 (0.1)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	1 (0.1)
	Missing	11 (0.7)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	7 (0.5)	18 (1.2)
	Total	1340 (87.0)	61 (4.0)	29 (1.9)	22 (1.4)	8 (0.5)	72 (4.7)	1532 (99.5)

Source: SCS Table 12.8.2.25.1

Table 59: Summary of Potential Drug-induced Liver Injuries

Criteria	001 (N=1452) n (%)	003 (N=88) n(%)	Total (N=1540) n (%)
Concurrent (ALT or AST) ≥ 3* x ULN and TBILI ≥ 2 x ULN	21 (1.4)	0 (0.0)	21 (1.4)
Concurrent (ALT or AST) ≥ 3 x ULN and TBILI ≥ 2 x ULN and ALP > 2 x ULN	19 (1.3)	0 (0.0)	19 (1.2)
Concurrent (ALT or AST) ≥ 3 x ULN and TBILI ≥ 2 x ULN and ALP ≤ 2 x ULN or missing	2 (0.1)	0 (0.0)	2 (0.1)

Source: SCS Table 12.8.3.1
Concurrent measurements are those occurring on the same date.
ALP = alkaline phosphatase, ALT = alanine aminotransferase, AST = aspartate aminotransferase, TBIL = total bilirubin, and ULN = upper limit of normal

Renal Function

Table 60: Summary of Creatinine by Worst on Treatment NCI-CTCAE Toxicity Grade

Parameter	Worst On Treatment Grade	001 (N=1452) N (%)	003 (N=88) N (%)	Total (N=1540) N (%)
Creatinine (µmol/L)	Any Grade ≥ 1	349 (24.0)	35 (39.8)	384 (24.9)
	Any Grade ≥ 3	6 (0.4)	0	6 (0.4)

Source: SCS Table 12.8.2.24.1

Table 61: Creatinine - Shift from Baseline to Highest NCI-CTCAE Grade on Treatment (Pooled Safety Set)

Study	NCI-CTC Grade at Baseline	Worst NCI-CTC Grade at Treatment						
		Grade 0	Grade 1	Grade 2	Grade 3	Grade 4	Missing	Total
Creatinine increased								
Total N=1540	Grade 0	1053 (68.4)	198 (12.9)	18 (1.2)	3 (0.2)	1 (0.1)	58 (3.8)	1331 (86.4)
	Grade 1	14 (0.9)	114 (7.4)	27 (1.8)	1 (0.1)	0 (0.0)	5 (0.3)	161 (10.5)
	Grade 2	0 (0.0)	1 (0.1)	19 (1.2)	0 (0.0)	0 (0.0)	1 (0.1)	21 (1.4)
	Missing	4 (0.3)	1 (0.1)	0 (0.0)	0 (0.0)	1 (0.1)	11 (0.7)	17 (1.1)
	Total	1071 (69.5)	314 (20.4)	64 (4.2)	4 (0.3)	2 (0.1)	75 (4.9)	1530 (99.4)

Source: SCS Table 12.8.2.25.1
Note: No Grade 3 or 4 events were reported.

Amylase and Lipase

Table 62: Summary of Amylase and Lipase by Worst on Treatment NCI-CTCAE Toxicity Grade

Parameter	Worst On-Treatment Grade	001 (N=1452) N (%)	003 (N=88) N (%)	Total (N=1540) N (%)
Amylase (IU/L)	Any Grade ≥ 1	158 (10.9)	6 (6.8)	164 (10.6)
	Any Grade ≥ 3	19 (1.3)	1 (1.1)	20 (1.3)
Triacylglycerol Lipase (U/L)	Any Grade ≥ 1	204 (14.0)	15 (17.0)	219 (14.2)
	Any Grade ≥ 3	58 (4.0)	4 (4.5)	62 (4.0)

Source: SCS Table 12.8.2.24.1

Table 63: Amylase and Lipase - Shift from Baseline to Highest NCI-CTCAE Grade on Treatment (Pooled Safety Set)

Study	NCI-CTC Grade at Baseline	Worst NCI-CTC Grade at Treatment						
		Grade 0	Grade 1	Grade 2	Grade 3	Grade 4	Missing	Total
Serum amylase increased								
Total N=1540	Grade 0	955 (62.0)	78 (5.1)	16 (1.0)	7 (0.5)	3 (0.2)	264 (17.1)	1323 (85.9)
	Grade 1	24 (1.6)	27 (1.8)	8 (0.5)	4 (0.3)	0 (0.0)	6 (0.4)	69 (4.5)
	Grade 2	1 (0.1)	4 (0.3)	5 (0.3)	1 (0.1)	0 (0.0)	1 (0.1)	12 (0.8)
	Grade 3	0 (0.0)	0 (0.0)	1 (0.1)	1 (0.1)	1 (0.1)	1 (0.1)	4 (0.3)
	Grade 4	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	2 (0.1)	0 (0.0)	2 (0.1)
	Missing	75 (4.9)	5 (0.3)	0 (0.0)	1 (0.1)	0 (0.0)	7 (0.5)	88 (5.7)
	Total	1055 (68.5)	114 (7.4)	30 (1.9)	14 (0.9)	6 (0.4)	279 (18.1)	1498 (97.3)
Lipase increased								
Total N=1540	Grade 0	875 (56.8)	71 (4.6)	29 (1.9)	22 (1.4)	7 (0.5)	246 (16.0)	1250 (81.2)
	Grade 1	27 (1.8)	20 (1.3)	10 (0.6)	6 (0.4)	2 (0.1)	13 (0.8)	78 (5.1)
	Grade 2	9 (0.6)	5 (0.3)	4 (0.3)	8 (0.5)	1 (0.1)	1 (0.1)	28 (1.8)
	Grade 3	6 (0.4)	2 (0.1)	5 (0.3)	11 (0.7)	3 (0.2)	4 (0.3)	31 (2.0)
	Grade 4	6 (0.4)	2 (0.1)	5 (0.3)	11 (0.7)	3 (0.2)	4 (0.3)	31 (2.0)
	Missing	74 (4.8)	6 (0.4)	5 (0.3)	1 (0.1)	1 (0.1)	21 (1.4)	108 (7.0)
	Total	991 (64.4)	104 (6.8)	53 (3.4)	48 (3.1)	14 (0.9)	285 (18.5)	1495 (97.1)

Source: SCS Table 12.8.2.25.1

Safety in special populations

Table 64: Treatment Emergent Adverse Events by Age Group – Safety Analysis Set

Event	001 (N=1452) n (%)	003 (N=88) n (%)	Total (N=1540) n (%)
Number of Subjects by Sub-Groups			
< 65	808 (55.6)	22 (25.0)	830 (53.9)
≥ 65 - < 75	435 (30.0)	35 (39.8)	470 (30.5)
≥ 75 - < 85	185 (12.7)	28 (31.8)	213 (13.8)
≥ 85	24 (1.7)	3 (3.4)	27 (1.8)
Subjects With At Least One Treatment Emergent Adverse Event			
< 65	751 (92.9)	22 (100.0)	773 (93.1)
≥ 65 - < 75	410 (94.3)	35 (100.0)	445 (94.7)
≥ 75	198 (94.7)	29 (93.5)	227 (94.6)
Treatment Emergent Adverse Events Grade ≥ 3			
< 65	427 (52.8)	14 (63.6)	441 (53.1)
≥ 65 - < 75	204 (46.9)	24 (68.6)	228 (48.5)
≥ 75	92 (44.0)	16 (51.6)	108 (45.0)
Related Treatment Emergent Adverse Events			
< 65	528 (65.3)	18 (81.8)	546 (65.8)
≥ 65 - < 75	271 (62.3)	26 (74.3)	297 (63.2)
≥ 75	143 (68.4)	18 (58.1)	161 (67.1)
Related Treatment Emergent Adverse Events Grade ≥ 3			
< 65	91 (11.3)	2 (9.1)	93 (11.2)
≥ 65 - < 75	37 (8.5)	2 (5.7)	39 (8.3)
≥ 75	18 (8.6)	0 (0.0)	18 (7.5)
Treatment Emergent Adverse Event Leading to Permanent Treatment Discontinuation			
< 65	97 (12.0)	0 (0.0)	97 (11.7)
≥ 65 - < 75	66 (15.2)	1 (2.9)	67 (14.3)
≥ 75	30 (14.4)	1 (3.2)	31 (12.9)
Related Treatment Emergent Adverse Events Leading to Permanent Treatment Discontinuation			
< 65	45 (5.6)	0 (0.0)	45 (5.4)
≥ 65 - < 75	32 (7.4)	1 (2.9)	33 (7.0)
≥ 75	13 (6.2)	0 (0.0)	13 (5.4)
Serious Treatment Emergent Adverse Events			
< 65	345 (42.7)	8 (36.4)	353 (42.5)
≥ 65 - < 75	165 (37.9)	18 (51.4)	183 (38.9)
≥ 75	68 (32.5)	10 (32.3)	78 (32.5)
Related Serious Treatment Emergent Adverse Events			
< 65	45 (5.6)	1 (4.5)	46 (5.5)
≥ 65 - < 75	27 (6.2)	4 (11.4)	31 (6.6)
≥ 75	13 (6.2)	0 (0.0)	13 (5.4)
Treatment Emergent Adverse Events Leading to Death			
< 65	99 (12.3)	1 (4.5)	100 (12.0)
≥ 65 - < 75	46 (10.6)	5 (14.3)	51 (10.9)
≥ 75	18 (8.6)	2 (6.5)	20 (8.3)
Related Treatment Emergent Adverse Events Leading to Death			
< 65	4 (0.5)	0 (0.0)	4 (0.5)
≥ 65 - < 75	2 (0.5)	0 (0.0)	2 (0.4)
≥ 75	0 (0.0)	0 (0.0)	0 (0.0)
Immune-related Adverse Events			
< 65	90 (11.1)	5 (22.7)	95 (11.4
≥ 65 - < 75	52 (12.0)	8 (22.9)	60 (12.8)
≥ 75	23 (11.0)	1 (3.2)	24 (10.0)
Treatment Related Immune-related Treatment Emergent Adverse Event			
< 65	77 (9.5)	3 (13.6)	80 (9.6)
≥ 65 - < 75	41 (9.4)	8 (22.9)	49 (10.4)
≥ 75	22 (10.5)	0 (0.0)	22 (9.2)
Infusion-related Reactions			
< 65	180 (22.3)	8 (36.4)	188 (22.7)
≥ 65 - < 75	110 (25.3)	7 (20.0)	117 (24.9)
≥ 75	62 (29.7)	4 (12.9)	66 (27.5)
Treatment Related Infusion-related Reactions			
< 65	175 (21.7)	8 (36.4)	183 (22.0)
≥ 65 - < 75	101 (23.2)	7 (20.0)	108 (23.0)
≥ 75	60 (28.7)	4 (12.9)	64 (26.7)

Source: SCS Table 12.6.1.2.2
Note: Immune-related adverse events and infusion-related reactions according to the updated case definitions.

No dose adjustment is proposed on the basis of age. An analysis of TEAEs in patients **> 65** years or **≤65**, as well as for patients >75 and ≤75 years, were provided for the avelumab treatment as showed in table above.

Table 65: TEAEs in elderly patients by age <65, 65-74, 75-84, ≥85

	Age			
	< 65 years N=929 n (%)	65-74 years N=533 n (%)	75-84 years N=249 n (%)	≥ 85 years N=27 n (%)
Any TEAE	906 (97.5)	521 (97.7)	244 (98.0)	26 (96.3)
Serious TEAE	427 (46.0)	238 (44.7)	100 (40.2)	12 (44.4)
• Fatal TEAE	127 (13.7)	72 (13.5)	25 (10.0)	4 (14.8)
• Leading to (prolongation of) hospitalization	392 (42.2)	216 (40.5)	89 (35.7)	10 (37.0)
• Life-threatening	32 (3.4)	21 (3.9)	9 (3.6)	1 (3.7)
• Disability/incapacitiy	14 (1.5)	7 (1.3)	4 (1.6)	0
• Other (medically significant)	53 (5.7)	26 (4.9)	16 (6.4)	0
TEAE leading to study termination	72 (7.8)	43 (8.1)	29 (11.6)	2 (7.4)
Psychiatric disorders (SOC)	166 (17.9)	86 (16.1)	61 (24.5)	3 (11.1)
Nervous system disorders (SOC)	277 (29.8)	147 (27.6)	87 (34.9)	10 (37.0)
Accidents and injuries (SMQ)	83 (8.9)	46 (8.6)	32 (12.9)	5 (18.5)
Cardiac disorders (SOC)	88 (9.5)	64 (12.0)	39 (15.7)	2 (7.4)
Vascular disorders (SOC)	176 (18.9)	102 (19.1)	51 (20.5)	8 (29.6)
Central nervous system vascular disorders (SMQ)	12 (1.3)	12 (2.3)	8 (3.2)	1 (3.7)
Infections and Infestations (SOC)	342 (36.8)	190 (35.6)	98 (39.4)	13 (48.1)
Anticholinergic syndrome (PT)	0	0	0	0
Quality of life decreased (PT)	0	0	0	0
Sum of postural hypotension, falls, black outs, syncope, dizziness, ataxia, fractures	126 (13.6)	78 (14.6)	56 (22.5)	4 (14.8)

Gender

TEAEs

TEAEs with at least a 5% difference between males and females respectively in the pooled safety set, were fatigue (27 % vs 32 %), weight decreased (18 % vs 13 %), nausea (18 % vs 27 %), diarrhoea (14 % vs 20 %), vomiting (12 % vs 17 %), abdominal pain (10 % vs 15 %), and urinary tract infection (3 % vs 12 %).

IrAEs and IRRs were similar between males and females.

Race

The vast majority of the patients were Caucasian (92 % and 82 % in 003 and 001 respectively).

Geographic Region

Table 66: Number of patients geographical region

	001 (N=1452) n (%)	003 (N=88) n (%)	Total (N=1540) n (%)
Number of Subjects by Sub-Groups			
North America	1108 (76.3)	51 (58.0)	1159 (75.3)
Western Europe	222 (15.3)	29 (33.0)	251 (16.3)
Eastern Europe	46 (3.2)	0 (0.0)	46 (3.0)
Australia	0 (0.0)	5 (5.7)	5 (0.3)
Asia	76 (5.2)	3 (3.4)	79 (5.1)

Table 67: Most Common TEAEs by PT (at Least 5% Difference Between Geographic Regions) – Pooled Safety Set

	Geographic Region	
Preferred Term	**North America N (%)**	**Western Europe N (%)**
Fatigue	384 (33.0)	49 (19.5)
Nausea	292 (25.1)	42 (16.7)
Constipation	221 (19.0)	29 (11.6)
Abdominal pain	162 (13.9)	21 (8.4)
Back pain	143 (12.3)	16 (6.4)
Arthralgia	129 (11.1)	15 (6.0)
Dehydration	85 (7.3)	5 (2.0)
Hyponatreamia	76 (6.5)	4 (1.6)
Asthenia	61 (5.2)	47 (18.7)
Abdominal pain upper	33 (2.8)	27 (10.8)

Source: SCS Table 12.6.2.2.5

Notes:

1. For subgroup illustrations of events, Australia was combined with North America.
2. Immune-related adverse events according to the updated case definitions.

Safety related to drug-drug interactions and other interactions

Table 68: Anti-drug Antibody Results

Subjects at Risk	EMR100070-001	EMR100070-003	Overall
Ever positive n/N0 (%)	53/1396 (3.8)	3/88 (3.4)	56/1484 (3.8)
Pre-existing positive n/N1 (%) Treatment boosted n/N2 (%)	7/1230 (0.6) 0/1142	0/85 0/79	7/1315 (0.5) 0/1221
Treatment emergent n/N3 (%)	46/1301 (3.5)	3/82 (3.7)	49/1383 (3.5)
Transient treatment emergent n/N3 (%)	15/1301 (1.2)	1/82 (1.2)	16/1383 (1.2)
Persistent treatment emergent n/N3 (%)	31/1301 (2.4)	2/82 (2.4)	33/1383 (2.4)

Source: SCS Table 12.9.2.1.1

N0: number of subjects with at least one valid result at any time point.

N1: number of subjects with a valid baseline result.

N2: Number of subjects with valid baseline result and at least one valid post-baseline result.

N3: Number of subjects with at least one valid post baseline result and without positive baseline result.

Treatment emergent: Not positive prior to treatment with avelumab and with at least 1 positive post-baseline result

Transient: If treatment-emergent subjects have a single positive evaluation, or duration between first and last positive result < 16 weeks and last assessment not positive.

Persistent: If treatment-emergent subjects have duration between first and last positive result ≥ 16 weeks or a positive evaluation at the last assessment.

Table 69: Safety Sub-analysis by Immunogenicity Status (pooled safety set)

Event	Ever- vs. Never-positive ADA[a]		Ever- vs. Never-positive nAb[b]		All Subjects[c] (N = 1738) n (%)
	Ever-positive (N = 106) n (%)	Never-positive (N = 1614) n (%)	Ever-positive (N = 41) n (%)	Never-positive (N = 1679) n (%)	
TEAE	105 (99.1)	1575 (97.6)	40 (97.6)	1640 (97.7)	1697 (97.6)
TEAE Grade ≥ 3	68 (64.2)	928 (57.5)	24 (58.5)	972 (57.9)	1008 (58.0)
TEAE leading to permanent treatment discontinuation	26 (24.5)	216 (13.4)	10 (24.4)	232 (13.8)	244 (14.0)
TEAEs excluding IRRs leading to drug interruption	31 (29.2)	330 (20.4)	9 (22.0)	352 (21.0)	363 (20.9)
Serious TEAE	57 (53.8)	708 (43.9)	16 (39.0)	749 (44.6)	777 (44.7)
TEAE Leading to Death	12 (11.3)	207 (12.8)	1 (2.4)	218 (13.0)	228 (13.1)
irAEs	15 (14.2)	231 (14.3)	4 (9.8)	242 (14.4)	251 (14.4)
IRR	41 (38.7)	393 (24.3)	18 (43.9)	416 (24.8)	439 (25.3)

Source: Module 5.3.5.3 90daySUR Table 12.6.1.1; Module 5.3.5.3 ADA NAB Table 12.6.1.2.11 and Table 12.20.1.1.1.

ADA: antidrug antibody; AE: adverse event; irAE: immune-related adverse event; IRR: infusion-related reaction; n: number of subjects meeting condition; N: number of subjects in category; nAb: neutralizing antibody; TEAE: treatment-emergent adverse event.

[a] Ever-positive ADA are all subjects with at least 1 positive ADA result, at either baseline or posttreatment. Never-positive are subjects negative at all time points.

[b] Ever-positive nAb are all subjects with at least 1 positive nAb result, at either baseline or posttreatment. Never-positive are subjects negative at all time points.

[c] All subjects includes the 18 subjects who do not have valid ADA results.

Discontinuation due to adverse events

Table 70: Subject Disposition and Primary Reason for Discontinuation from Study – Safety Analysis Set

	Status / Reason	001 (N=1452) n (%)	003 (N=88) n (%)	Total (N=1540) n (%)
Safety Analysis Set Subjects	n	1452 (100.0)	88 (100.0)	1540 (100.0)
	Still on Treatment	406 (28.0)	26 (29.5)	432 (28.1)
	Off Treatment	1046 (72.0)	62 (70.5)	1108 (71.9)
	Re-initiated Treatment	1 (0.1)	0 (0.0)	1 (0.1)
	Off Treatment after Re-initiation	1 (0.1)	0 (0.0)	1 (0.1)
	Still in Follow-Up	396 (27.3)	15 (17.0)	411 (26.7)
	Off Study	651 (44.8)	47 (53.4)	698 (45.3)
Reason Off Treatment	Adverse Event	146 (10.1)	2 (2.3)	148 (9.6)
	Lost to Follow-Up	5 (0.3)	1 (1.1)	6 (0.4)
	Protocol Non-Compliance	9 (0.6)	1 (1.1)	10 (0.6)
	Death	62 (4.3)	7 (8.0)	69 (4.5)
	Disease Progression	724 (49.9)	44 (50.0)	768 (49.9)
	Withdrew Consent	56 (3.9)	4 (4.5)	60 (3.9)
	Other	44 (3.0)	3 (3.4)	47 (3.1)
Reason Off Study	Lost to Follow-Up	28 (1.9)	1 (1.1)	29 (1.9)
	Death	533 (36.7)	39 (44.3)	572 (37.1)
	Withdrew Consent	78 (5.4)	7 (8.0)	85 (5.5)
	Other	11 (0.8)	0 (0.0)	11 (0.7)
	Study Reached Predefined End	1 (0.1)	0 (0.0)	1 (0.1)

Source: SCS Table 12.1.1

Note: Subject 100070-001-154-0007 in Study EMR100070-001 was listed as meeting predefined study end on 06 October 2015.It was subsequently determined that the subject had died; the database has been corrected for the final analysis.

Table 71: Most Common TEAEs Leading to Permanent Treatment Discontinuation (at Least 2 Subjects in the Total Group) by SOC and PT – All Subjects; abbreviated by the assessor

Body System or Organ Class Preferred term	001 Subjects (N=1452) n (%)	003 Subjects (N=88) n (%)	Total Subjects (N=1540) n (%)
Number of Subjects With At Least One Event	193 (13.3)	2 (2.3)	195 (12.7)
Blood and lymphatic system disorders	5 (0.3)	0 (0.0)	5 (0.3)
Anaemia	2 (0.1)	0 (0.0)	2 (0.1)
Cardiac disorders	5 (0.3)	1 (1.1)	6 (0.4)
Cardiac tamponade	2 (0.1)	0 (0.0)	2 (0.1)
Gastrointestinal disorders	23 (1.6)	0 (0.0)	23 (1.5)
Diarrhoea	4 (0.3)	0 (0.0)	4 (0.3)
General disorders and administration site conditions	41 (2.8)	0 (0.0)	41 (2.7)
Disease progression	29 (2.0)	0 (0.0)	29 (1.9)
Fatigue	6 (0.4)	0 (0.0)	6 (0.4)
Hepatobiliary disorders	5 (0.3)	0 (0.0)	5 (0.3)
Autoimmune hepatitis	2 (0.1)	0 (0.0)	2 (0.1)
Immune system disorders	5 (0.3)	0 (0.0)	5 (0.3)
Autoimmune disorder	2 (0.1)	0 (0.0)	2 (0.1)
Infections and infestations	10 (0.7)	0 (0.0)	10 (0.6)
Lung infection	2 (0.1)	0 (0.0)	2 (0.1)
Sepsis	2 (0.1)	0 (0.0)	2 (0.1)
Injury, poisoning and procedural complications	33 (2.3)	0 (0.0)	33 (2.1)
Infusion related reaction	31 (2.1)	0 (0.0)	31 (2.0)
Investigations	26 (1.8)	1 (1.1)	27 (1.8)
Aspartate aminotransferase increased	7 (0.5)	0 (0.0)	7 (0.5)
Gamma-glutamyltransferase increased	7 (0.5)	0 (0.0)	7 (0.5)
Alanine aminotransferase increased	4 (0.3)	0 (0.0)	4 (0.3)
Musculoskeletal and connective tissue disorders	10 (0.7)	0 (0.0)	10 (0.6)
Arthralgia	3 (0.2)	0 (0.0)	3 (0.2)
Myositis	2 (0.1)	0 (0.0)	2 (0.1)
Neoplasms benign, malignant and unspecified (incl cysts and polyps)	9 (0.6)	0 (0.0)	9 (0.6)
Myelodysplastic syndrome	2 (0.1)	0 (0.0)	2 (0.1)
Renal and urinary disorders	3 (0.2)	0 (0.0)	3 (0.2)
Acute kidney injury	2 (0.1)	0 (0.0)	2 (0.1)
Respiratory, thoracic and mediastinal disorders	18 (1.2)	0 (0.0)	18 (1.2)
Vascular disorders	2 (0.1)	0 (0.0)	2 (0.1)
Embolism	2 (0.1)	0 (0.0)	2 (0.1)

Source: SCS Table 12.6.6.1

Updated: Overall, 6.2% of the patients discontinued avelumab due to treatment-related TEAEs. In Study 003, 3 additional subjects were reported to have treatment-related TEAEs leading to treatment discontinuation (total of 4 subjects, PTs of GGT increased, ALT increased, blood CPK increased, ileus, and transaminase increased [the latter was already reported]). One additional case of treatment discontinuation, already described in the previous SCS, was reported beyond the on-treatment period (increased creatinine due to treatment-related interstitial nephritis).

Dose reductions and Dose delays

Dose reductions where not foreseen in the **003** and **001**studies as this was not allowed as per clinical study protocols. Dose reductions in this analysis were defined as a dose of < 90% of the planned dose. Potential reasons for a dose < 90% of the planned dose included premature discontinuations of an infusion due to occurrence of an IRR and an increased weight not taken into account for calculation of the planned dose at a given visit. Maximum deviations in the schedule of 1 or 2 days were not considered as a delay.

Table 72: Summary of Dose Reductions and Dose Delays - Safety Analysis Set

Characteristic	Statistics	001 (N=1452)	003 (N=88)	Total (N=1540)
At least one Dose Reduction, N (%)	0	1405 (96.8)	88 (100.0)	1493 (96.9)
	1	40 (2.8)	0 (0.0)	40 (2.6)
	2	6 (0.4)	0 (0.0)	6 (0.4)
	3	1 (0.1)	0 (0.0)	1 (0.1)
	≥ 4	0 (0.0)	0 (0.0)	0 (0.0)
No Dose Delay, N (%)[a]	No delay	1151 (79.3)	49 (55.7)	1200 (77.9)
	0 day	786 (54.1)	23 (26.1)	809 (52.5)
	1 day	278 (19.1)	19 (21.6)	297 (19.3)
	2 days	87 (6.0)	7 (8.0)	94 (6.1)
Subjects with at least one Dose Delay, N (%)		301 (20.7)	39 (44.3)	340 (22.1)
	Worst dose delay 3-6 days delay	63 (4.3)	10 (11.4)	73 (4.7)
	3 days	39 (2.7)	8 (9.1)	47 (3.1)
	4 days	18 (1.2)	2 (2.3)	20 (1.3)
	5 days	4 (0.3)	0 (0.0)	4 (0.3)
	6 days	2 (0.1)	0 (0.0)	2 (0.1)
	≥ 7 days delay	238 (16.4)	29 (33.0)	267 (17.3)
	7-13 days	61 (4.2)	7 (8.0)	68 (4.4)
	14-20 days	151 (10.4)	18 (20.5)	169 (11.0)
	21-27 days	5 (0.3)	1 (1.1)	6 (0.4)
	≥ 28 days	21 (1.4)	3 (3.4)	24 (1.6)

Source: SCS Table 12.5.1

[a] For completeness, the number of subjects with no delay (0 days), 1 or 2 days deviation from the planned 14-day infusion schedule are presented but are not considered as a delay.

Post marketing experience

The applicant did not submit post marketing experience as the product has not yet been marketed.

2.6.1. Discussion on clinical safety

Adverse reactions were reported for 88 patients with metastatic MCC treated with avelumab 10 mg/kg and adverse reactions reported for 1,650 patients in a phase I study in other solid tumours. As it is not anticipated that the safety profile of a mAb would differ to any major extent relative to different types of solid tumours, hence, the pooling of the safety data is acceptable as it allows for a more comprehensive evaluation of the short term safety.

The mean/median treatment duration was 23 w/17 weeks and 15 w/12 weeks in 003 and 001 respectively with a maximum number of avelumab infusions of 35 and 51 (median 7 infusions) respectively. The vast majority achieved a relative dose intensity of > 90 % with a fair consistency throughout cycles.

The TEAEs reported were similar across the studies. TEAEs reported in ≥ 15% of patients in study 003 were (in decreasing rates) fatigue, diarrhoea, nausea, decreased appetite, peripheral oedema, constipation, cough and arthralgia whilst in study 001 it was fatigue, nausea, constipation, diarrhoea, infusion related reaction, decreased appetite and weight decreased.

The vast majority of the TEAEs reported in both studies were considered related to study drug (71 % in the 003 study and 65 % in 001). Almost all patients experienced at least one TEAE and whilst there was quite a high proportion Grade ≥3 and serious TEAEs reported (61 % and 41 % in 003; 50 % and 40 % in

001), the treatment was relatively well tolerated was demonstrated by the low rate of permanent treatment discontinuations (2.3 % in 003 and 13 % in 001). Dose escalation or reduction is not recommended. Dosing delay or discontinuation may be required based on individual safety and tolerability; see Table 1 of the SmPC.

A high proportion of Grade ≥3 TEAEs occurred in the respective studies (61 % in study 003 and 50 % in study 001). In study 003, TEAEs with Grade ≥ 3 reported in ≥ 3% of subjects were anaemia, hypertension, disease progression, lymphopenia, GGT increased and lipase increased whilst in study 001, it was disease progression, anaemia, dyspnoea, abdominal pain and hyponatremia.

The most frequently reported serious TEAE was disease progression in both studies (4.5% in study 003 and 8 % in study 001). In the 003 study serious TEAEs reported in > 1 subject each were acute kidney injury, anaemia, abdominal pain, asthenia, cellulitis, and general physical health deterioration. In study 001, other serious TEAEs reported > 2% of subjects each were dyspnoea, abdominal pain, pleural effusion, and pneumonia. Relatedness to study drug was considered in about 6 % of the patients in study 003 (enterocolitis, infusion related reaction, transaminases increased, chondrocalcinosis, synovitis and tubule-interstitial nephritis in one patient each). In study 001, a similar proportion was reported.

In a safety update (data cut-off date 09 June 2016) in the combined 001 + 003 study population for a total of 1738 patients, Grade 3 events (N=650, 37 %) were observed for anaemia (5.8%), hypertension (4.3%), hyponatremia (3.7%) and dyspnoea (3.5%), Grade 4 events were observed for sepsis (1.3%), lipase increased (0.6%) and respiratory failure (0.6%). Other TEAEs with at least 5 subjects reporting included hypokalaemia and dyspnoea (each 0.5%), acute kidney injury (0.3%), and hyponatremia (0.3%). Grade 5 events (13 %) were predominantly due to disease progression (8.4%) and 0.5% died due to respiratory failure, 0.3% from pneumonia and 0.3% from sepsis.

Immune related adverse events were not considered higher than what is expected for an immunotherapy agent (overall 16 % in study 003 and 11 % in study 001 with the rate of Grade ≥3 rather low [1 % in 003 and 1.6 % in 001]. Detailed guidelines for the management of immune-related adverse reactions (immune-related hepatitis, immune-related colitis, immune-related endocrinopaties, immune-related nephritis) are described in section 4.4 of the SmPC. A description of relevant immune-related ADRs (immune related pneumonitis, immune related hepatitis, immune related colitis, immune related endocrinopathies, immune-related nephritis and renal dysfunction, are presented in section 4.8 of the SmPC.

Avelumab is most frequently associated with immune-related adverse reactions. Most of these, including severe reactions, resolved following initiation of appropriate medical therapy or withdrawal of avelumab (see “Description of selected adverse reactions” below).

The safety of avelumab has been evaluated in 1,738 patients with solid tumours including metastatic MCC receiving 10 mg/kg every 2 weeks of avelumab in clinical studies. In this patient population, the most common adverse reactions with avelumab were fatigue (32.4%), nausea (25.1%), diarrhoea (18.9%), decreased appetite (18.4%), constipation (18.4%), infusion-related reactions (17.1%), weight decreased (16.6%), and vomiting (16.2%).

The most common Grade ≥ 3 adverse reactions were anaemia (6.0%), dyspnoea (3.9%), and abdominal pain (3.0%). Serious adverse reactions were immune-related adverse reactions and infusion-related reaction (see section 4.4).

Approximately 7 % and 6 % of the patients in **003** and **001** respectively, required steroid treatment for irAEs whereof 4.5% and 3.4% received high-dose corticosteroid therapy. Systemic high-dose corticosteroid treatment was given for 1.1% and 1.2% of the patients with Grade ≥ 3 irAEs in **003** and **001** respectively.

A similar proportion of patients experienced at least one infusion-related reaction (IRR) in the 003 and 001 study (22 % and 24 % respectively). The IRR events were mainly of Grade 1 or 2. Overall, a small proportion of subjects (14 subjects [1.0%]) experienced Grade ≥ 3 IRRs (all in study **001**) and only 27 subjects (~ 2 %) permanently discontinued study drug administration due to an IRR (likewise all in study **001**). No fatal IRR was reported in either study and no Grade 4 IRR was reported in the **003** study. A warning and recommendation on how to manage IRRs have been included in the SmPC in 4.4. Patients have to be premedicated with an antihistamine and with paracetamol prior to the first 4 infusions of Bavencio. If the fourth infusion is completed without an infusion-related reaction, premedication for subsequent doses should be administered at the discretion of the physician.

Of 1,738 patients treated with avelumab 10 mg/kg as an intravenous infusion every 2 weeks, 1,627 were evaluable for treatment-emergent anti-drug antibodies (ADA) and 96 (5.9%) tested positive. In ADA positive patients, there may be an increased risk for infusion-related reactions (about 40% and 25% in ADA ever-positive and ADA never-positive patients, respectively). Based on data available, including the low incidence of immunogenicity, the impact of ADA on pharmacokinetics, efficacy and safety is uncertain, while the impact of neutralizing antibodies (nAb) is unknown.

There were no reports of fatal events due to AEs in study 003. In the larger 001 study, there were rather few AE-related fatalities reported (3 % in total regardless of related or not).

A safety update was provided with additional safety information of 3 months for study 003 and about 7 months for study 001 (cut-off 9th June 2016). No major concerns were raised on the safety data provided, providing further reassurance of the consistency of the safety data.

There is a lack of data on patients with active central nervous system (CNS) metastasis, active or with a history of any autoimmune disease, a history of other malignancies within the last 5 years, organ transplant, conditions requiring therapeutic immune suppression or active infection with HIV or hepatitis B or C, as these patients were excluded from the clinical trial (as reflected in sections 4.4 and 5.1 of the SmPC). Hence there is lack of data regarding immunocompromised patients. In the absence of clinical safety data, avelumab should be used with caution in these populations after careful consideration of the potential risk-benefit on an individual basis. An Early Access Programme and the German real-world cohort study has been requested as an additional PhV activity (PASS) to address the missing information of safety and efficacy in immune compromised patients (patients with autoimmune disease, HIV, Hepatitis B or C infections or organ transplants).

For patients with adrenal insufficiency, patients should be monitored for signs and symptoms of adrenal insufficiency during and after treatment. Corticosteroids should be administered (1 to 2 mg/kg/day prednisone intravenously or oral equivalent) for Grade ≥ 3 adrenal insufficiency followed by a taper until a dose of less than or equal to 10 mg/day has been reached. Avelumab should be withheld for Grade 3 or Grade 4 symptomatic adrenal insufficiency (see section 4.2).

Avelumab can cause Type 1 diabetes mellitus, including diabetic ketoacidosis (see section 4.8). Patients should be monitored for hyperglycaemia or other signs and symptoms of diabetes. Initiate treatment with insulin for Type 1 diabetes mellitus. Avelumab should be withheld and anti-hyperglycaemics in patients with Grade ≥ 3 hyperglycaemia should be administered. Treatment with avelumab should be resumed when metabolic control is achieved on insulin replacement therapy.

The effect of avelumab on male and female fertility is unknown. Although studies to evaluate the effect of avelumab on fertility have not been conducted, there were no notable effects in the female reproductive organs in monkeys based on 1 month and 3 month repeat dose toxicity studies (see section 5.3). In addition, human IgG1 immunoglobulins are known to cross the placental barrier. Therefore, avelumab has the potential to be transmitted from the mother to the developing foetus. It is not recommended to use avelumab during pregnancy unless the clinical condition of the woman requires treatment with avelumab.

Avelumab has negligible influence on the ability to drive and use machines. Fatigue has been reported following administration of avelumab (see section 4.8). Patients should be advised to use caution when driving or operating machinery until they are certain that avelumab does not adversely affect them.

Reporting suspected adverse reactions after authorisation of the medicinal product is important. It allows continued monitoring of the benefit/risk balance of the medicinal product. Healthcare professionals are asked to report any suspected adverse reactions via the national reporting system listed in Appendix V.

Three patients were reported to be overdosed with 5% to 10% above the recommended dose of avelumab. The patients had no symptoms, did not require any treatment for the overdose, and continued on avelumab therapy. In case of overdose, patients should be closely monitored for signs or symptoms of adverse reactions. The treatment is directed to the management of symptoms.

Special precautions for disposal and handling instructions are presented in section 6.6 of the SmPC.

2.6.2. Conclusions on the clinical safety

The safety data collected at an earlier data cut remains consistent with longer follow up. No major concerns have been identified in the updated analysis. The safety of avelumab in the proposed indication appears to be acceptable and manageable with the recommendations as proposed in the SmPC and the RMP. The safety risks associated with immune related adverse reactions are managed through additional risk minimisation activities implemented in the form of educational materials that will inform HCPs and patients on how to identify and properly handle suspected immune related ADRs.

The CHMP considers the following measures necessary to address issues related to safety:

- PASS: German real-world cohort study should be submitted as additional PhV activity to address the missing information of safety and efficacy in immune compromised patients.

2.7. Risk Management Plan

Safety concerns

Table 73: Summary of the Safety Concerns

Summary of safety concerns	
Important identified risks	• Immune-related pneumonitis • Immune-related hepatitis • Immune-related colitis • Immune-related endocrinopathies (thyroid disorders, adrenal insufficiency, type 1 diabetes mellitus, pituitary disorders) • Other immune-related events (myositis, myocarditis, Guillain-Barre Syndrome, uveitis)

	• Immune-related nephritis and renal dysfunction • Severe infusion-related reactions (grade ≥ 3)
Important potential risks	• Other immune-related events (encephalitis, myasthenic syndrome, pancreatitis) • Severe cutaneous reactions • Immunogenicity • Embryofoetal toxicity
Missing information	Safety in patients • With Autoimmune disease • With HIV, Hepatitis B or C infections • With Organ transplants Use during lactation Long-term treatment Safety and efficacy in immune compromised patients

Pharmacovigilance plan

Table 74: Ongoing and planned studies in the post-authorisation pharmacovigilance plan

Study / Activity Type, Title and category	Objectives	Safety Concerns addressed	Status (planned, started)	Date for submission
Non-interventional cohort study to assess characteristics and management of patients with Merkel Cell Carcinoma in Germany (category 3)	5-year open cohort study of patients with MCC in Germany to 1) describe patient characteristics (including co-morbidities and concomitant medications), 2) estimate background rates of potential safety events (including immune mediated events), 3) describe treatment patterns, and 4) characterize disease outcomes (effectiveness and safety). Objectives related to effectiveness/ safety outcomes will also be assessed in the sub-group of immune compromised patients treated	Safety and efficacy in immune compromised patients	Planned	Study protocol to be submitted to PRAC within 2 months after granting of marketing authorisation Final study report: Q1 2024

	with avelumab, and an exploratory objective (due to expected limited sample size) will compare these outcomes in immune compromised patients with the ones in immune competent patients.			

Risk minimisation measures

Table 75: Summary table of Risk Minimisation Measures

Safety Concern	**Routine risk minimization measures**	**Additional risk miminimization measures**
	Important Identified Risks	
Immune-related pneumonitis	SmPC wording in sections 4.2, 4.4 and 4.8	Educational material to be provided to HCPs and to patients
Immune-related hepatitis	SmPC wording in sections 4.2, 4.4 and 4.8	Educational material to be provided to HCPs and to patients
Immune-related colitis	SmPC wording in sections 4.2, 4.4 and 4.8	Educational material to be provided to HCPs and to patients
Immune-related endocrinopathies (thyroid disorders)	SmPC wording in sections 4.2, 4.4 and 4.8	Educational material to be provided to HCPs and to patients
Immune-related endocrinopathies (adrenal insufficiency)	SmPC wording in sections 4.2, 4.4 and 4.8	Educational material to be provided to HCPs and to patients
Immune-related endocrinopathies (type 1 diabetes mellitus)	SmPC wording in sections 4.2, 4.4 and 4.8	Educational material to be provided to HCPs and to patients
Immune-related endocrinopathies (pituitary disorder)	SmPC wording in sections 4.2, 4.4 and 4.8	Educational material to be provided to HCPs and to patients.
Other Immune-related events – myositis	SmPC wording in sections 4.2, 4.4 and 4.8	Educational material to be provided to HCPs and to patients.
Other immune-related events – myocarditis	SmPC wording in sections 4.2, 4.4 and 4.8	Educational material to be provided to HCPs and to patients
Other immune-related events - Guillain-Barre Syndrome	SmPC wording in sections 4.2, 4.4 and 4.8	Educational material to be provided to HCPs and to patients

Safety Concern	Routine risk minimization measures	Additional risk miminimization measures
Other immune-related event (uveitis)	SmPC wording in sections 4.2, 4.4 and 4.8	Educational material to be provided to HCPs and to patients
Immune-related nephritis and renal dysfunction	SmPC wording in sections 4.2, 4.4 and 4.8	Educational material to be provided to HCPs and to patients
Severe infusion-related reactions	SmPC wording in sections 4.2, 4.4 and 4.8	Educational material to be provided to HCPs and to patients
Important Potential Risks		
Other immune-related events (encephalitis)	SmPC wording in section 4.2	None
Other immune-related events (myasthenic syndrome)	SmPC wording in section 4.2	None
Other immune-related event (pancreatitis)	SmPC wording in section 4.2	None
Severe cutaneous reactions	SmPC wording in section 4.2	None
Immunogenicity	SmPC wording in section 4.8	None
Embryofetal toxicity	SmPC wording in sections 4.6 and 5.3	None
Missing information		
Safety in patients with autoimmune disease	SmPC wording in sections 4.4 and 5.1	None
Safety in patients with HIV, Hepatitis B or C	SmPC wording in sections 4.4 and 5.1	None
Safety in patients with organ transplants	SmPC wording in sections 4.4 and 5.1	None
Use during lactation	SmPC wording in sections 4.6	None
Long-term safety	None	None
Safety and efficacy in immune compromised patients	SmPC wording in sections 4.4	None

Conclusion

The CHMP and PRAC considered that the risk management plan version 1.6 is acceptable.

2.8. Pharmacovigilance

Pharmacovigilance system

The CHMP considered that the pharmacovigilance system summary submitted by the applicant fulfils the requirements of Article 8(3) of Directive 2001/83/EC.

Periodic Safety Update Reports submission requirements

The requirements for submission of periodic safety update reports for this medicinal product are set out in the Annex II, Section C of the CHMP Opinion. The applicant requested alignment of the PSUR cycle with the international birth date (IBD). The IBD is 23 March 2017. The new EURD list entry will therefore use the IBD to determine the forthcoming Data Lock Points.

2.9. New Active Substance

The applicant declared that avelumab has not been previously authorised in a medicinal product in the European Union.

The CHMP, based on the available data, considers avelumab to be a new active substance as it is not a constituent of a medicinal product previously authorised within the Union.

2.10. Product information

2.10.1. User consultation

The results of the user consultation with target patient groups on the package leaflet submitted by the applicant show that the package leaflet meets the criteria for readability as set out in the *Guideline on the readability of the label and package leaflet of medicinal products for human use.*

2.10.2. Additional monitoring

Pursuant to Article 23(1) of Regulation No (EU) 726/2004, Bavencio (avelumab) is included in the additional monitoring list as it contains a new active substance and is to be approved under a conditional marketing authorisation.

Therefore the summary of product characteristics and the package leaflet includes a statement that this medicinal product is subject to additional monitoring and that this will allow quick identification of new safety information. The statement is preceded by an inverted equilateral black triangle.

3. Benefit-Risk Balance

3.1. Therapeutic Context

3.1.1. Disease or condition

Merkel cell carcinoma is a rare, aggressive, neuroendocrine cancer associated with UV exposure, Merkel cell polyomavirus, immunosuppression (8-10% of the MCC patients, mainly in relation to CLL, organ transplant and HIV infection, associated with a bleak prognosis) and elderly Caucasians (≥65 years old).

Its incidence is 0.2-0.4 cases/100 000 individuals / year in Europe, while in the US it is 0.79 and in Australia 1.6 (where it is mainly linked to ultraviolet exposure). The median age at diagnosis is around 75 years. A minority of cases are metastatic at presentation, 5-12%. The overall 5-year survival for node-negative disease is about 60%, in regional nodal disease at presentation about 40% and drops to <20% in the metastatic setting.

3.1.2. Available therapies and unmet medical need

There are currently no approved therapies for metastatic Merkel cell carcinoma where the standard of care is chemotherapy and radiotherapy. The most commonly used first-line chemotherapy regimen in disseminated disease is a platinum compound ± etoposide, resulting in high response rates (60-70%), but poor duration of response. With respect to relapsed disease, study data are sparse, but the ORR is much lower than in the first-line setting and duration is brief.

There is an unmet medical need for both first-line and in treatment experienced patients with metastatic disease.

3.1.3. Main clinical studies

The pivotal study EMR100070-003 is an ongoing multicentre, single-arm study designed in 2 parts to evaluate the efficacy and safety of avelumab in subjects with metastatic Merkel cell carcinoma (mMCC):

- Part A tested avelumab in patients with mMCC previously treated with at least one line of chemotherapy and progressed after the most recent regimen – second line or later therapy (n=88)
- Part B is ongoing and includes chemotherapy treatment-naïve mMCC patients – first line therapy (n=39).

3.2. Favourable effects

The results from Part A in previously treated mMCC patients showed an overall ORR of 33%, formally rejecting the null hypothesis of ORR≤20%. The majority of responses (22/29) were observed early, around week 6-7 and these responses proved to be durable: in the pre-planned analysis at 12 months after the last patient started treatment (cutoff 3rd of September 2016), from 29 patients with a confirmed response, 25 had a DOR>6 months. The longest ongoing response durations included one CR of 23.3

months and 3 patients with 18 months response, 2 of which were CRs. The median PFS was 2.7 months (95% CI 1.4-6.9) and the immune-related median PFS was 4 months (95% CI 2.3, -).

The prelimiary results for Part B in treatment-naïve patients are consistent with the results in pre-treated patients and non-confirmed BOR was about 70%, i.e. comparable to what is achievable with intensive chemotherapy. Based on the tumour activity observed, a detrimental effect on OS is not expected.

3.3. Uncertainties and limitations about favourable effects

As the 003 study was designed as a single arm trial, there is no comparator arm to determine the true effect size observed for avelumab in terms of ORR, DOR and PFS for both chemotherapy-treated and naive patients. The retrospective observational study included few patients and only descriptive analyses were performed. Hence, as the data is not deemed to be robust, no clear conclusion could be drawn from the study and the data can only be considered as supportive. As chemotherapy-treated patients have few options after relapse, avelumab appears to be at least as good, if not better than chemotherapy for 2nd line treatment. For 1st line treatment, while ORR and DOR appear promising, more mature PFS data are required from Part B in order to better estimate patient benefit. Data from the final analysis are required to confirm the magnitude of the effect of the treatment in terms of PFS, but also in terms of possible patient survival. Therefore the CHMP recommends a conditional approval with the specific obligation to submit the final results from Study EMR 100070-003 Part B.

A relationship between PDL-1 expression and ORR has been demonstrated and MCV expression seems to also impact on ORR. Therefore, the CHMP recommends the applicant to try to identify biomarkers that will help select patients that are likely to benefit from avelumab.

3.4. Unfavourable effects

The pooling of the safety from both studies and of the different indications is acceptable as the safety profile of avelumab would not differ relative to other types of solid tumours. The safety database is considered acceptable and of sufficient size in order to identifying most of the safety risks associated with avelumab treatment. The safety update provided an additional follow up of 3 months for study 003 and about 7 months for study 001. No new major concerns regarding the safety profile of avelumab was raised although long term safety has been included in the RMP as missing information. The safety of avelumab has been evaluated in 1,738 patients with solid tumours including metastatic MCC receiving 10 mg/kg every 2 weeks of avelumab in clinical studies. In this patient population, the most common adverse reactions with avelumab were fatigue (32.4%), nausea (25.1%), diarrhoea (18.9%), decreased appetite (18.4%), constipation (18.4%), infusion-related reactions (17.1%), weight decreased (16.6%), and vomiting (16.2%).

The ADRs identified are consistent between the two studies submitted. No concerns were raised in terms of tolerability for avelumab from an exposure perspective and the ADRs identified consistent with those expected for an immunotherapy. Immune-related ADRs such as pneumonitis, colitis, hepatitis, endocrinopathies (thyroid disorders, adrenal insufficiency, type 1 diabetes mellitus, pituitary disorders), nephritis and renal dysfunction as well as other immune-related events (myositis, myocarditis, Guillain-Barre Syndrome, uveitis) have been included in the RMP as important identified risks and will be managed through routine as well as additional risk minimisation activities. Educational material will be provided to patients and HCP in order to increase awareness and provide information concerning the signs and symptoms of certain important identified risks of avelumab.

The most common Grade ≥ 3 adverse reactions were anaemia (6.0%), dyspnoea (3.9%), and abdominal pain (3.0%). Serious adverse reactions were immune-related adverse reactions and infusion-related reaction (see section 4.4).

Approximately 7 % and 6 % of the patients in **003** and **001** respectively, required steroid treatment for irAEs whereof 4.5% and 3.4% received high-dose corticosteroid therapy. High-dose corticosteroid treatment was given for 1.1% and 1.2% of the patients with Grade ≥ 3 irAEs in **003** and **001** respectively. Recommendation on the use of steroids to manage immune related ADRs are included in the SmPC section 4.4.

Infusion related reactions (IRR) were observed in the **003** and **001** study (22 % and 24 % respectively). No fatal IRR was reported in either study. Recommendation on the management of IRR is included in the SmPC section 4.4. Therefore, severe infusion-related reactions (grade ≥ 3) have been included in the RMP as an important identified risk.

No deaths occurred in the studies that have been attributed to avelumab. At the time of the data cut-off date in the **003** study, about half of the patients had died (49 %) with the majority due to disease progression
(46 %).

3.5. Uncertainties and limitations about unfavourable effects

From the safety database, there is uncertainty as to the potential relationship of avelumab with some of the observed immune-related events such as encephalitis, myasthenic syndrome, pancreatitis and severe cutaneous reactions. These safety concerns have been included in the RMP as important potential risks and will be managed through routine PhV. Embryo-foetal toxicity is also an important potential risk that has been described in non-clinical models, however there is no data in humans (SmPC section 4.6 and 5.3). There is recommendation in the SmPC for women to use effective contraception during treatment. There is also missing information on the safety of patients with autoimmune disease, HIV, Hepatitis B or C infections and organ transplants as these patients have been excluded from the entry criteria for the study EMR100070-003. These will be monitored through routine PhV plan.

It is unknown whether avelumab is excreted in human milk. Since it is known that antibodies can be secreted in human milk, a risk to the newborns/infants cannot be excluded. Breast-feeding women should be advised not to breast-feed during treatment and for at least 1 month after the last dose due to the potential for serious adverse reactions in breast-fed infants. The use of bavencio in lactation was included as missing information in the RMP.

3.6. Effects Table

Table 76: Effects Table for Bavencio in metastatic Merkel cell carcinoma

Effect	Short Description	Unit	Outcome of treatment	Uncertainties/ Strength of evidence
Favourable effects study EMR100070-003				
PART A				
ORR	CR+PR	%	33 (95% CI 23; 44%)	12 months minimum follow-up

Effect	Short Description	Unit	Outcome of treatment	Uncertainties / Strength of evidence
Durable response rate	At least 6 months response duration among all subjects treated	%	31 (95% CI 21; 40%)	12 months minimum follow-up
PFS	Median, range	mo	2.7 (95% CI 1.4-6.9) (0.03-24.5)	Underestimate due to "plateau" irPFS: 4 (0.03-22.1) 95% CI 2.3, -
PART B				
ORR	CR+PR	%	62 (95% CI 42; 79)	13 weeks minimum follow-up
Duration of response for subjects with confirmed response	Proportion of duration of response with at least 6 months duration	%	83 (95% CI 46; 96)	13 weeks minimum follow-up
Study 100070OBS-001 (comparator arm - qualified patients)				
ORR 2nd line Part A Part B 1st line Part A	CR+PR	%	 20.0% (95%CI 5.7, 43.7) 8.8% (95%CI 1.9, 23.7) 31.3% (95%CI 20.6, 43.8)	Sample size is small to draw conclusions Part A N= 67 Part B N= 34
DOR 2nd line Part A Part B 1st line Part A	duration of time from first documented CR/PR, to the earliest date of first progression or recurrent disease, or date of death (median)	months	 1.7 months (95%CI 0.5, 0.3) 1.9 months (95%CI 1.3, 2.1) 5.7 months (95%CI 2.6,8.7)	
PFS 2nd line Part A Part B 1st line Part A	date of treatment to the date of progression, or date of death due to any cause, or date of initiation of new regimen (median)	months	 2.1 months (95%CI1.0, 3.2) 3.0 months (95%CI 2.6, 3.1) 4.6 months (95%CI 3.0, 7.0)	
Unfavourable effects				
Treatment duration PART A Mean Median	n	Weeks	 23 17	

Effect	Short Description	Unit	Outcome of treatment	Uncertainties / Strength of evidence
	TEAEs Any Fatigue Diarrhoea Nausea Appetite ↓ Oedema periph Constipation Cough Arthralgia	%	 98 38 23 21 19 18 17 17 16	
	IrAEs[a] Any Hypothyroidism Arthritis Hyperthyroidism		 16 5 3 2	None led to treatment discontinuation. One patient (1 %) had a Grade ≥3 IrAE (immune-mediated hepatitis).
	IRR Any IRR Chills Pyrexia	%	 22 17 5 3	There were no Grade ≥3 reports and none led to treatment discontinuation. One patient (1 %) was reported with a Grade 2 SAE of IRR.
	Related TEAEs Any Fatigue IRR Nausea Diarrhoea Asthenia Rash Appetite ↓	%	 71 24 15 9 9 8 7 6	
	Grade ≥ 3 TEAE Any Anaemia Lymphopenia hypertension	%	 61 10 7 6	
	SAEs Any AKI PD	%	 41 5 5	
	Deaths All PD TEAE Unknown	%	 49 46 0 3	Of note, 8 patients (9%) were reported with TEAEs with a fatal outcome although for all subjects, the primary cause of death was considered due to PD.
	Discontinuations due to AEs Any	%	2.3	2 patients permanently discontinued treatment including one patient with pericardial effusion (un-related) and one patient with a SAE of Grade 3 transaminases increased (related).
N=85	**HAHA** Pre-existing Treatment-emergent positive	%	0 5.9%	

Abbreviations: AKI= Acute Kidney Injury; TEAEs= Treatment emergent adverse events; IrAEs= Immune-related Adverse Events ; SAEs= Serious Adverse Events ; HAHA= Human Anti-Human Antibodies

3.7. Benefit-risk assessment and discussion

3.7.1. Importance of favourable and unfavourable effects

mMCC is a rare and aggressive cancer where tumour progression is related to symptom worsening and death. There is no consensus on the best standard therapy for treating mMCC patients. Patients with metastatic disease treated with chemotherapy have shown good tumour responses but responses are mostly of short duration and patients progress within a few months.

After progression on chemotherapy, there are no active next-lines of therapies available. The primary endpoint of the pivotal study EMR100070-003 was met and a clinically relevant benefit has been demonstrated with avelumab in the treatment of patients that have been previously treated with chemotherapy. The reported ORR (33%) for avelumab in the next-line treatment of mMCC is not considered outstanding, however, durability of responses is convincing. Preliminary efficacy data seems to indicate that chemotherapy naïve patients also respond to treatment, which may result in a better outcome than previously treated patients, as patients with fewer lines of treatment seem to respond better to treatment. The data is still immature and a further update of the efficacy data is required in order to better characterise the treatment effect. The safety of avelumab is as expected for an immunotherapy product with immune adverse reactions which will be monitored through routine and additional risk minimisation activities. In general the safety is considered acceptable and manageable.

3.7.2. Balance of benefits and risks

The favourable results with avelumab monotherapy in both naïve and pretreated mMCC patients terms of ORR and DOR outweigh the safety risks observed and are considered acceptable.

3.7.3. Additional considerations on the benefit-risk balance

The indication is based on the submitted data from Part A of the pivotal EMR100070-003 study which included chemotherapy-pretreated mMCC patients as well as Part B which enrolled 112 treatment-naïve subjects. The applicant has provided interim data of Part B, including 25 treatment-naïve patients followed for a minimum of 6 weeks (among them, 16 have ≥ 13 weeks follow up and 5 ≥ 6 months follow up). The preliminary data provided from Part B showed high response rates to avelumab in treatment-naïve patients, as well as preliminary evidence of prolonged duration of responses in some patients. This could be seen as a confirmation of the results in 2+ line setting and as such, it is expected that mMCC patients treated in first line would also derive a clinically meaningful benefit in a rare disease where there are no approved treatment options. There was no evidence of a detrimental effect in naïve patients in terms of PFS or survival, although the follow up data is currently limited. Therefore, the CHMP considered that there was enough clinical evidence to not restrict the line of treatment in the indication.

Conditional marketing authorisation

As comprehensive clinical data on the safety and efficacy of the medicinal product are not available, a conditional marketing authorisation was requested by the applicant in the initial submission.

The product falls within the scope of Regulation (EC) No 507/2006 concerning conditional marketing authorisations, as it aims at the treatment of a life-threatening disease and is designated as an orphan medicinal product.

Furthermore, the CHMP considers that the product fulfils the requirements for a conditional marketing authorisation for the reasons detailed below:

- The benefit-risk balance is positive: A positive benefit risk has been demonstrated in treatment naïve patients as well as pretreated patients, as discussed above.
- It is likely that the applicant will be able to provide comprehensive data: The applicant is in a position to provide further updated data in the ongoing Part B of study EMR100070-003 to confirm the observed clinical benefit. The CHMP considers that the totality of data available after submission of these results will be comprehensive for this condition.
- Unmet medical needs will be fulfilled: There are no approved treatments for this highly aggressive condition. Chemotherapy is currently the *de facto* standard of care (although not authorised for this condition), with a good ORR but patients relapse quickly as the durability of the response is short. Avelumab, based on the available scientific data, is expected to provide non-inferior ORR, longer duration of response and more favourable safety profile, as compared with the current standard of care.
- The benefit to public health of the medicinal product's immediate availability on the market outweighs the risks due to need for further data: The safety of avelumab appears consistent with other targeted immunotherapeutic products such as anti-PD-1/PDL-1 targeting medicinal products and is superior to safety of current *de facto* standard of care chemotherapy (but not authorised for this condition). The safety risks will be mitigated by routine PhV as well as educational material as additional risk minimisation measures. In the absence of approved therapies in this life threatening condition, and taking into account the positive benefit-risk balance of avelumab, the CHMP considered that it would be appropriate to allow immediate availability of avelumab on the market.

It is proposed that the final results of study part B would be considered as specific obligations. The applicant is required to provide the study report of the primary analysis which will be conducted 15 months after the accrual of the last subject (Q2 2019), by 30 January 2020.

3.8. Conclusions

The overall B/R of Bavencio is positive.

Divergent positions are appended to this report.

4. Recommendations

Outcome

Based on the CHMP review of data on quality, safety and efficacy, the CHMP considers by majority decision that the risk-benefit balance of Bavencio is favourable in the following indication:

Bavencio is indicated as monotherapy for the treatment of adult patients with metastatic Merkel cell carcinoma (MCC).

The CHMP therefore recommends the granting of the conditional marketing authorisation subject to the following conditions:

Conditions or restrictions regarding supply and use

Medicinal product subject to restricted medical prescription (see Annex I: Summary of Product Characteristics, section 4.2).

Other conditions and requirements of the marketing authorisation

Periodic Safety Update Reports

The requirements for submission of periodic safety update reports for this medicinal product are set out in the list of Union reference dates (EURD list) provided for under Article 107c(7) of Directive 2001/83/EC and any subsequent updates published on the European medicines web-portal.

The marketing authorisation holder shall submit the first periodic safety update report for this product within 6 months following authorisation.

Conditions or restrictions with regard to the safe and effective use of the medicinal product

Risk Management Plan (RMP)

The MAH shall perform the required pharmacovigilance activities and interventions detailed in the agreed RMP presented in Module 1.8.2 of the marketing authorisation and any agreed subsequent updates of the RMP.

An updated RMP should be submitted:

- At the request of the European Medicines Agency;
- Whenever the risk management system is modified, especially as the result of new information being received that may lead to a significant change to the benefit/risk profile or as the result of an important (pharmacovigilance or risk minimisation) milestone being reached.

Additional risk minimisation measures

Prior to launch of Bavencio in each Member State the marketing authorisation holder (MAH) must agree about the content and format of the educational programme, including communication media, distribution modalities, and any other aspects of the programme, with the National Competent Authority.

The educational programme is aimed at increasing awareness and providing information concerning the signs and symptoms of certain important identified risks of avelumab, including immune-related pneumonitis, hepatitis, colitis, thyroid disorders, adrenal insufficiency, type 1 diabetes mellitus, nephritis and renal dysfunction, myocarditis, myositis, hypopituitarism, uveitis, Guillain-Barre syndrome and infusion related reactions, and how to manage them.

The MAH shall ensure that in each Member State where Bavencio is marketed, all healthcare professionals and patients/carers who are expected to prescribe and use Bavencio have access to/are provided with the following educational package:

- Healthcare Professional / Frequently Asked Question Brochure
- Patient Information Brochure

- Patient Alert Card

The physician educational material should contain:

  - o The Summary of Product Characteristics
  - o Healthcare professionals brochure

- **The healthcare professional / Frequently Asked Question brochure** shall contain the following key elements:

  - o Relevant information (e.g. seriousness, severity, frequency, time to onset, reversibility as applicable) of the following safety concerns associated with the use of Bavencio:
    - Immune-Related Pneumonitis
    - Immune-Related Hepatitis
    - Immune-Related Colitis
    - Immune-Related Endocrinopathies (diabetes mellitus, thyroid disorders, adrenal insufficiency)
    - Immune-related nephritis and renal dysfunction
    - Other immune-related adverse reactions including myocarditis, myositis, hypopituitarism, uveitis and Guillain-Barre Syndrome
    - Infusion-Related Reactions
  - o Description of the signs and symptoms of immune-related adverse reactions.
  - o Details on how to minimise the safety concerns through appropriate monitoring and management.
  - o Reminder to distribute the patient brochure with the patient alert card to all patients receiving treatment with Bavencio and to advise them to carry the patient alert card at all times and show it to any healthcare professional who may treat them.
  - o Reminder to educate patients/caregivers about the symptoms of immune-related adverse reactions and of the need to report them immediately to the physician.

The patient educational material should contain

  - o The package leaflet
  - o Patient Information brochure
  - o Patient Alert Card

- **The Patient Information brochure** shall contain the following key messages:

  - o Brief introduction to the tool and its purpose
  - o Brief introduction to Bavencio treatment
  - o Recommendation to consult the package leaflet
  - o Information that avelumab can cause serious side effects during or after treatment, that need to be treated right away and warning message on the importance of being aware of signs and symptoms while receiving avelumab treatment
  - o Reminder of the importance to consult their doctor before any change of treatment or in case of side effect

- **The Patient Alert Card** shall contain the following key messages:
  - o Brief introduction to avelumab (indication and purpose of this tool)
  - o Description of the main signs and symptoms of the following safety concerns and reminder of the importance of notifying their treating physician immediately if symptoms occur, persist or worsen:
    - Immune-Related Pneumonitis
    - Immune-Related Hepatitis
    - Immune-Related Colitis
    - Immune-Related Endocrinopathies (diabetes mellitus, thyroid disorders, adrenal insufficiency)
    - Immune-related nephritis and renal dysfunction
    - Other immune-related adverse reactions including myocarditis, myositis, hypopituitarism, uveitis and Guillain-Barre Syndrome
    - Infusion-Related Reactions
  - o Warning message for patients on the importance of consulting their doctor immediately in case they develop any of the listed signs and symptoms and on the important not attempting to treat themselves.
  - o Reminder to carry the Patient Alert Card at all times and to show it to all healthcare professionals that may treat them.
  - o The card should also prompt to enter contact details of the physician and include a warning message for healthcare professionals treating the patient at any time, including in conditions of emergency, that the patient is using Bavencio.

Specific Obligation to complete post-authorisation measures for the conditional marketing authorisation

This being a conditional marketing authorisation and pursuant to Article 14(7) of Regulation (EC) No 726/2004, the MAH shall complete, within the stated timeframe, the following measures:

Description	**Due date**
In order to confirm the efficacy for chemotherapy-naïve treated patients, the MAH should submit the final results of study EMR 100070-003 – Part B.	30th January 2020.

Conditions or restrictions with regard to the safe and effective use of the medicinal product to be implemented by the Member States

Not applicable.

New Active Substance Status

Based on the CHMP review of the available data, the CHMP considers that avelumab is a new active substance as it is not a constituent of a medicinal product previously authorised within the European Union.

APPENDIX 1

Divergent position dated 20 July 2017

Divergent position – Bavencio (EMEA/H/C/004338)

It is the opinion of the undersigned members:

The evidence available to date regarding efficacy and safety of Bavencio (avelumab) in the chemotherapy-naïve mMCC population is considered insufficient to support a positive benefit/risk.

Current data on the treatment-naïve mMCC patients from the pivotal study EMR100070-003 (Cohort B) indicate an ORR of 71% in 14 patients with at least 6 months follow up and an ORR of 65% in the 29 pts with at least 13 weeks follow-up. From these data no superiority in ORR can be concluded when compared to chemotherapy (ORR up to 70%). The median DoR of chemotherapy is 6 months, however, the DoR in Bavencio-treated patients is immature and early evidence of durability is observed in only 6 subjects reported to have a DoR of at least 6 months. Further concerns are raised over the very small number of patients enrolled in Part-B of Study 003 (n=29/112) and the limited follow-up duration taking into consideration that alternative treatment options (i.e., chemotherapy) are available with relatively high rates of response. Moreover, indirect comparison with sparse historical data does not support superiority of Bavencio in terms of efficacy in comparison with chemotherapy.

Furthermore, extrapolation of data from the second to the first line is hampered, since the first line population compared to the second line population may have different prognostic characteristics e.g. less indolent disease, lower mutational loads etc. Moreover, in second line patients part of the susceptibility to immunotherapy could be related to the higher mutational load induced by platinum containing therapy, whereas part of the susceptibility in first line may be more related to disruption of the immune response after DNA integration of MCV.

The unmet need in this setting is acknowledged, as well as the rarity of the disease. However, whether and to what extent Bavencio can fulfill the unmet medical in the mMCC treatment naïve population is unclear at this time.

In view of all the above, the B/R of Bavencio in a mMCC chemotherapy naïve population is considered negative at this time and approval cannot be granted for this population.

London, 20 July 2017

Johann Lodewijk Hillege (The Netherlands)

Natalja Karpova (Latvia)

Romaldas Maciulaitis (Lithuania)